Agentic Trading: When LLM Agents Meet Financial Markets

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Abstract

The integration of Large Language Models (LLMs) with autonomous agent architectures is transforming quantitative finance, shifting algorithmic trading from isolated predictive modeling toward fully autonomous, closed-loop trading agents. These systems perceive multimodal market signals, maintain active context, associative vector stores, and structured knowledge bases, execute complex strategic reasoning, emit executable trading actions, and adapt continuously to non-stationary market regimes. This survey provides a comprehensive, systematic, and forward-looking synthesis of the rapidly evolving landscape of agentic trading. We structure the literature through a unified **Architecture–Capability–Adaptation (A-C-A)** conceptual framework: (i) *Architecture* deconstructs the foundational cognitive loop into perception (textual, numerical, and multimodal), memory systems (working memory, associative vector storage, and structured knowledge bases), cognitive reasoning (reactive, reflective, and planning-driven), and execution mechanisms (order routing and friction management); (ii) *Capability* examines primary financial functions, including automated alpha discovery, portfolio optimization, and multi-tier risk control; and (iii) *Adaptation* investigates how agents evolve through in-context learning, supervised fine-tuning, reinforcement learning, multi-agent coordination, and architectural self-evolution. Drawing on 185 systematically selected studies (including a core empirical subset of $n = 80$ systems evaluated under closed-loop market conditions), we analyze design spaces, operational trade-offs, and emerging paradigms. Furthermore, as an essential cross-cutting dimension, we examine empirical evaluation practices across the literature—including backtesting split schemes, transaction-cost realism, survivorship-bias handling, and execution fidelity—and distill methodological best practices to guide future research. By bridging artificial intelligence, market microstructure, and quantitative finance, this survey aims to serve as a foundational roadmap for designing, benchmarking, and deploying reliable agentic trading systems.

Keywords: Large language models, Trading agents, Autonomous financial systems, Algorithmic trading, Multi-agent systems, System architecture

1. Introduction

The integration of Large Language Models (LLMs) into financial trading marks a shift from traditional quantitative prediction models toward more autonomous, agentic systems. Conventional quantitative finance approaches typically employ black-box machine learning models that generate price or return predictions without explicit reasoning chains or adaptive capabilities. In contrast, LLM-based trading agents often implement an explicit loop—perceiving market information, storing experiences, reasoning about decisions, executing trades, and learning from outcomes Zhang et al. (2024); Xiao et al. (2024); Yu et al. (2024).

Figure 1 situates representative papers in the chronological development of the field, illustrating the evolution from domain-adapted financial language models to multimodal and multi-agent trading systems. To provide a structured and comprehensive navigation of this design space, this survey synthesizes **185** included studies screened through **2026-09-30**, combining an initial multi-source ledger (cutoff 2026-03-09) with a formal extension window covering 2026-03-10 to 2026-09-30. To help readers distinguish full trading systems from component modules, the literature is organized into a primary empirical cohort ($n = 80$) evaluated under closed-loop market conditions (satisfying *Action Output* + *Closed-Loop Evaluation*) and an informative background tier (**105** studies) covering foundation models, isolated predictors, and specialized tools. Furthermore, to provide an objective view of current empirical practices, our synthesis also systematically documents how key experimental protocols are reported across the literature—including time-consistent split designs (**50/80**), explicit transaction costs (**47/80**), universe survivorship controls (**48/80**), and execution semantics (**47/80**).

Realizing the potential of agentic trading systems involves distinct engineering and architectural considerations. First, LLM-based agents generate natural-language rationales and multi-step reasoning traces; ensuring that these explanations faithfully reflect execution requires grounded observation spaces and structured decision logs. Second, agents can adapt their strategies under changing market regimes by drawing on episodic memory and dynamic reflection, demanding robust

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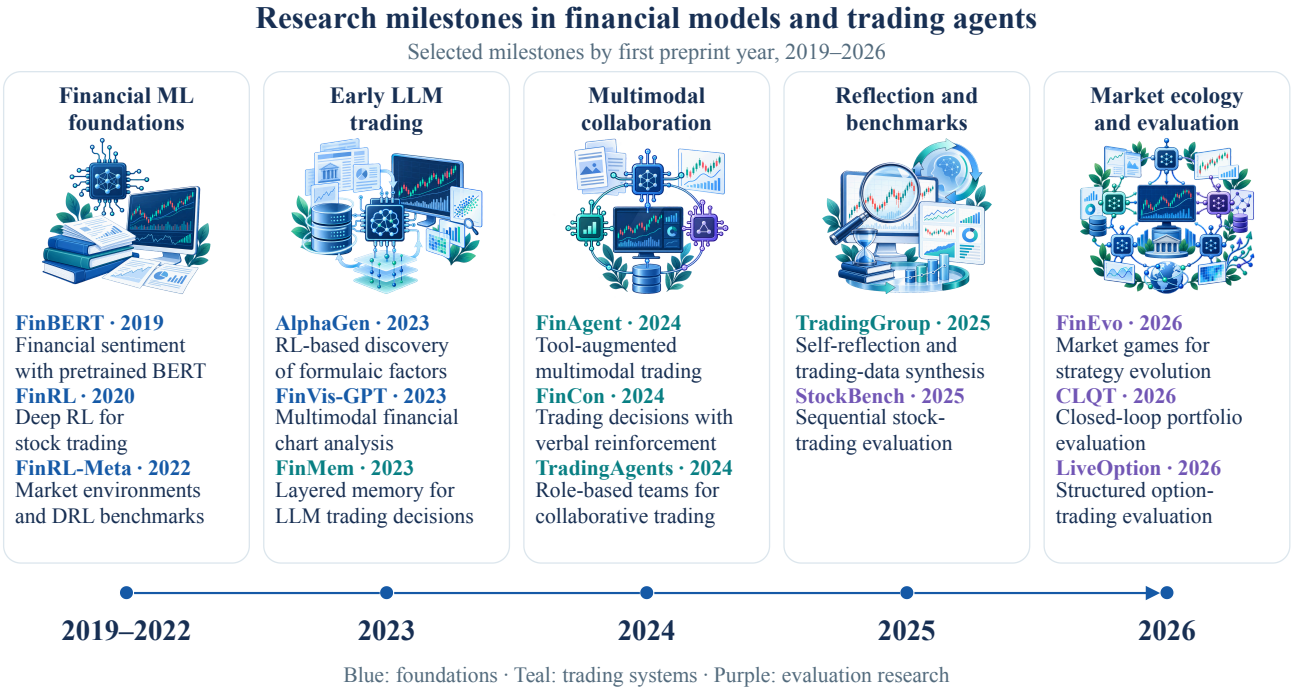

Figure 1: Selected research timeline of financial language models and agentic trading, 2019–2026. Five thematic periods connect financial language modeling and reinforcement-learning foundations with early LLM trading, multimodal collaboration, reflection and evaluation, and market ecology with expanded evaluation. Each paper is labeled by its first arXiv submission year. Panel widths are for readability and do not represent period duration. Foundation papers provide historical context rather than expanding the empirical review corpus Araci (2019); Liu et al. (2020, 2022b); Yu et al. (2023a); Wang et al. (2023b); Yu et al. (2023b); Zhang et al. (2024); Yu et al. (2024); Xiao et al. (2024); Tian et al. (2025); Chen et al. (2025); Zou et al. (2026); Qu and Chen (2026); Luo et al. (2026a).

risk boundaries against catastrophic drift. Third, modular designs allow specialized components for perception, memory, reasoning, and execution (see Figure 2), enabling systems to be decoupled, tested, and enhanced incrementally.

Trading agents are essentially autonomous financial decision-support pipelines: they transform raw, multimodal market observations into executable actions subject to real-world friction, liquidity constraints, and supervisory controls. A comprehensive survey of this domain must therefore examine the complete decision loop: what information channels are perceived, how memory is indexed, which reasoning paradigms generate trading hypotheses, how orders are routed, and how agents adapt to non-stationary market feedback.

1.1. Related Surveys and Complementary Positioning

As artificial intelligence intersects with financial engineering, several surveys and benchmark suites have mapped distinct facets of this evolving domain (summarized in Table 1). We position our work as complementary to these existing contributions across four main lines:

General Autonomous Agent Architectures. Foundational surveys such as Wang et al. (2024) provide overarching taxonomies for LLM-based autonomous agents across general domains, structuring the field around agent profiling, memory systems, planning capabilities, and action spaces. While essential for establishing general cognitive concepts, these frameworks naturally do not address domain-critical financial realities such as order execution semantics, market frictions, slippage, and temporal causality. The present survey builds upon these general agent concepts and specializes them to the financial market environment.

Financial Machine Learning and Econometric Factor Modeling. Methodological surveys in quantitative finance, such as Giglio et al. (2022), synthesize machine learning in asset pricing, focusing on expected return estimation, factor discovery, stochastic discount factor modeling, and econometric testing. These works provide statistical and theoretical grounding for alpha discovery. In contrast, our survey centers on the *agentic* dimension—how automated models transition from statistical predictors into active, autonomous decision-makers that manage portfolios and execute orders.

Financial Foundation Models and Standardized Task Benchmarks. Initiatives such as FinBen Xie et al. (2024), PIXIU Xie et al. (2023), and InvestorBench Li et al. (2025), alongside open models like FinGPT Wang et al. (2023a), provide vital infrastructure and standardized benchmarks for evaluating LLMs on financial NLP and isolated forecasting tasks. Our survey complements these task-level benchmarks by examining end-to-end, closed-loop agent systems where perception, reasoning, and execution operate together in live or simulated trading

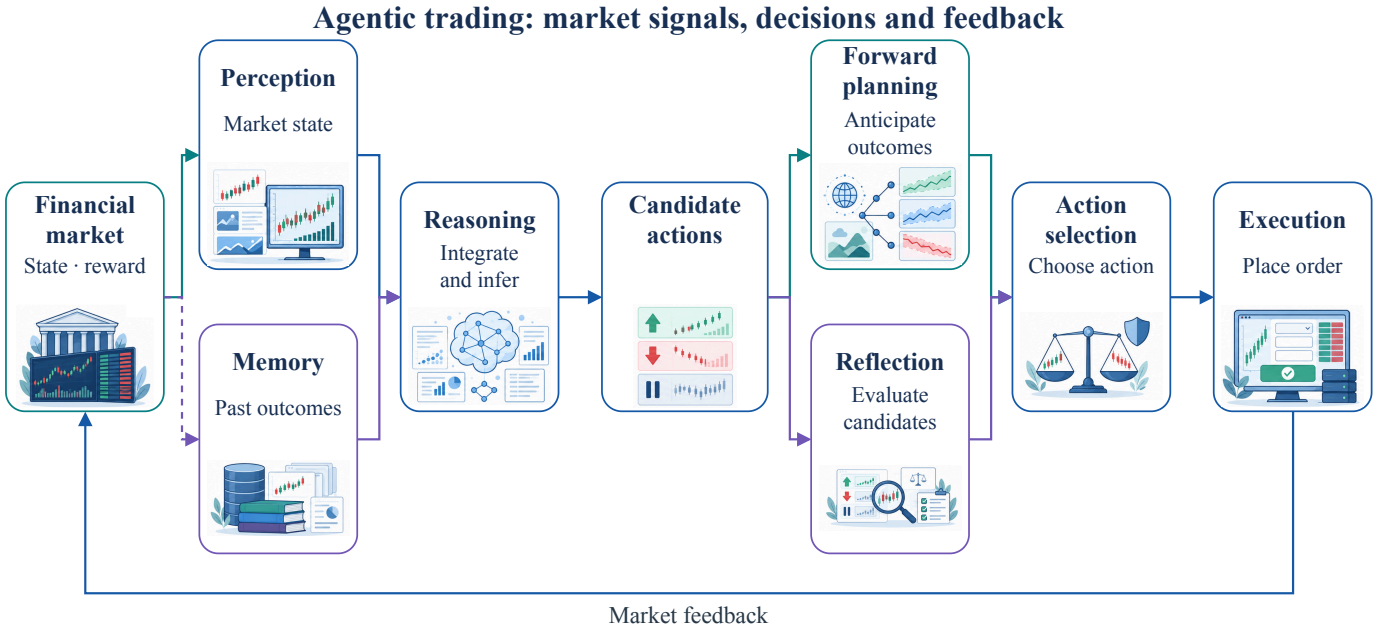

Figure 2: End-to-end decision and execution loop in agentic trading. Market observations and historical experiences are gathered via perception and memory modules; cognitive reasoning produces candidate actions that are validated through forward planning or reflection; the execution module maps decisions to market orders under risk constraints; and market outcomes provide feedback for continuous adaptation.

Table 1: Scopes and organizing perspectives of related surveys, benchmarks, and financial AI frameworks.

Work	Type	Target Domain	Primary Review Focus	System Decomposition	Evaluation Perspective
This survey	Survey	Trading agents	End-to-end trading pipelines	Perception, memory, reasoning, execution	A-C-A framework and empirical protocols
Wang et al. (2024)	Survey	General AI agents	Cognitive agent architectures	Profile, memory, planning, action	General capabilities and benchmarks
Giglio et al. (2022)	Survey	Asset pricing and ML	Factor models and econometrics	Statistical estimation and pricing kernels	Econometric testing and predictability
FinBen (Xie et al., 2024)	Benchmark	Financial NLP and quant tasks	Standardized task evaluation	Model-level task mapping	Financial tasks across difficulty tiers
FinGPT (Wang et al., 2023a)	Model/tool	Financial NLP	Open financial AI models	Models and data frameworks	Text and sentiment benchmarks
InvestorBench (Li et al., 2025)	Benchmark	Trading decisions	Investor persona evaluation	Simulated decision agents	Performance across investor styles
Ding et al. (2024)	Survey	LLM trading agents	Trading agent architectures	Agent modules, inputs, challenges	Backtest patterns and open questions
Dong et al. (2025)	Survey	LLM agents in finance	Broad financial applications	Multi-role workflows and deployment	Research, advisory, and compliance

Note: Categorizations describe each work's primary organizing scope and core objectives. Benchmarks (FinBen, InvestorBench) and foundation frameworks (FinGPT) establish essential task-level foundations, while complementary surveys focus on general agent principles Wang et al. (2024), asset-pricing econometrics Giglio et al. (2022), trading agent fundamentals Ding et al. (2024), and broad financial application domains Dong et al. (2025).

environments.

Existing Surveys on LLM Agents in Finance and Trading. Recently, Ding et al. (2024) presented a pioneering survey on LLM agents in financial trading, systematically categorizing agent architectures, data inputs, and broad backtesting performance. Concurrently, Dong et al. (2025) provided an extensive survey of LLM agents across diverse financial sectors, highlighting real-world deployment scenarios across research, advisory, and market operations. Our survey is designed to be deeply complementary to these works: rather than focusing solely on broad financial workflows or individual algorithmic setups, we offer an in-depth, architecture-centric decomposition of the end-

to-end trading loop, systematically analyze execution and risk mechanisms, and provide a structured synthesis of empirical evaluation practices across the literature.

1.2. A Conceptual Framework: Disentangling Architecture, Capability, and Adaptation

To analyze the wide diversity of trading agents cohesively, this survey introduces the **Architecture–Capability–Adaptation (A-C-A)** framework as an analytical lens. By separating *how systems are constructed* from *what financial objectives they pursue* and *how they evolve*, the A-C-A lens clarifies design choices across three orthogonal dimensions:

Dimension 1: Architectural Mechanisms (How the System is Built). The architectural layer deconstructs an agent into its underlying engineering blocks: multimodal perception (text, time-series, charts), memory hierarchies (working memory, associative vector storage, structured knowledge bases), cognitive reasoning paradigms (reactive, reflective, planning-based), and action execution modules (order slicing, venue routing, latency and cost modeling). This decomposition identifies the inputs, retained state, decision procedures, and execution assumptions that can be compared across systems (Part I, Sections 3 to 6).

Dimension 2: Functional Capabilities (What the System Accomplishes). The capability layer describes the financial tasks executed by the agent, including automated alpha generation (symbolic factor discovery and mathematical expression mining), portfolio allocation (multi-asset weighting and dynamic rebalancing), and proactive risk management (drawdown controls, stop-loss limits, and supervisory compliance checks). Distinguishing capabilities from architectures clarifies that diverse architectural setups (e.g., reactive vs. strategic) can be deployed across identical financial goals with differing operational trade-offs (Part II, Sections 7 to 9).

Dimension 3: Adaptive Pathways (How the System Evolves). The adaptation layer characterizes changes to agent behavior, parameters, and organization over time. It covers in-context learning, supervised fine-tuning, reinforcement learning, and multi-agent coordination and adaptive interactions. Self-evolution concerns agent-managed updates retained across decisions; structural self-evolution is the subset that changes strategy code, update procedures, or system organization. Memory consolidation and reflection can support persistent adaptation within an otherwise fixed architecture (Part III, Sections 10 to 12).

Table 2 illustrates how representative trading systems in the literature map across this three-dimensional A-C-A framework, highlighting how distinct architectural building blocks combine with functional goals and adaptation mechanisms. Appendix C provides additional system-level mappings.

1.3. Contributions and Organization

This survey provides a comprehensive and structured synthesis of the agentic trading literature. The primary contributions are:

1. **The A-C-A Analytical Framework.** We organize system comparisons by architecture, financial capability, and adaptation. Table 2 and the illustrative mappings in Appendix C show how the lens applies to representative systems.
2. **A Closed-Loop Mechanism Synthesis.** Parts I–III connect perception, memory, reasoning, and execution with alpha discovery, portfolio management, risk control, and adaptation. The technical chapters describe module interfaces, design alternatives, and operational trade-offs across this decision loop.
3. **Transparent Literature Mapping.** Based on 185 systematically mapped studies, with a formal search window of 2022–2026 and earlier foundational references (including a core empirical subset of $n = 80$ closed-loop trading systems), we link representative mechanisms to their literature context. Section 2 and Table 3 document the selection boundary and corpus composition.
4. **Empirical Evaluation and Methodological Landscape.** We characterize reported evaluation practices in the empirical subset across temporal splitting (50/80), transaction-cost modeling (47/80), survivorship-bias controls (48/80), and execution semantics (47/80). Table 4 reports the initial and extension cohorts separately to contextualize the pooled counts.
5. **Forward-Looking Challenges and Best-Practice Guidelines.** We identify key engineering and financial bottlenecks (including latency, market impact, hallucination cascades, and strategy degradation) and connect them to methodological recommendations (MR-1–MR-7). Section 13 and Table 20 organize these directions and the recommendations' applicability by study type.

The remainder of this survey is organized as follows. Section 2 details the literature selection protocol and synthesis scope. **Part I: Architecture** (Sections 3 to 6) deconstructs the core architectural pillars: perception, memory, reasoning, and action execution. **Part II: Capability** (Sections 7 to 9) examines the primary financial capabilities: alpha discovery, portfolio optimization, and risk control. **Part III: Adaptation** (Sections 10 to 12) explores how agents learn, collaborate, and evolve. Section 13 discusses open challenges and methodological guidelines, and Section 14 concludes the survey.

We begin by defining what constitutes an agentic trader, drawing on cognitive science and artificial intelligence literature to identify the essential architectural components.

Part I addresses the fundamental question: *What constitutes an agentic trader?* We characterize an agentic trader by four core architectural components—perception, memory, reasoning, and action/execution—that together support autonomous decision-making in financial markets. Whereas traditional trading systems focus primarily on prediction, agentic systems typically include an explicit cognitive loop that interacts with and learns from the market environment through decisions mapped to tradable actions under market constraints.

Table 2: Illustrative mapping of representative trading systems under the A-C-A conceptual framework, showing how architectural components, functional capabilities, and adaptation pathways intersect.

System	Architecture Dimension	Capability Dimension	Adaptation Pathway
FinAgent Zhang et al. (2024)	Multimodal Perception (Text+Visual) + Execution Layer	Direct Stock Action Selection & Position Execution	In-Context Learning from market feedback
AlphaAgent Tang et al. (2025)	Reflective/Strategic Reasoning + Feedback Validation	Automated Code-Based Alpha Discovery & Factor Pruning	Hypothesis-Factor Iterative Refinement
TradingAgents Xiao et al. (2024)	Hierarchical Role-based Multi-Agent Coordination	Joint Alpha Generation, Portfolio Sizing & Risk Control	Hierarchical Feedback & Memory Updating

2. Review Protocol and Evidence Scope

To synthesize this rapidly expanding literature with conceptual clarity and methodological rigor, this review adopts a structured literature selection and corpus mapping methodology. We define transparent inclusion boundaries to differentiate end-to-end trading agent implementations from broader financial tools, and map the design space along well-defined architectural, functional, and empirical evaluation dimensions.

The formal search for contemporary trading-agent work covered records published from **2022-01-01** through **2026-09-30**; the 2022 start date marks the period in which LLM-based systems began to appear as end-to-end trading agents rather than as isolated NLP or forecasting components. Citation chaining also retained nine pre-2022 foundational background works, including classical portfolio and learning methods; these references belong to the included ledger but fall outside the contemporary search window. The initial-window search covered 2022-01-01 through **2026-03-09** (last query date: 2026-03-09), using ACM Digital Library (including ICAIF), IEEE Xplore, arXiv, SSRN, and Google Scholar, complemented by backward and forward citation chaining. For this revision we ran a pre-specified extension window over **2026-03-10 to 2026-09-30** on arXiv reusing the three initial query families and adding trading-agent, multi-agent, and alpha-mining patterns (agentic trading, LLM finance agents, evaluation/validity, trading-agent phrases, multi-agent markets, and alpha mining); the extension retrieved 1,539 unique records, of which 154 entered eligibility assessment under the same inclusion boundary (Appendix A). After deduplication and screening, the current ledger contains 185 included studies from 246 candidate records. The windows are not matched search samples: the initial window spans multiple databases and citation chaining, whereas the extension uses arXiv only. Window-level differences therefore describe the assembled ledger and should not be interpreted as a controlled comparison of search coverage.

To make the review boundary reproducible, we distinguish three fixed denominators throughout the manuscript. The *candidate registry* contains 246 unique records after deduplication. The *included evidence-mapping ledger* contains 185 records retained for either primary evidence or background/design context. The *primary empirical subset* contains 80 records that satisfy all three conditions: Action Output, Closed-Loop Evaluation, and an eligible empirical evaluation type (`backtest`, `simulation`, `live`, or `benchmark`); all protocol-field and reproducibility percentages use this 80-record subset unless explicitly stated otherwise; the design-theme synthesis draws on representative primary and background work. Table 3 summarizes the resulting selection flow.

The inclusion boundary for the *primary empirical subset* is deliberately restrictive: a study must satisfy all three conditions: *(i) Action Output*, meaning that the system emits tradable actions such as orders, position changes, or portfolio allocations; *(ii) Closed-Loop Evaluation*, meaning that those actions are evaluated in backtesting, simulation, or live trading; and *(iii) an eligible evaluation type* coded as `backtest`, `simulation`, `live`, or `benchmark`. The third condition keeps narrative-only descriptions outside empirical denominators even when action output and a closed-loop claim are coded present. In the current S0 v2 registry, 102 included records carry the raw `primary` role tag; 22 of these are narrative-only and are assigned to the background tier for empirical denominators. Thus the 80-study primary empirical subset plus 105 background/context records partitions the 185 included studies. Under this rule, the present snapshot yields 80 primary empirical studies and 105 background studies.

The screening workflow proceeded in four steps. First, database and preprint searches were run with agent/trading/finance keyword families and then expanded through backward and forward citation chaining. Second, duplicate records were removed by title, arXiv identifier, DOI, and manual inspection. Third, title/abstract screening removed records outside financial trading, portfolio construction, execution, or risk-management settings. Fourth, full-text screening assigned included records to the empirical primary subset only when all three criteria above were met; finance-relevant records failing any criterion were retained as background/context, not final exclusions. Accordingly, the initial window contributes 19 primary empirical and 58 background records, while the extension contributes 61 and 47, respectively. These cohorts also differ in coding depth: the initial-window coding was followed by a targeted extraction audit, whereas the extension cohort was coded in a single pass with focused full-text extraction of protocol paragraphs. Table 4 reports both cohorts alongside pooled coverage. Search composition and coding depth can both affect these rates; the

Table 3: Study selection summary using the canonical candidate registry, combining the initial-window ledger (cutoff 2026-03-09) with the extension window (2026-03-10 to 2026-09-30). The denominator is the 246-record deduplicated registry. Final exclusions are outside the survey domain; finance-relevant records that lack Action Output, Closed-Loop Evaluation, or eligible evaluation type remain in the included background tier. Detailed logs and per-record decisions are provided in Supplementary Material S1.

Stage	Count
Registry candidate records[a]	246
Included in evidence mapping set	185
Excluded at screening/eligibility	61
Final exclusion reason	
Outside finance/trading/portfolio/risk-management scope	61
Included evidence-map partition	
Primary empirical subset	80
Background/context tier	105

[a]The canonical registry currently contains 246 unique records after deduplication (185 included in the evidence mapping set and 61 excluded at screening/eligibility). The included evidence mapping set is partitioned into 80 primary empirical studies and 105 background/context records.

ledger does not identify their separate contributions.

The technical chapters synthesize architectural scope through representative systems and the illustrative A-C-A notes in Appendix C. The current registry does not contain populated A-C-A assignments for all 80 empirical studies, so it supports corpus-level protocol statistics rather than numerical chapter-level taxonomy prevalence. Protocol-critical fields include time-consistent split discipline, execution timing or semantics, transaction-cost modeling, universe/survivorship handling, and reproducibility artifacts. Missing or unclear reporting is coded conservatively as *NR/Unknown* rather than imputed. This structured coding enables a transparent, multifaceted synthesis that contextualizes reported findings with their respective execution realism, temporal validity controls, and methodological transparency.

We audited the screening stage on a stratified double-screening sample of 182 raw screening records in Supplementary Material S1. Because that sample predates the final initial-window canonical registry, we retain it only as a screening-boundary quality check; it does not inform extraction-level protocol fields, reproducibility tiers, or A-C-A classification robustness. To address the more consequential coding decisions, we added a targeted extraction-level audit over 15 fixed-seed initial-window records sampled from the legacy 41-record primary-role frame (seed = 20260426). Seven of the 15 sampled records are narrative-only under the current eligible-evaluation-type rule. This audit therefore checks coding agreement on its sampled fields; it is not a representative reliability estimate for the current 80-record primary empirical subset. The audit covers action output, closed-loop evaluation, evidence role, evaluation type, split protocol, cost model, execution semantics, universe/survivorship handling, artifacts, and R0–R3 reproducibility (Table B.24). The per-study coding sheet and adjudication notes are provided in Supplementary Material S3. The extension-window cohort (154 records) was coded in a single pass from abstract-level screening plus focused full-text extraction of protocol paragraphs; its per-record decisions are released in the v2 registry, and its field-level codings should be read as more preliminary than the audited initial-window subset. Table 5 summarizes the two audit layers and their scope.

A source check on 2026-10-03 corrected Alpha^2 from R2 to R1 because its official repository provides pseudocode rather than a runnable implementation. The journal version of the explainable portfolio-policy study also links public code, correcting its old R0 assignment to R1 (de la Rica Escudero et al., 2025). Its complete runnable reproduction is not established here. The current registry and distributions incorporate these amendments; historical S3 records are retained as coding snapshots. Neither amended work was in the 15-record reliability sample, so the recorded 150/150 agreement is unchanged.

The A-C-A framework organizes representative trading systems by architecture, financial capability, and adaptation. Corpus-level descriptive statistics for protocol reporting, reproducibility, and peer-review status are summarized in Appendix B; boundary decisions and final exclusions are summarized in Appendix D. Detailed query strings, initial-window screening logs, the combined registry, current protocol tables, targeted reliability-audit files, and coding notes are provided in Supplementary Materials S1–S3. Historical mapping and coding snapshots are identified separately from the current registry and protocol table.

2.1. Agentic Trading as an Expert-System Decision Pipeline

Within ESWA's expert-systems framing, we interpret LLM-based trading agents as expert-system decision pipelines, which makes the evaluation problem explicit. Perception modules perform knowledge acquisition from prices, news, filings, social media, and market microstructure signals. Memory, retrieval-augmented generation, and tool-mediated context stores form a knowledge base whose provenance and update rules must be inspectable. Reasoning modules act as inference engines that convert state representations into hypotheses, rationales, and candidate decisions. Action and execution modules form the external action interface, where target weights, orders, fills, latency assumptions, and market frictions must be made explicit. Risk controls, compliance filters, and portfolio limits

Table 4: Protocol-field coverage by retrieval and coding cohort. Each cell gives the number reporting the field, cohort denominator, and percentage. The initial cohort received a targeted extraction audit; the extension received single-pass coding. Differences are descriptive and combine search composition with coding depth.

Field	Initial (n = 19)	Extension (n = 61)	Pooled (n = 80)
Time split	2/19 (10.5%)	48/61 (78.7%)	50/80 (62.5%)
Transaction costs	1/19 (5.3%)	46/61 (75.4%)	47/80 (58.8%)
Universe / survivorship	1/19 (5.3%)	47/61 (77.0%)	48/80 (60.0%)
Execution semantics	11/19 (57.9%)	36/61 (59.0%)	47/80 (58.8%)

Table 5: Reliability and coding-quality audits reported with the submission package.

Audit layer	**Sample**	**Observed agreement**	**Agreement note**	**Scope**
S1 screening-boundary audit	182 raw records	Reported in S1	Boundary only	Inclusion/exclusion and screening-stage labels
S3 extraction-level audit	15 legacy primary-role records	All 150 field comparisons agreed pre-adjudication (100.0%)	field-dependent; see Supplementary Material S3	Extraction fields used for protocol and reproducibility statistics

serve as rule constraints on the path from inference to action. Finally, logs, immutable artifacts, and human oversight provide the explanation and accountability layer expected of decision-support systems.

This expert-system framing provides a cohesive conceptual spine for our synthesis. A trading agent's real-world viability depends not only on its conceptual design but also on the rigor of its empirical evaluation—including temporal split integrity, transaction-cost realism, universe survivorship controls, and execution fidelity. The A-C-A framework organizes this multifaceted design space systematically: architecture describes how market knowledge is acquired, stored, inferred, and acted upon; capability delineates which core financial functions are realized; and adaptation examines how the decision pipeline evolves under non-stationary market feedback.

3. Perceptual Architecture

The preceding section introduced the cognitive architecture of agentic traders. This section addresses the question: "How do agents perceive financial markets?"

In this survey, we define **Perception** as the mechanism by which an agent acquires a state observation O_t from the environment state S_t at decision time t. Under the primary-subset boundary specified in Section 2, standalone financial NLP or time-series forecasting models (e.g., isolated BERT or LSTM predictors) are treated as **Background (BG)** unless they form, or are explicitly integrated into, an evaluated action-producing loop: component models lack the closed-loop integration required for agentic classification, although they may still supply perceptual capabilities that end-to-end agents incorporate.

A perception module must therefore do more than predict; it must provide actionable state representations—such as sentiment scores, market regime indicators, or visual summaries—that the agent's reasoning component can use to formulate valid actions.

Architectural Role and Design Dimensions. In an autonomous trading agent, perception functions as the critical observation interface between external market feeds and internal cognitive reasoning. Rather than functioning as an isolated prediction engine, the perception layer in an end-to-end agentic architecture must construct a structured, temporally synchronized observation O_t that is immediately actionable for memory retrieval, factor computation, and order formulation. Current literature explores this design space across three core modalities:

- **Textual Perception** (Section 3.1): Parsing unstructured financial news, earnings announcements, filings, and social sentiments.
- **Time-Series Perception** (Section 3.2): Encoding numerical sequences, including order book microstructure, price trends, and macroeconomic fundamentals.
- **Multimodal Perception** (Section 3.3): Aligning and fusing graphical representations (e.g., candlestick charts, technical patterns) with textual and numerical signals.

Across these modalities, the central design trade-offs revolve around feature representation density, cross-modal synchronization, and latency constraints, balanced against the critical requirement of preventing temporal information leakage.

Figure 3 compares three complementary perceptual modalities by information source and processing mechanism.

3.1. Textual Perception

Textual perception refers to the acquisition of state observations O_t from linguistic information sources (news articles, earnings reports, analyst commentary, social media posts, regulatory filings) at decision time t. The module encodes textual inputs into representations—such as sentiment scores, event embeddings, or contextualized vectors—that the reasoning component can use to formulate actions.

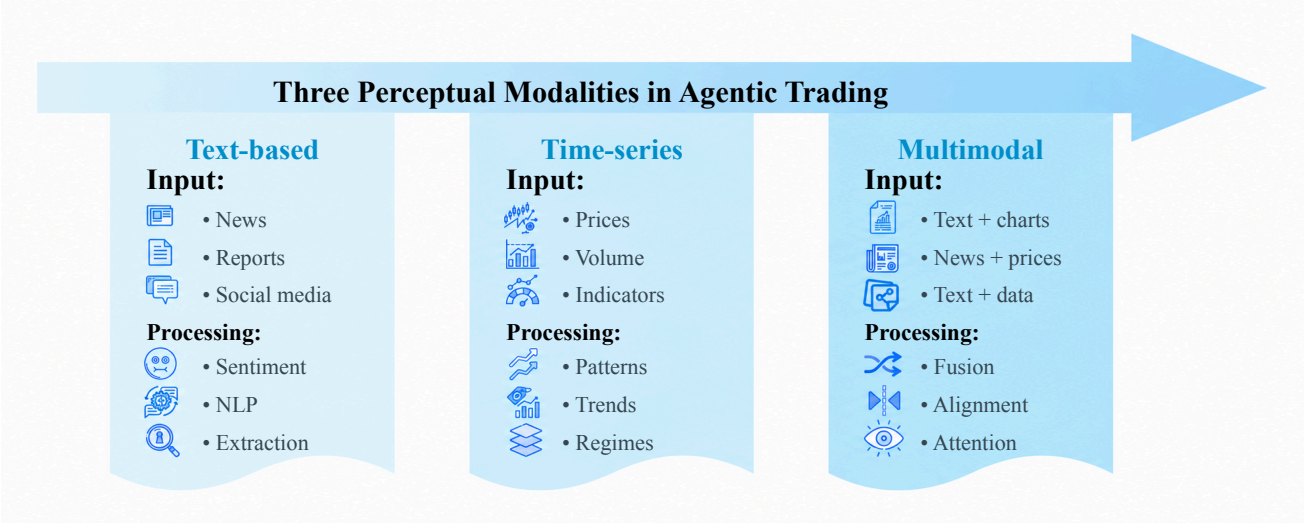

Figure 3: Three perceptual modalities in agentic trading. Textual perception processes linguistic information from news and reports. Time-series perception analyzes numerical patterns in price and volume data. Multimodal perception integrates heterogeneous data sources through cross-modal fusion mechanisms.

Representative Approaches. Several lines of work have developed specialized architectures for financial text understanding. Du et al. (2024) provide a comprehensive survey of financial sentiment analysis techniques, reviewing prior evidence that sentiment signals are often associated with subsequent market movements. Iacovides et al. (2024) introduce FinLlama, an LLM-based approach fine-tuned for financial sentiment analysis in algorithmic trading scenarios. Zhang et al. (2023a) propose Instruct-FinGPT, which uses instruction tuning to adapt general-purpose LLMs to financial sentiment tasks, while Zhang et al. (2023b) explore retrieval-augmented generation to enhance sentiment analysis with external knowledge.

Technical Mechanisms. Textual perception typically follows a pipeline of tokenization, encoding, and extraction. Domain-specific language models such as FinBERT (Araci, 2019) provide contextualized representations of financial text, capturing terminology and semantic nuances that are often underrepresented in general-purpose models. Rodriguez Inserte et al. (2024) show that adapting large language models through task-specific fine-tuning can improve performance on financial sentiment tasks. Sentiment classification models then map these representations to polarity labels (e.g., positive, negative, and neutral) or to more fine-grained affective states. Xing (2025) further studies heterogeneous LLM agents for financial sentiment analysis and reports improved benchmark performance relative to selected baselines.

Critical Challenge: Temporal Alignment. A pervasive risk in text perception is look-ahead bias. News timestamps often record publication time rather than the moment an item actually becomes available to a trading pipeline, creating a mismatch between historical records and executable decision contexts; agents trained on historical news corpora without explicit ingestion-lag modeling or embargo enforcement are therefore likely to exhibit inflated backtest performance. A related risk arises from post-publication revisions: some financial disclosures and structured reports are updated after initial release, so using retrospectively cleaned or final versions in backtesting can introduce serious data leakage López de Prado (2018).

Table 6 summarizes representative textual perception methods, highlighting the diversity of data sources and technical approaches. Because these methods are effective on textual information but cannot directly process numerical price data or visual chart patterns, agents also require the complementary perceptual modalities discussed next.

3.2. Time-Series Perception

Time-series perception refers to the acquisition of state observations O_t from numerical sequences (price, volume, Limit Order Books—LOB) and structured fundamental data (e.g., balance sheet tables) at decision time t. This modality encodes temporal patterns—such as trend indicators, volatility estimates, or regime classifications—into representations the reasoning component can use. Its inputs range from familiar price and volume sequences (Open-High-Low-Close-Volume—OHLCV) to deeper market structures such as LOB and options-implied state variables. Because these inputs evolve over time, agents must model temporal dependence, seasonality, structural breaks, and non-stationarity when constructing actionable state representations.

Representative Approaches. Representative work in this area studies how sequence models encode information from market data streams into trading-relevant state representations. Recent systems use transformer-style or attention-based encoders to summarize price dynamics, while classical econometric approaches remain useful for regime identification and structural-change analysis. In the current agentic trading literature, however, most implemented systems still rely on readily accessible

Table 6: Textual perception methods. **Audit Status**: BG=Background Model (not integrated into agent loop); R0-R3=Reproducibility Level (see Appendix B). **Time Disc.**=Time Discipline (Static=offline batch processing; RT=real-time streaming with specified latency).

Reference	Input Source	Representation	Time Disc.	Audit
FinBERT Araci (2019)	News Headlines	Sentiment Score (Scalar)	Static	BG
FinGPT Wang et al. (2023a)	News/Social	Summary/Embedding	Static	BG
BloombergGPT Wu et al. (2023)	Financial Corpus	General LM	Static	BG
FinLlama Iacovides et al. (2024)	Reports	Instruction Tuned	Static	BG
Instruct-FinGPT Zhang et al. (2023a)	Financial Text	Instruction Tuned	Static	BG
RAG-FinGPT Zhang et al. (2023b)	Mixed Sources	Retrieval-Augmented	Static	R0
Heterogeneous Agents Xing (2025)	Multi-source	Specialized Architectures	Static	R0

OHLCV-style inputs, while explicit use of high-frequency LOB data and richly structured fundamentals remains limited.

Technical Mechanisms. Key challenges include non-stationarity, noise, and sensitivity to temporal misalignment. Successful approaches often employ attention mechanisms to focus on informative time points, regime-switching models to capture structural breaks Hamilton (1989); Bai and Perron (2003), and representation-learning objectives that improve robustness under changing market conditions. Standard positional encodings and related sequence-ordering schemes can also help models capture periodic structure and temporal context in sequential financial data Vaswani et al. (2017).

Table 7 compares representative time-series perception methods. Methods that operate on numerical sequences alone cannot directly incorporate the rich contextual information available in text; vision-language variants such as VISTA that render series as charts already illustrate the cross-modal direction developed next.

3.3. Multimodal Perception

Multimodal perception refers to the acquisition of state observations O_t from heterogeneous sources (chart images, text, numerical series, audio) at decision time t, integrating these into unified representations through cross-modal alignment mechanisms. This mirrors trading practice, in which traders synthesize visual technical patterns with news flow and price action.

Representative Approaches. For visual perception, Wang et al. (2023b) introduce FinVis-GPT for financial chart analysis, which treats chart-like market representations as visual inputs amenable to language-based interpretation. For multimodal fusion, Zhang et al. (2024) introduce FinAgent, which combines textual news and numerical market information through a cross-modal pipeline designed for trading decisions. More broadly, multimodal systems differ in where fusion occurs along the pipeline, a distinction examined under Technical Mechanisms below.

Critical Challenge: Cross-Modal Asynchrony. Fusing modalities with different update frequencies creates synchronization risks. News arrives sporadically, whereas market and LOB data may update at much higher frequencies. Naive framing strategies such as pairing the latest available news item with the current price can therefore misrepresent the true decision context when the text signal is stale. Moreover, visual interpretation and multimodal inference introduce nontrivial latency, so the perceived market state may already be outdated by the time the agent acts in live settings. Protocol reporting should therefore state perception-latency assumptions and alignment procedures explicitly.

Technical Mechanisms. Multimodal fusion poses significant challenges because modalities differ in dimensionality, temporal resolution, and noise characteristics. Early fusion concatenates raw features from all modalities before processing, while late fusion maintains separate processing pipelines and combines high-level representations. Cross-attention mechanisms provide a flexible alternative by enabling one modality to attend selectively to another during representation learning Vaswani et al. (2017); Tsai et al. (2019). The temporal-alignment problem noted above is especially acute in financial settings, since textual events arrive discretely while prices update continuously; preventing look-ahead bias therefore requires strict timestamp discipline, explicit embargo handling, and clearly reported latency budgets López de Prado (2018).

Figure 4 illustrates the multimodal fusion pipeline, showing how text encoders process news and reports while time-series encoders analyze price movements, with a cross-modal fusion layer integrating these representations.

Table 8 summarizes representative visual and multimodal perception approaches. Taken together, these studies suggest that richer market state estimation often benefits from combining heterogeneous signals rather than relying on a single modality in isolation.

Extension-window additions. The extension cohort adds perception-side infrastructure rather than new modalities: mechanism-based decomposition of market dynamics for downstream agents Chen (2026b) and synthesized financial tool-use data that couples perception to callable tools Huang et al. (2026a). Both remain background-level designs in our ledger, but they illustrate how the perception layer is being engineered as a service for agent pipelines rather than as a standalone predictor.

Synthesis and Design Considerations. Perceptual architecture turns heterogeneous market feeds into state observations O_t. Textual, time-series, and multimodal inputs differ in source and processing, while all require temporal alignment; multimodal fusion adds cross-modal synchronization, and each design must control look-ahead bias and latency. These observations feed the memory and retrieval systems examined next.

Table 7: Time-series and structured perception methods. Audit and Time Disc. tags follow the convention in Table 6.

Reference	Input Data	Mechanism	Time Disc.	Audit
VISTA Khezresmaeilzadeh et al. (2025)	Chart Images + Text	Zero-Shot VLM Inference	Static	RO
Regime-Switching Hamilton (1989)	Returns	Hidden Markov Model	Static	BG

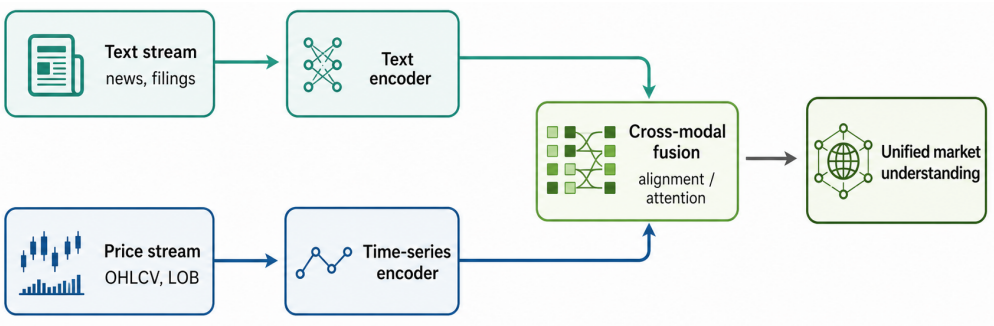

Figure 4: Multimodal fusion flow in agentic trading. The diagram illustrates how text encoders and time-series encoders process their respective inputs before a cross-modal fusion layer integrates the representations into a unified market understanding.

Table 8: Visual and multimodal perception methods. **Latency**: Estimated inference latency (critical for alignment). **Note**: Latency estimates are author-reported; see Supplementary Material S2 for detailed benchmarking protocols. Audit tags follow the convention in Table 6.

Reference	Modalities	Fusion Strategy	Latency	Audit
FinVis-GPT Wang et al. (2023b)	Chart Images	Vision-Language Analysis	Low	BG
FinAgent Zhang et al. (2024)	News + Prices + Kline Charts	Cross-Attention	High	RO
Multimodal Transformer Tsai et al. (2019)	Text + Audio + Vision	Cross-Modal Attention	Med	BG

Table 9: Summary of perceptual architectures in trading agents. The **Protocol Risk** column highlights specific alignment challenges for each modality.

Modality	**Primary Inputs**	**Mechanism**	**Protocol Risk**	**Examples**
Textual	News, Reports, Social Media	LLM Fine-tuning, Prompting, Sentiment Analysis	Publication Lag, Embargo Violations	FinBERT, FinGPT
Time-Series	OHLCV, LOB, Fundamentals	Temporal Attention, Regime Switching	Look-ahead Bias, Data Revisions	VISTA
Multimodal	Charts, Text+Price+Audio	Vision-Language Analysis, Cross-Modal Attention	Synchronization Latency, Causal Misalignment	FinVis-GPT, FinAgent

4. Memory Architecture

The preceding section introduced perceptual mechanisms for gathering market information. This section asks how agents store, retain, and retrieve that information. It focuses on memory as an organized state store; Learning Paradigms examines how agents use feedback to update behavior, while Self-Evolution examines how retained state and update policies change over time.

Memory serves three operational roles: **working memory** maintains active trading context, **associative vector storage** retrieves historical episodes by similarity, and **structured knowledge bases** preserve explicit financial information. The core design challenges are balancing retrieval relevance against context-window saturation, mitigating retrieval drift, and maintaining temporal causality across long trading horizons. These categories are examined in Section 4.1, Section 4.2, and Section 4.3, respectively. Earlier agent literature also uses cognitive terminology: working memory follows Baddeley and Hitch (1974), episodic and semantic memory follow Tulving (1972), and multi-store models draw on Atkinson and Shiffrin (1968). These cognitive descriptions provide context for memory-augmented designs Zong et al. (2024); the categories below identify the storage and retrieval mechanisms rather than biological memory processes.

4.1. Working Memory

In agentic trading, working memory is often conflated with a generic "context window."For robust agent design, we distinguish between the **Deterministic State Store** (Layer A) and the **Generative Context** (Layer B).

Layer A: Deterministic State Store (Audit-Truth). This layer holds the ground-truth state of the trading system, including current positions, open orders, account balances, and risk limits. In a reliable trading architecture, we argue that it should be **read-only** to the LLM and updated solely by the environment (e.g., EMS/OMS feedback). Treating this state as a "memory"subject to forgetting or hallucination is a critical vulnerability; in practice, it is better implemented as a structured store exposed through tools or state queries, as seen in tool-using trading agent designs Xiao et al. (2024).

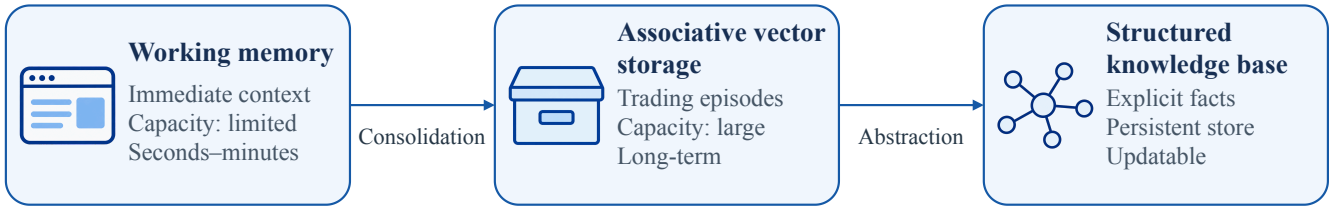

Figure 5: Three memory categories in agentic trading. Working memory maintains active trading context, associative vector storage supports retrieval of historical trading episodes, and structured knowledge bases preserve explicit financial information. Arrows denote consolidation from working context into stored experience and abstraction from experience into reusable structured knowledge.

Layer B: Generative Context (Inference Scope). This layer corresponds to the LLM's active context window, containing the sliding history of observations, chain-of-thought reasoning traces, and recent reflections. Unlike Layer A, information here is transient and subject to eviction.

Technical Mechanisms. To manage Layer B's limited capacity, agents employ eviction policies such as **Summary-Compression** (condensing old steps into a narrative) or **hierarchical memory management** (using layered storage and retrieval priorities to retain higher-value fragments) Packer et al. (2023); Sun and Zeng (2025). FinMem Yu et al. (2023b) adopts the same layered pattern in a trading setting, buffering recent context in a working layer atop its deeper memory layers; it is therefore listed under working memory in Table 11. A key design trade-off is the *context-to-noise ratio*: maintaining too much history can dilute attention and bury decision-relevant content in long contexts, a risk consistent with the broader "Lost in the Middle"finding Liu et al. (2024), while aggressive pruning may drop reasoning that is still relevant to a live position or pending order.

Table 10 summarizes the three memory categories, highlighting their differing capacity constraints and temporal scales. Working memory enables rapid decision-making, while associative vector storage and structured KBs support longer-term retrieval and knowledge accumulation.

4.2. Associative Vector Storage

Associative vector storage provides a way to index and retrieve stored experience through learned or engineered similarity. Trading episodes may be represented this way, but not every episodic store uses vector embeddings. To keep retrieved episodes auditable, we define a valid episode as a tuple $(S_t, A_t, O_{t+k}, \tau)$, where S_t is the state, A_t the action, O_{t+k} the outcome realized at step $t+k$, and τ the timestamp at which this outcome becomes observable.

Information Leakage via Memory. We refer to a key vulnerability in trading agents as the "Oracle Fallacy,"in which an agent retrieves a similar past episode that contains a *post-hoc* narrative (e.g., "this trade failed due to news X released tomorrow").

Protocol Implication

To prevent such leakage, we suggest that reproducible trading-agent protocols enforce an **Outcome Embargo**: an episode recorded at t cannot expose its outcome field to retrieval until current time $t_{now} \geq \tau$.

Complementary work addresses deterministic replay in financial agents, providing assurance frameworks for regulatory audit Khatchadourian (2026).

Representative Approaches. FinAgent (Zhang et al., 2024) uses layered memory and diversified retrieval, with temporal control primarily at the prompt level rather than through explicit database constraints. FinMem (Yu et al., 2023b) organizes memory in layers, while Reflexion (Shinn et al., 2023) stores unsuccessful trajectories as lessons. These examples illustrate storage and retrieval choices; their cross-time update governance is considered in Section 12.

Technical Mechanisms. Vector databases Johnson et al. (2017); Malkov and Yashunin (2020) enable similarity search, but naive cosine similarity is often insufficient for financial time series, where relevance also depends on recency and regime. "Time-Aware Retrieval"therefore becomes a useful design principle.

Protocol Implication

We suggest a relevance score $f(q,k)$ that includes a decay term $e^{-\lambda(t_{now}-t_k)}$ to prioritize recent regimes.

Furthermore, **Drift Handling** is required: as market regimes shift (e.g., from low to high volatility), old episodes with high feature similarity may still yield misleading expectations, a risk consistent with recent retrieval-augmented time-series forecasting evidence Xiao et al. (2025a).

4.3. Structured Knowledge Bases

Structured knowledge bases store explicit domain knowledge (e.g., "high interest rates hurt growth stocks"). Their auditability depends on provenance and update controls, not only on format (graph vs. text). Model weights can also encode financial knowledge, but that parametric knowledge is a related, implicit mechanism rather than a structured KB: it is not an external source-addressable store and should not be classified as one.

Table 10: Memory system comparison.

Memory Category	Capacity	Volatility	Key Characteristics
Working Memory	Limited (Context)	Volatile	Fast cache, eviction policies, sliding window
Associative Vector Storage	Large (Disk)	Persistent	Vector embeddings, similarity search, temporal indexing
Structured Knowledge Base	Source-dependent	Persistent	Knowledge graphs, sourced records, explicit relations

Related mechanism: parametric knowledge (implicit). This knowledge is encoded in model weights via pre-training or fine-tuning (e.g., FinGPT (Wang et al., 2023a)). While efficient, it is **opaque**—it is difficult to trace *why* the model knows a fact—and **static**, since updates usually require additional training or alignment. For trading, this raises the risk of "outdated priors,"in which older market regularities persist in model behavior after the regime that produced them has changed.

Structured KBs (explicit, non-parametric) retrieve information from an external source. We distinguish:

- **Curated KBs**: Verified documents (Rulebooks, SEC Filings) where source integrity is high Wang et al. (2025b).
- **Uncurated KBs**: Open-web data (Social Media, Forums) susceptible to "Noise Injection"or adversarial poisoning.

Technical Mechanisms. To preserve source integrity, "Source Tracking"is required: every retrieved retrieved fact should carry a provenance log (Document ID + Version Timestamp). Knowledge Graphs encode explicit entity-relation structure (e.g., $Entity_A \xrightarrow{supplier_of} Entity_B$) and can therefore support provenance-oriented reasoning and constraint checking more directly than unstructured text alone.

4.4. Design Trade-offs

A systematic architectural analysis of these systems reveals key design trade-offs that performance-only benchmarks tend to obscure.

Latency vs. Recall Depth. Deeper retrieval in high-fidelity RAG systems can introduce latency that is incompatible with high-frequency or rapid mid-frequency strategies. Agents therefore need to balance *retrieval depth* against *decision latency* and *execution speed*.

Context Length vs. Distraction. Although newer LLMs support 1M+ token windows, loading all retrieved memories into context degrades reasoning through the "Lost in the Middle" phenomenon Liu et al. (2024). A disciplined memory design therefore favors *less* but *higher-relevance* context over an unfiltered dump of history.

Noise Injection Risk. Allowing uncurated knowledge sources (e.g., scraping Reddit) increases the risk of "Noise Injection,"in which irrelevant or manipulated public sentiment overrides the agent's underlying strategy. We therefore adopt a "Hierarchy of Truth"as a design principle: internal Layer A state should override external noisy signals when the two conflict.

Extension-window additions. The extension cohort contributes several memory- and retrieval-facing designs. On the knowledge side, LLM-enhanced dynamic financial knowledge graphs support cross-entity retrieval for trading decisions Zhang (2026b), and a benchmark for event-grounded personalized memory evaluates retention of investor-specific context across sessions Wang et al. (2026a). On the retrieval side, document-routed hybrid retrieval Cheng et al. (2026b) and time-series augmented generation Kolonin et al. (2026) target the financial RAG layer that can feed vector stores and structured KBs. A cautionary complement: information-fidelity analyses show that aggressive summarization can distort the very inputs a memory layer is meant to preserve Lee et al. (2026)—reinforcing the "Hierarchy of Truth" principle above.

Synthesis and Design Considerations. Memory architecture provides temporal depth and experiential continuity: working memory tracks active trading state, associative retrieval selects historical episodes, and structured knowledge bases preserve explicit information. Retrieval depth must be balanced against latency and context distraction, while temporal rules and provenance checks protect against hindsight leakage and preserve source traceability. The retrieved context feeds the reasoning engine examined in Section 5.

5. Reasoning Architecture

Having examined how agents perceive markets and store information, we now address the question: "How do agents make trading decisions?" Building on the notation from Section 3 and Section 4, we write O_t for the state observation at time t and M_{assoc} for retrieved associative memory.

We treat reasoning as the decision function that transforms perceptions and memories into candidate actions. Because decision horizon, computational budget, and search depth are more directly auditable than broad cognitive analogies, we organize reasoning architectures along a **Time-Scale Reasoning Paradigm**. The three paradigms below loosely echo the System 1/System 2 distinction of dual-process theory, although that framework offers no consensus definitions for long-horizon planning; we therefore use the analogy strictly as a shorthand for latency and compute budget, not as a cognitive claim.

Architectural Role and Reasoning Spectrum. Reasoning is the cognitive engine of an agentic trading system, converting state representations O_t and retrieved memories into structured, actionable trading hypotheses. Financial markets present a unique cognitive challenge characterized by non-stationary dynamics,

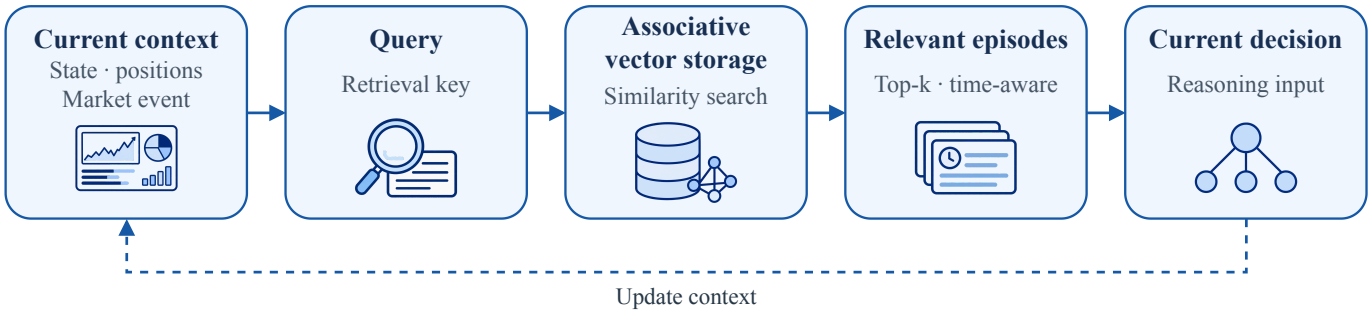

Figure 6: Memory retrieval mechanism. The agent formulates a query based on current context, searches an associative vector store using similarity-based retrieval, and selects relevant episodes to inform the current decision.

Table 11: Comparison of memory retrieval mechanisms in trading agents. Tags follow the convention in Table 6 for the representative works in each row. NR = not reported; N/A = not applicable.

Memory Category	Representative Ref	Timestamp Rule	Embargo	Drift Check	Audit Status
Working Memory	FinMem Yu et al. (2023b)	N/A	N/A	N/A	R0
Episodic memory (paper term)	FinAgent Zhang et al. (2024)	NR	NR	No	R0
Parametric knowledge (related)	FinGPT Wang et al. (2023a)	Implicit	No	No	BG
Structured Knowledge Base	FinSage Wang et al. (2025b)	Versioned	Yes (Doc Date)	Yes	BG

low signal-to-noise ratios, and severe latency constraints. To address these demands, reasoning architectures operate across a spectrum of decision timescales:

- **Reactive Control** (Section 5.1): Sub-second rule-based or specialized policy execution, optimized for immediate order adjustments and safety stops.
- **Reflective Reasoning** (Section 5.2): Second-scale deliberative inference utilizing chain-of-thought, self-critique, and tool execution to formulate structured trade rationales.
- **Strategic Planning** (Section 5.3): Multi-minute or daily planning loops employing tree search (e.g., Monte Carlo Tree Search) and multi-step scenario simulation over extended horizons.

The central engineering trade-off lies in balancing cognitive depth and reasoning faithfulness against computational latency and the risk of search-based overfitting.

Table 12 contrasts the three paradigms in terms of time-scale, mechanism, reproducibility level, and evaluation considerations.

Terminological Clarification: Reactive Reasoning

We use "reactive" strictly for *non-LLM* latency-constrained control. LLM-based methods (including ReAct adaptations) operate at second-scale or slower, placing them in the **reflective** category. This distinction matters because it prevents the misconception that LLMs can support high-frequency trading decisions.

Figure 7 illustrates the time-scale comparison.

Reasoning I/O Contract. Formally, we model the agentic reasoning module as a structured decision function $f(O_t, M_{assoc}, P_{portfolio}) \rightarrow (A_{cand}, \alpha, V_{check})$, where:

- **Inputs**: timestamped observation O_t, retrieved associative memory M_{assoc}, and read-only portfolio state $P_{portfolio}$ (positions, risk limits).
- **Outputs**: a set of candidate actions A_{cand}, a confidence score $\alpha \in [0, 1]$, and a validity check V_{check} (e.g., "holds < max_leverage").

The contract is designed to reduce the risk of "hallucinated actions" by enforcing risk constraints *during* or *immediately after* the reasoning step, rather than leaving them entirely to the final execution layer.

5.1. Reactive Reasoning

Reactive reasoning, the System 1 end of the analogy above, denotes a **constrained, simplified decision** optimized for latency budgets measured in microseconds to milliseconds. **Importantly**, this paradigm *does not* involve LLM-mediated deliberation—the latency of even lightweight LLMs (hundreds of milliseconds) exceeds the budgets of high-frequency trading. Instead, reactive trading systems employ rule-based controllers, lightweight neural networks (often <100K parameters), or pre-compiled decision tables that execute in constant time.

Representative Approaches. ReAct (Yao et al., 2023b) exemplifies **reflective** (not reactive) reasoning in general agent architectures—its "think" step involves explicit LLM inference that operates at second-scale latency. In trading contexts requiring millisecond response times, ReAct-style deliberation must be *decoupled from the critical path* and confined to offline analysis or non-latency-sensitive decision support. Rule-based systems encode expert knowledge as if-then rules (e.g., "if

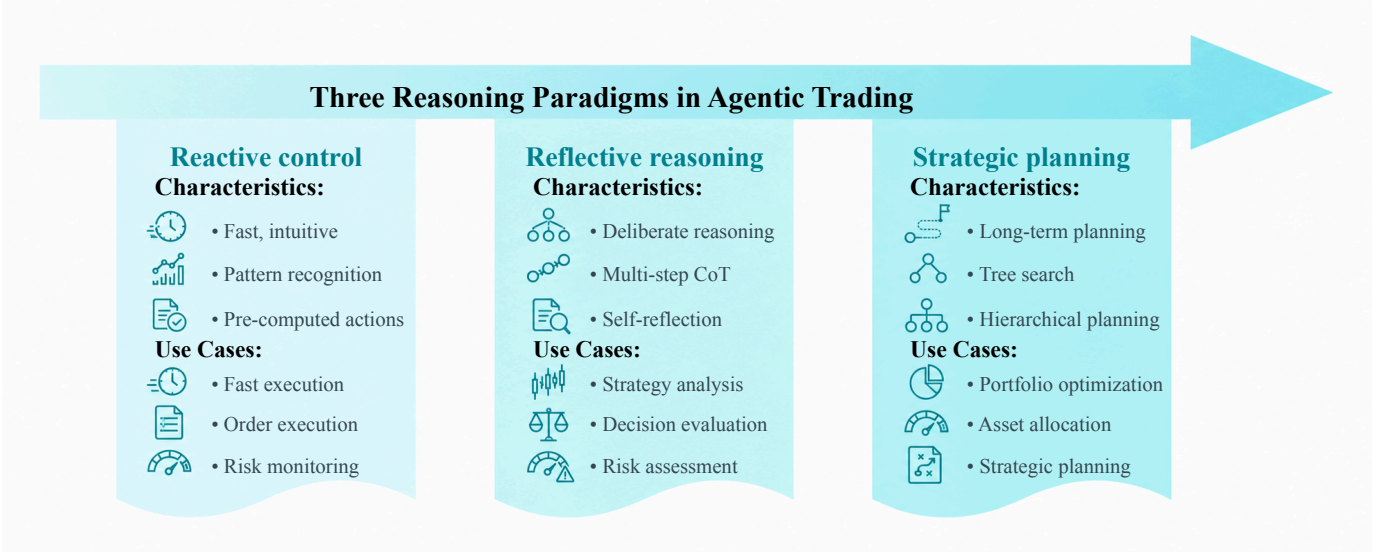

Figure 7: Three complementary reasoning roles: reactive control supports fast operational responses, reflective reasoning evaluates decisions through deliberation and self-reflection, and strategic planning explores longer decision horizons. Their latency depends on the model, hardware, tools, and search budget.

Table 12: Comparison of reasoning paradigms. **R-Level**: Reproducibility level (R0–R3, see Appendix B). **Evaluation considerations** summarize protocol dimensions relevant to interpreting reported results.

Paradigm	Time-Scale	Mechanism	R-Level	Representative Works	Evaluation considerations
Reactive	Microsecond–millisecond budgets	Rule-based / Lightweight NN (often <100K params) / pre-compiled decision tables	R0–R1	ReAct Yao et al. (2023b) (reflective, not reactive), reactive components in FinAgent Zhang et al. (2024)	Execution Latency often ignored
Reflective	Seconds	CoT / Self-Critique	R0–R1	FinCoT Nitarach et al. (2025), FinCon Yu et al. (2024)	Inference Cost not reported
Strategic	Minutes/Hours	MCTS / Planning	R0–R1	Alpha Jungle Shi et al. (2025), RAPTOR Almon et al. (2025)	Search Budget & Overfitting Risk

price crosses moving average, buy") and execute immediately once the conditions are met, sacrificing flexibility for speed because they rely on pre-defined patterns rather than reasoning from first principles. Consequently, the reactive paradigm is most effective in relatively stable market regimes where previously learned patterns remain actionable. In agentic trading settings, this shallow-loop structure typically appears as a reactive component inside larger systems—for example, through tool-augmented signal retrieval coupled with constrained actions (Zhang et al., 2024), or through role-specialized teams that decompose perception while preserving a structured final decision pipeline (Xiao et al., 2024). Gradient-based policy optimization can further couple reactive decisions to reward-driven updates in interactive markets (Xiong et al., 2025).

Technical Mechanisms. Reactive systems employ feature extractors that map raw market data to action logits and then select and execute the highest-logit action. Shallow neural architectures with limited reasoning depth are common because they balance prediction speed against accuracy; typical choices include temporal convolutional networks for processing time-series data Bai et al. (2018) and attention mechanisms for weighting relevant features Vaswani et al. (2017). A further advantage is *relative* interpretability: because the mapping runs from a small feature set to actions, it can be inspected and constrained Lundberg and Lee (2017), although inspection alone does not guarantee faithful explanations of the decision logic. The same simplicity, however, limits adaptability: these systems struggle with novel situations outside their training distribution, and when market structure shifts abruptly they may continue applying outdated patterns, potentially leading to significant losses.

Beyond latency, the integration of tool use into financial agents requires explicit evaluation along at least two dimensions: AgentGuard studies safety-oriented evaluation for tool orchestration Chen and Cong (2025), whereas FinToolBench measures the capability of LLM agents to use financial tools in realistic tasks

Lu et al. (2026). Autonomous agent frameworks can likewise strengthen complex reasoning over structured knowledge Jiang et al. (2024).

5.2. Reflective Reasoning

Reflective reasoning, the System 2 counterpart in the same analogy, produces slow, deliberate decisions through multi-step chains of thought. This paradigm enables agents to reason through complex scenarios, consider multiple hypotheses, and reflect on potential mistakes before acting. Unlike reactive systems that respond immediately to stimuli, reflective reasoning allocates computational resources to explore the decision space more thoroughly, which makes it suitable for medium-frequency trading where execution latency on the order of seconds is acceptable.

Representative Approaches. Chain-of-Thought (CoT) prompting (Wei et al., 2022) guides LLMs to explicitly reason through trading decisions, producing intermediate steps such as "the stock price dropped because of worse- than-expected earnings, but the long-term fundamentals remain strong, so this is a buying opportunity." FinCoT (Nitarach et al., 2025) adapts CoT to finance through domain-grounded prompting that encourages financial reasoning in trading tasks. FinCon (Yu et al., 2024) operationalizes reflective workflows through multi-agent or role-structured coordination that updates investment beliefs over time. CryptoTrade (Li et al., 2024c) and TradingGroup (Tian et al., 2025) further instantiate this reflective pattern by revising trading decisions based on prior outcomes and synthesizing multi-source signals. Reflexion (Shinn et al., 2023) extends this paradigm with self-reflection, in which agents critique their own reasoning and refine their conclusions before acting. A Reflexion-based trader, for example, might generate an initial trading rationale, identify potential flaws in its logic (such as overlooking sector-wide risks), and revise its decision accordingly.

Technical Mechanisms: Protocol Implication

We synthesize five stages for reflective trading decisions: (1) analyze the current situation; (2) retrieve relevant memories; (3) consider multiple hypotheses; (4) assess each candidate action's benefits and risks; and (5) select an action and justify the choice.

Temperature sampling controls the trade-off between creativity and consistency, with lower temperatures producing more deterministic reasoning. A recurring challenge is maintaining coherence across long reasoning chains, since LLMs may lose context or contradict themselves over dozens of steps; checkpointing (saving intermediate reasoning states) and consistency checks (verifying that conclusions follow from premises) help mitigate these issues Wei et al. (2022); Wang et al. (2022); Madaan et al. (2023). The computational cost of reflective reasoning scales linearly with the number of reasoning steps, so it is more expensive than reactive approaches but also significantly more capable of handling novel situations that require adaptation.

A further caveat concerns feedback timing: Reflexion-style systems assume relatively prompt feedback, whereas trading outcomes often materialize only after meaningful market delay Shinn et al. (2023). Beyond self-critique loops, modern risk-sensitive frameworks provide structured decision support for financial markets, although their exact internal decomposition varies by implementation Liu and Dang (2025b).

5.3. Strategic Reasoning

Strategic reasoning extends beyond immediate decisions to plan across extended time horizons, weighing sequences of actions and their long-term consequences. This paradigm addresses questions such as "what portfolio allocation will maximize risk-adjusted returns over the next quarter?" rather than "should I buy this stock right now?" It operates at time scales of minutes to hours, which suits portfolio optimization, risk management, and longer-term investment strategies where thorough planning matters more than immediate execution.

Representative Approaches. Tree-of-Thought (ToT) (Yao et al., 2023a) frames reasoning as search through a tree of possible thoughts, exploring multiple reasoning paths in parallel and selecting the most promising through explicit evaluation. Each node in the tree represents a partial reasoning state, and the algorithm systematically explores alternatives rather than committing to the first plausible solution. Monte Carlo Tree Search (MCTS) methods such as Navigating Alpha Jungle (Shi et al., 2025) apply the same idea to the space of possible trading strategies, using rollouts to estimate the long-term value of each candidate strategy. These methods balance exploration (trying new strategies) against exploitation (refining known good strategies) through upper confidence bounds applied to financial domains. Planning ideas from classical AI and reinforcement learning continue to inform strategic trading design: state spaces represent market conditions and actions represent trading decisions. For portfolio-scale planning, RAPTOR (Almon et al., 2025) integrates deliberative portfolio reasoning with an orchestrated rebalancing pipeline to produce interpretable, risk-aware allocations.

Technical Mechanisms. Strategic reasoning requires models of the environment—predicting how market prices will evolve in response to trading actions. Model-based methods learn explicit transition models that capture causal relationships between actions and market states, which then allow agents to simulate future scenarios and evaluate action sequences without executing them. Model-free methods instead learn value functions directly from experience, estimating the long-term return of each state-action pair without an explicit environment model Sutton and Barto (2018). To keep planning tractable, hierarchical schemes decompose long horizons into shorter sub-problems: a strategic planner sets high-level targets (e.g., "reduce tech exposure by 20%"), and tactical planners execute specific trades to achieve these targets. Abstraction is critical here—planning over individual stocks is intractable because the space of possible portfolios explodes combinatorially, whereas planning over sectors or factors reduces the state space to manageable dimensions. A strategic planner might therefore operate at the level of sector allocations (technology, healthcare,

energy), while tactical planners handle individual stock selection within each sector, so that agents can reason about long-term objectives while remaining computationally feasible Kocsis and Szepesvári (2006); Sutton et al. (1999).

Critical Challenge: Search-Based Overfitting. Unlike game environments (e.g., Go, Chess) where simulated rollouts are exact, financial planning relies on noisy historical data. Search algorithms (MCTS, ToT) that evaluate thousands of candidate strategies therefore face the *multiple testing* problem—a profitable-looking trajectory may emerge purely by chance. To mitigate this *data snooping* risk, reproducible evaluation requires reporting the **Search Budget** (number of visited nodes) and enforcing strict **Walk-Forward Validation**, in which the plan is frozen before being applied to out-of-sample data Bailey et al. (2014).

Hybrid Architecture: Cascaded Controller. To balance these trade-offs, advanced systems can employ what we call a **Cascaded Controller** architecture (cf. Figure 7 for the paradigms it routes among): a master intent router classifies the market regime (e.g., "Flash Crash" vs. "Stable Trend") and routes execution to the appropriate paradigm, triggering lower-level reactive stops in volatile conditions while scheduling strategic re-allocations during calm periods. This architectural recommendation synthesizes principles observed in hierarchical trading systems Yu et al. (2023b); Xiao et al. (2024).

Extension-window addition. Complementing the paradigm taxonomy, work in the extension cohort examines how market state is *represented* for reasoning: representation signatures that align internal features with risk-feedback signals Xue (2026) probe whether an agent's reasoning is grounded in decision-relevant structure rather than surface patterns.

Synthesis and Design Considerations. Reactive control, reflective reasoning, and strategic planning trade decision speed against deliberation and search depth; reflective methods depend on alignment with market timescales, while strategic search requires explicit exploration budgets and walk-forward verification to guard against data snooping. Candidate decisions then pass to the action execution layer examined next.

6. Action and Execution Architecture

The preceding sections examined perception, memory, and reasoning. This section addresses the complementary question: "How do agents turn decisions into market actions?"

Architectural Role and Execution Scope. The action and execution architecture bridges cognitive reasoning with the physical reality of financial markets, marking the definitive boundary between an abstract forecasting model and a genuine trading agent. While forecasting models evaluate directional accuracy on paper, an autonomous trading agent must transform high-level intentions into executable market orders subject to microstructure frictions, liquidity limits, and latency budgets. The execution design space encompasses three key dimensions:

- **Decision-to-Order Mapping**: Translating target weights or directional convictions into discrete order specifications (order side, quantity, limit price, time-in-force).
- **Execution Policies and Cost Modeling**: Implementing algorithmic order slicing (e.g., TWAP, VWAP, POV) and explicitly modeling fees, spreads, and market impact.
- **Microstructure and Latency Management**: Incorporating order book dynamics, fill probabilities, and exchange round-trip latency into trade scheduling.

Action I/O Contract. Formally, we model the action execution module as a structured function $f(D_t) \rightarrow (A_{order}, A_{algo})$ (see Figure 8), where:

- **Input**: Decision output D_t (e.g., target portfolio weights w_t^* or signal s_t).
- **Output**: A set of executable orders A_{order} (with strictly typed fields: side, size, limit price, time-in-force (TIF)) and an execution algorithm selection A_{algo} (e.g., "submit immediately" or "TWAP over 5 min").
- **Failure Modes**: The interface typically handles *Rejection* (risk check failure), *Partial Fill* (liquidity shortage), and *Timeout* (latency violation), feeding these states back to memory.

The contract is designed to prevent "implicit execution," in which an agent is assumed to trade at the close price without order-generation delays or constraints.

Comparability note. Protocol-comparable action/execution claims typically report MR-3 (action semantics), MR-4 (execution & costs), and MR-6 (artifacts & logs); see Section 13.

6.1. Decision-to-Order Mapping

First Principle: Action Definition Risk. A critical failure mode in agent design is the ambiguity between *target weights* (portfolio intent) and *executable orders* (market instructions). If an agent emits only target weights without explicitly modeling order timing, fill constraints, and costs, evaluation quietly collapses intent into execution and thereby underestimates the *execution gap* and liquidity constraints Zhang et al. (2024); Chen et al. (2022); Cheridito and Weiss (2025); Liu and Dang (2025a); Papadakis et al. (2025); Olby et al. (2025).

In the surveyed systems, action modules translate high-level decisions (e.g., *increase exposure*) into concrete order instructions. This mapping depends on explicit choices about order type (limit versus market) and queue position, because microstructure theory models execution probability as a function of order aggressiveness and market state Foucault (1999); Parlour (1998). Constraints such as pre-trade validation and kill switches then act as safeguards that keep theoretical alpha from becoming realized losses once market impact is accounted for Kyle (1985); Glosten and Milgrom (1985).

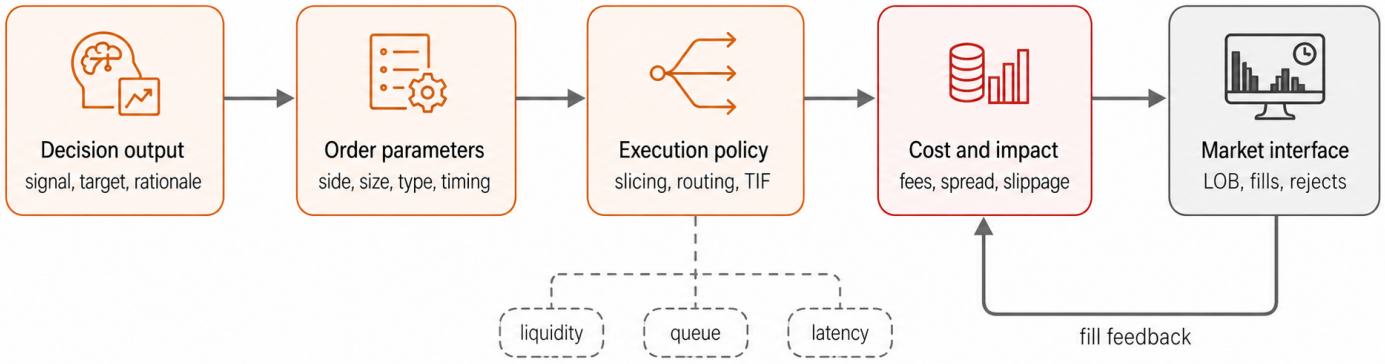

Figure 8: From decision to execution. Agent outputs map into order parameters and are executed with explicit cost and impact modeling under market microstructure constraints.

Table 13: Representative action/execution approaches and evaluation dimensions. **Type**: E2E = end-to-end agent literature; Algo = algorithmic-execution literature. Note that *Agent* literature often adopts simplified assumptions (Sim-based), whereas *Algo* literature focuses on execution quality but lacks agentic reasoning. **Evaluation considerations** summarize execution details relevant to interpreting reported results. **Tags** follow the R0–R3 reproducibility-level convention (see Appendix B).

Category	Type	Action Space	Core Mechanism	Cost Assumption	Evaluation considerations	Tags
Schedule-based	Algo	Order schedule	TWAP/VWAP-style scheduling	Explicit fees	Slippage Calib.	R0
RL Execution	Algo	Market Orders	Policy optimization in simulator	Impact-aware reward	Latency Budget	R0
Sim-based RL	E2E	Order-level	Market simulator + RL	Simulator-dependent	Decision Timing	R0
Microstructure	Algo	Limit Orders	Fill/LOB modeling	Implicit via LOB	Realized Spread	R0
Hierarchical	E2E	Meta-Actions	Hierarchical RL for order types	Explicit/implicit	Failure Modes	R0

6.2. Execution and Cost Modeling

Execution policies determine how orders are placed over time and across venues. Typical approaches include schedule-based execution (e.g., time-weighted average price (TWAP) and volume-weighted average price (VWAP)), cost-aware optimization that trades urgency against market impact Almgren and Chriss (2001); Bertsimas and Lo (1998); Perold (1988), and learning-based execution that adapts to market conditions. Recent studies apply deep learning and reinforcement learning to execution and impact-aware trading Genet (2025); Byun (2023); Cheridito and Weiss (2025); Hafsi and Vittori (2024); Chen et al. (2022); Olby et al. (2025). Within this line, FinPos Liu and Dang (2025a) introduces a position-aware trading agent designed for real financial markets, emphasizing continuous portfolio adjustment, while StockSim Papadakis et al. (2025) provides a dual-mode order-level simulator for evaluating multi-agent LLMs in financial markets and thereby enables realistic execution modeling. Table 13 contrasts representative action and execution approaches across these categories, summarizing their action spaces, cost assumptions, and common audit gaps.

6.3. Microstructure and Latency Awareness

Microstructure signals (order book imbalance, spread, queue position) act not merely as inputs for optimization but as **hard feasibility constraints** that filter the action space. Throughout, **LOB** denotes the real-time queue of outstanding buy and sell limit orders; its features include imbalance (bid versus ask volume), spread, and depth.

1. **Data Availability**: Agents can incur "look-ahead bias" when they act on LOB features (e.g., $Imbalance_t$) that are not yet available at decision time t because of processing latency; feasible actions must therefore be conditioned on $O_{t-\delta}$.
2. **Latency Budget**: In high-frequency regimes, the *latency budget*—the time from perception to exchange acknowledgment—determines whether an arbitrage opportunity still exists Budish et al. (2015): if $T_{compute} > T_{decay}$, the action a_t is invalid regardless of its theoretical alpha.

Models of order book dynamics Gould et al. (2013); Briola et al. (2024) then allow these constraints to be validated before execution.

Protocol Implication

Beyond execution optimization, LLMs can support **post-trade narratives** by summarizing execution logs. To reduce "ex post rationalization" (in which the LLM invents a strategic reason for a random failure), we recommend grounding these narratives in **immutable logs** and replayable evidence Tan et al. (2025a); Khatchadourian (2026).

- **Recommendation**: Narrative generation can cite the specific *Order_ID* and cryptographic *Snapshot_Hash* produced by the execution layer.
- **Verification**: Auditors can replay the

Snapshot_Hash to check whether the market condition claimed by the narrative (e.g., "high spread") actually held.

Without this *Log-to-Narrative Verification* link, LLM explanations remain qualitative summaries rather than audit artifacts.

Extension-window additions. The extension cohort sharpens the execution side along two directions. First, evaluation infrastructure: systematic benchmarking of learning-based equity trading Wang and Zhang (2026) and inference-time optimization that plans before committing orders Go et al. (2026) both treat the decision-to-order mapping as a measurable component rather than an implementation detail, while realistic market-impact modeling for RL traders Abbade and Costa (2026) supplies the cost layer that MR-4 asks papers to report; auditability-oriented execution frameworks such as AutoQuant Deng (2025) reinforce the same principle from the engineering side by treating execution constraints as an auditable system boundary. Second, a crypto-native execution niche: multi-agent MEV generation and cross-chain arbitrage systems Wen et al. (2026); Ye et al. (2026b) operate directly on order-flow and settlement mechanics, a setting in which execution semantics constitute the strategy rather than a downstream assumption.

Synthesis and Practical Considerations. The action execution layer grounds cognitive decisions in market dynamics by accounting for transaction costs, slippage, fill timing, and microstructure constraints. In multi-agent or institutional deployments, structured order logging supports audits that link execution traces to strategic reasoning. With these architectural foundations established, Part II turns to enabled capabilities, beginning with automated alpha discovery.

7. Alpha Discovery Capability

The preceding sections established the architectural foundations of agentic traders. This section addresses the question: "How do agents discover trading signals?"

Here, alpha discovery refers to generating and testing candidate trading signals; consistent with the evidence-map framing, we count autonomous alpha search as supported only when the reported system connects discovery to out-of-sample or closed-loop evaluation, rather than merely proposing factors.

Functional Scope and Discovery Paradigms. Automated alpha discovery represents one of the most prominent applications of LLM agents in quantitative finance, transitioning formulaic factor generation from manual trial-and-error to autonomous, hypothesis-driven exploration. By combining financial reasoning with programmatic code generation, agents search through vast functional hypothesis spaces to uncover predictive trading signals. Across the literature, alpha discovery is realized through three core paradigms:

- **Code-Based Generation** (Section 7.1): Expressing economic hypotheses in executable code and refining them through automated syntax and runtime feedback.
- **Retrieval-Augmented Discovery** (Section 7.2): Grounding factor exploration in academic literature, financial corpora, and historical factor libraries to seed domain-consistent hypotheses.
- **Evolutionary and Genetic Search** (Section 7.3): Guiding symbolic expression trees through LLM-directed mutation, crossover, and fitness selection.

The central methodological considerations include managing search budgets to prevent multiple-testing data snooping, enforcing factor diversity to mitigate crowding, and verifying that statistical predictive power survives net of turnover and execution frictions.

Classical Baseline: Factor Models and Information Coefficient

Before LLM-based approaches, alpha generation relied on manually engineered factor models (e.g., the Fama–French three-factor model, momentum, value), in which factor validity is measured by the information coefficient (IC), the correlation between factor scores and subsequent returns, and statistical rigor required controlling for multiple testing (White's Reality Check, Hansen's SPA) to distinguish genuine signals from data mining Fama and French (1993); Grinold and Kahn (2000). LLM-based agents automate this discovery process but inherit the same fundamental challenge: ensuring that discovered patterns generalize out-of-sample rather than exploit historical noise.

Comparability note. Comparable alpha-search evidence should report MR-2 (time split + sealed test), MR-4 (execution & costs), factor definitions, and MR-6 (artifacts & logs); see Section 13 for detailed protocol requirements and reporting checklists.

Author Framework

Alpha-to-Trade Contract. We introduce the Alpha-to-Trade Contract as a conceptual framework for separating statistical significance from tradable signals; the conversion contract specifies three checks:

1. **Frequency Compatibility**: Does the signal update frequency match the execution latency budget?
2. **Turnover Constraints**: Does the theoretical gain survive realistic transaction costs and market impact?
3. **Liquidity Reality**: Is the factor universe liquid enough for the assumed capital capacity?

Papers omitting this contract risk conflating theoretical IC with realized alpha return, a common pitfall in agentic trading literature López de Prado (2018).

Figure 9 illustrates these discovery paradigms.

7.1. Code-based Alpha Discovery

Code-based alpha generation draws on the code-synthesis capabilities of LLMs to translate natural-language hypotheses about market behavior into executable factor definitions. Because it lowers the programming barrier, this paradigm lets domain experts express and test their trading ideas directly.

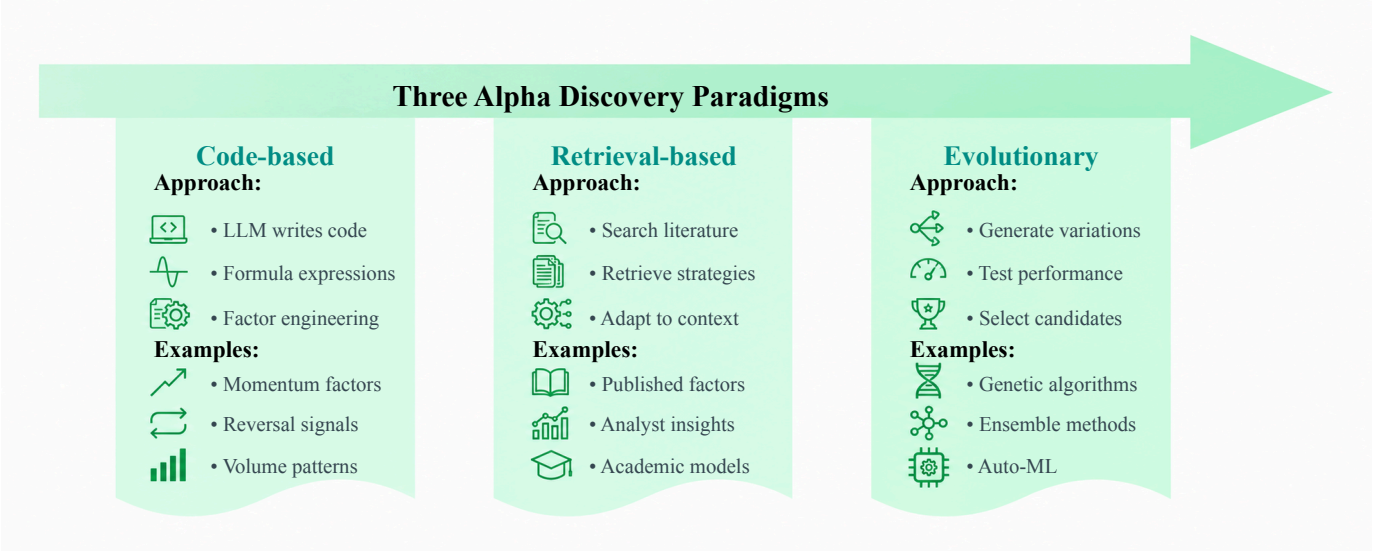

Figure 9: Three alpha discovery paradigms. Code-based discovery translates natural language hypotheses into executable code. Retrieval-based discovery finds and adapts existing factors from a library. Evolutionary discovery searches the factor space through mutation and selection.

Representative Approaches. CogAlpha (Liu et al., 2025) extends code-based generation with structured prompting templates that guide the LLM toward syntactically correct and semantically meaningful factor definitions. AlphaAgent (Tang et al., 2025) further introduces regularization mechanisms (originality enforcement, hypothesis alignment, and complexity control) to counteract alpha decay. FactorMAD (Duan et al., 2025) uses a multi-agent debate framework in which LLM agents critique and refine factor proposals to enhance interpretability. Automate Strategy Finding (Kou et al., 2025) implements a three-stage pipeline with risk-aware multi-agent validation and ensemble strategy selection, and reports 53.17% cumulative returns on SSE50 under its backtest setup (*Split*: `NR`; *Execution*: `NR`; *Cost*: `NR`; *Universe*: SSE50; *Tier*: `BG`); because protocol-critical fields are `NR`/`Unknown`, we read such results as illustrative rather than protocol-comparable evidence.

Beyond factor synthesis, recent work combines LLM reasoning with reinforcement learning for structured trading decision-making Xiao et al. (2025b), and ensemble strategies built on deep reinforcement learning report strong performance on stock trading tasks Yang et al. (2025).

Technical Mechanisms. The generation pipeline operates as an *adaptive search process* susceptible to overfitting: it (1) parses the natural-language hypothesis to identify key components, (2) maps them to code templates and APIs, (3) generates executable code, (4) validates it through sandboxed execution, and (5) backtests it on historical data; Figure 10 illustrates this generation–validation loop. Crucially, the "validation chain" feeds performance feedback into the next generation step, so the loop risks **adaptive overfitting**: as the agent iterates to maximize the validation Sharpe ratio, it effectively "mines" the validation set Bailey et al. (2014); López de Prado (2018). A rigorous protocol therefore requires reporting the **Search Budget** (the number of generated hypotheses) and ensuring that final performance is measured on a sealed *test set* that is never exposed to the agent during the refinement loop; without this separation, reported "validation" metrics are statistically inflated. Sandboxed execution, standard practice for mitigating security risks, addresses a different concern and does not by itself enforce split integrity. Whatever form the output takes—synthesized factors include momentum and reversal signals—its validity depends entirely on the integrity of the data split.

7.2. Retrieval-based Alpha Discovery

Retrieval-based alpha generation adapts existing factor knowledge rather than generating factors from scratch: the agent retrieves related signals or financial context that can inform the current situation.

Representative Approaches. RAG-based financial retrieval systems (Cook et al., 2025) index domain knowledge as embeddings and retrieve contextually similar items using semantic search. PRISM (Ng et al., 2025) refines this retrieval through prompt-optimized in-context modeling for financial domains, and retrieval-augmented frameworks for financial time-series forecasting (Xiao et al., 2025a) illustrate how LLM generation can be combined with historical pattern retrieval.

Technical Mechanisms. Retrieval relies on vector embeddings that encode factor semantics (formula structure) and market context. The validity of this paradigm, however, hinges on **Factor Library Governance**, a data-integrity requirement that is easy to overlook.

- **Survivorship Bias**: If the library collects only "historically successful" factors, the retrieval process implicitly conditions on future outcomes, inflating backtest performance.

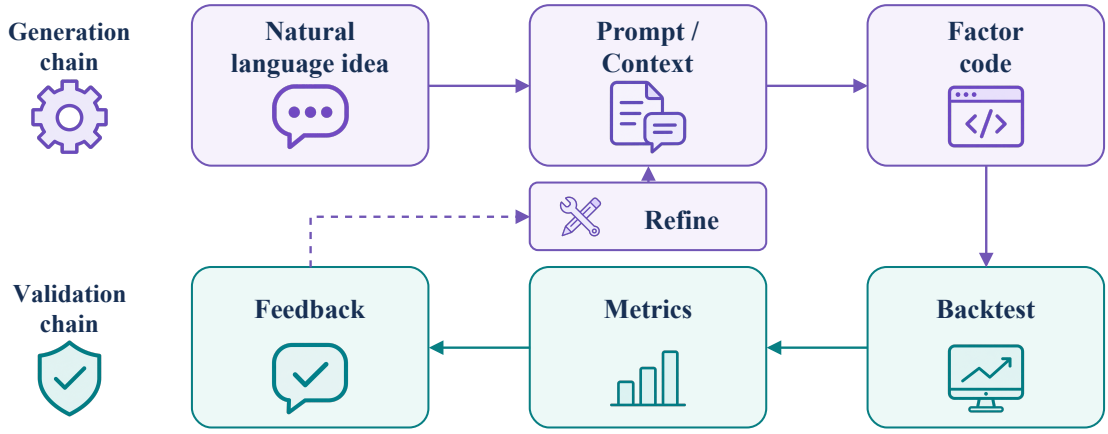

Figure 10: Chain-of-Alpha workflow. The generation chain (top) translates natural language into factor code. The validation chain (bottom) backtests the factor, computes performance metrics, and provides feedback for refinement. The two chains interact iteratively until a satisfactory factor is produced.

- **Look-ahead Leakage in Embeddings**: If the embedding model is trained on textual descriptions containing future knowledge (e.g., "a successful high-volatility factor"), the retrieval query q_t effectively leaks information from $t + k$.

Protocol Implication

We recommend **Point-in-Time (PIT) Governance** for verifiable evidence: the retrieval corpus L_t available at time t must contain only documents and performance metadata known at t Cook et al. (2025); Ng et al. (2025).

7.3. Evolutionary Alpha Discovery

Evolutionary alpha generation searches the factor space through iterative mutation, crossover, and selection, treating factor discovery as an optimization problem. This paradigm can discover novel factor combinations that human experts might not conceive.

Representative Approaches. Navigating Alpha Jungle (Shi et al., 2025) employs Monte Carlo Tree Search (MCTS) to efficiently explore the combinatorial space of possible factors: each node in the search tree represents a partial factor definition (e.g., a specific transformation applied to a price series), and MCTS balances exploration (trying new factor components) against exploitation (refining promising factors). Alpha^2 (Xu et al., 2024a) uses deep reinforcement learning, in which an agent learns to construct factors by sequencing primitive operations and receives rewards based on backtest performance. EFS (Evolutionary Factor Searching) (Luo et al., 2025) reformulates asset selection as a top-m ranking task guided by LLM-generated factors, incorporating an evolutionary feedback loop for iterative refinement.

Technical Mechanisms. Evolutionary methods define a search space of primitive operations (price transformations, technical indicators) and combination rules, which MCTS and Genetic Programming (GP) navigate by balancing exploration (new operations) against exploitation (refining promising factors). The paradigm, however, faces a severe **Multiple Testing Scalability** risk: as the search space grows exponentially, the probability of finding a false positive approaches 1.0 unless rigorous statistical corrections are applied White (2000). A valid protocol must therefore report (1) a **Primitive Ops Audit** (the exact set of allowed functions), (2) the **Total Search Depth/Budget**, and (3) statistical tests for data snooping (e.g., White's Reality Check or Hansen's SPA) Hansen (2005). At present, few agent papers report these quantities, which makes their results difficult to distinguish from noise; regularization techniques (complexity penalties, ensemble methods) are useful heuristics but do not substitute for formal multiple-hypothesis testing.

Open Question: Mathematical Bounds for Agent-Driven Search. While classical multiple-testing corrections (White's Reality Check, Hansen's SPA, and the Deflated Sharpe Ratio Bailey et al. (2014)) remain the statistical gold standard, a critical question emerges for LLM-based agents: can the agent's internal reflection/evolution loop provide novel mathematical bounds on overfitting risk? Current evidence suggests **not yet**. The reflection budget (Section 12.3) limits search iterations but offers no statistical guarantee on the false discovery rate; the sealed test set protocol prevents adaptive overfitting but relies on experimental discipline rather than closed-form bounds; and factor originality constraints (e.g., AST-based similarity in AlphaAgent) shrink the effective search space without quantifying the residual multiple-testing penalty. *This represents a significant research gap*: future work could explore whether LLM reasoning traces, factor genealogy logs, or Bayesian model averaging over discovered factors can yield principled, agent-native overfitting controls that complement or extend classical methods. Until such methods exist, evolutionary alpha systems should report both (1) search-budget and protocol constraints and (2) classical multiple-testing statistics (e.g., deflated Sharpe ratio Bailey et al. (2014)) to enable rigorous evaluation.

Table 14 organizes representative mechanisms and their design

Table 14: Representative alpha-discovery mechanisms and design considerations. The considerations identify review dimensions discussed in this survey. Tags describe artifact availability (R0–R1) or background literature (BG); see Table B.24.

Paradigm	Mechanism	Search object	Dimensions to examine	Works	Tags
Code-based	LLM → Code	Semantic factors	Structured prompting; hypothesis-to-code mapping	CogAlpha Liu et al. (2025)	R0
Code-based	Multi-Agent Debate	Interpretable factors	Debate-based critique and refinement	FactorMAD Duan et al. (2025)	BG
Retrieval	Vector Search	Factor library	Point-in-time (PIT) library governance	RAG-Fintech Cook et al. (2025)	R0
Retrieval	Prompt-Refined RAG	Financial corpus	Point-in-time (PIT) corpus governance	PRISM Ng et al. (2025)	BG
Evolutionary	MCTS / Deep RL	Combinatorial operations	Allowed operations; search budget; multiple testing	Alpha Jungle (R0), Alpha^2 (R1) Shi et al. (2025); Xu et al. (2024a)	R0-R1
Evolutionary	LLM-Guided Evolution	Sparse portfolios	Search budget and sealed-test separation	EFS Luo et al. (2025)	BG

considerations.

Extension-window additions (2026-03-10 to 2026-09-30). Additional studies automate factor or strategy discovery through agentic generation–backtest loops (Weng et al., 2026; Yuan et al., 2026b; Yu et al., 2026a; Li et al., 2026e; Liu et al., 2026; Yi et al., 2026; Zhao and Sha, 2026; Wang et al., 2026d; Li et al., 2026d; Zhao et al., 2026; Kim et al., 2026a; Ye et al., 2026a; Li et al., 2026g; Kvasiuk et al., 2026; Sharma and Shroff, 2026; Kim et al., 2026b), with further variants pushing toward trajectory-level self-iteration (Zhu et al., 2026a), constrained hypothesis-to-factor pipelines in cryptocurrency markets (Huang et al., 2026c), self-evolving coding agents for quantitative research (Li et al., 2026h), and stratified strategy-alignment objectives (Huang and Yu, 2026).

The reported evaluation settings span held-out or out-of-sample windows, per-trade or turnover-based costs, and named index universes such as CSI 300/500 or S&P 500. Interpreting results across these settings requires attention to the split, cost, and universe conventions used by each study.

A parallel mini-cluster interrogates the evaluation layer itself: closed-loop benchmarks and referee protocols for mining agents (Qu and Chen, 2026; Qu et al., 2026; Wang et al., 2026c; Deng et al., 2026; Cao et al., 2026; Zhang et al., 2026a), together with leakage and validity analyses (Ye et al., 2026c; Zhang and Stadie, 2026; Yao et al., 2026), now make evaluation design an explicit research object rather than an implementation detail. Two further studies deserve note: a direct assessment of LLMs on technical market-analysis tasks (Ntale, 2026), and evidence that nominally valid LLM signals can fail across regime boundaries (Yang, 2026)—a useful reminder that alpha-mining output is regime-conditional even when protocol fields are reported.

Synthesis and Empirical Considerations. Code-based generation, retrieval-augmented discovery, and evolutionary search explore different parts of the factor hypothesis space. Their empirical value depends on out-of-sample survival: recursive search can overfit noisy backtests, so walk-forward testing, factor-correlation checks, and turnover-adjusted transaction costs remain essential. The next section examines how discovered signals are aggregated into portfolios in Section 8.

8. Portfolio Management Capability

The preceding section examined how agents generate alpha signals. This section addresses the question: "How do agents manage multi-asset portfolios?"

The evidence is strongest where papers specify the action object—weights, orders, constraints, or rebalancing rules—and thinnest where portfolio management appears only as a conceptual layer.

Functional Scope and Allocation Paradigms. Portfolio management represents the functional layer where individual asset signals are synthesized into constrained, multi-asset capital allocations. While single-asset trading models emit directional buy/sell signals in isolation, portfolio-level agents must account for cross-asset covariance, leverage limits, sector exposure budgets, and rebalancing turnover. Across the surveyed literature, portfolio agents operate through three primary approaches:

- **Strategic Asset Allocation** (Section 8.1): Synthesizing macro market regimes and factor signals into baseline target asset distributions.
- **Tactical Position Sizing** (Section 8.2): Dynamically modulating individual asset exposures based on volatility, conviction, and liquidity.
- **Dynamic Rebalancing** (Section 8.3): Managing execution turnover and rebalancing intervals under transaction-cost and market-friction constraints.

Key empirical considerations include the specification of rebalancing calendars, explicit universe filtering (accounting for

delistings and liquidity), and turnover-induced transaction cost modeling.

Classical Baseline: Mean-Variance Optimization and Kelly Criterion

Classical portfolio theory casts allocation as mean-variance optimization (Markowitz 1952), which maximizes risk-adjusted returns given covariance estimates, or as growth-optimal betting (Kelly 1956), which maximizes the log-utility of wealth. The Black-Litterman model incorporates investor views into equilibrium returns. These frameworks yield closed-form solutions under Gaussian assumptions but struggle with non-stationary higher-order moments and tail risks—precisely where LLM-based approaches may add value through adaptive reasoning and regime-aware dynamic allocation.

Comparability note. Protocol-comparable portfolio claims typically report MR-1 (data/universe), MR-2 (time split), MR-3 (I/O contracts + constraints), and MR-4 (execution & costs); see Section 13.

8.1. Asset Allocation

Asset allocation sets portfolio weights across asset classes and broad themes (sectors, regions, styles), shaping the long-horizon risk/return envelope.

Representative Approaches. Classical baselines include mean-variance optimization Markowitz (1952) [BG] and Black-Litterman Black and Litterman (1992) [BG]; both hinge on estimates (returns, covariances, or implied priors) that can drift quickly. Risk-based allocations (risk parity / HRP-style clustering) are often preferred when estimates are noisy. In agentic systems, LLMs commonly act as *view generators* Lee et al. (2025); Kirtac and Germano (2025) conditioned on news and macro releases. Lee et al. (2025) show how LLM-generated views can be fed into the Black-Litterman framework, while Kirtac and Germano (2025) use LLM-based sentiment analysis to drive portfolio optimization decisions.

A parallel line of work pursues end-to-end neural allocation at scale: end-to-end networks have been proposed for variance minimization in large portfolios Bongiorno et al. (2025), and deep architectures combining 3D-CNNs with BiLSTMs target medium-term covariance forecasting Reis et al. (2025). Graph attention-based heterogeneous multi-agent reinforcement learning Zhang (2026a) offers adaptive portfolio optimization through multi-agent coordination, and pair-trading work that couples graph attention networks for pair selection with deep reinforcement learning for execution provides a non-LLM background example of relation-aware portfolio construction Xia et al. (2025) [BG]. HARLF Coriat and Benhamou (2025) integrates hierarchical reinforcement learning with lightweight LLM-driven sentiment analysis for financial portfolio optimization. These neural approaches complement mean-variance optimization by capturing non-linear dependencies that classical methods may miss Zouaoui and Naas (2025); Uysal et al. (2024).

These pipelines, however, introduce severe leakage risks wherever they ingest text and macro data. The time-series evaluation literature therefore requires **Point-in-Time (PIT) Textual Alignment**: macro inputs must use the unrevised values available at the exact release timestamp, and validation pipelines must prevent future information from leaking through preprocessing or feature construction Liu et al. (2022a); Albelali and Ahmed (2025) [BG]. For the same reason, *regime tagging* is best generated online from $t-k$ lookback windows, since full-sample regime clustering can leak future volatility states Hamilton (1989); Bai and Perron (2003) [BG].

Technical Mechanisms. HRP-style allocation reduces sensitivity to covariance noise by allocating at the cluster level before distributing within clusters. Regime-aware allocation (statistical or LLM-driven) can switch risk budgets when volatility/liquidity conditions change, but it is best treated as a hypothesis to test rather than a guarantee. For agentic trading, the practical core is constraint-aware optimization (exposure, leverage, turnover) with explicit transaction-cost assumptions López de Prado (2016); Ledoit and Wolf (2004); Jagannathan and Ma (2003) [BG].

8.2. Position Sizing

Position sizing determines how much capital to allocate to each trading signal, balancing the upside of strong signals against the risk of outsized losses. This tactical decision directly shapes portfolio volatility and drawdown.

Representative Approaches. The Kelly criterion Kelly (1956) [BG] links size to estimated edge and uncertainty, but is fragile to estimation error; most implementations therefore use fractional Kelly. Recent work revisits portfolio optimization from complementary angles Boyd et al. (2024); Salo et al. (2024) [BG], highlighting the growing importance of constraint-aware sizing in modern markets. Alternatives include risk-budget schemes (risk parity-style) and decision-informed neural methods Hwang et al. (2025) that integrate learned forecasts and LLM inputs into portfolio decisions.

Beyond these rules, recent formulations widen the sizing design space. ODE-based dynamic optimization Alkhudaydi (2025) offers continuous-time solutions to the mean-variance problem, while cardinality- and bound-constrained formulations Li et al. (2024b); Caner and Fan (2024) address practical trading constraints that theoretical models often ignore. On top of these formulations, agentic systems may inject qualitative risk flags extracted from text; such inputs, however, require strict time alignment and auditability.

Technical Mechanisms. In practice, sizing layers use conservative shrunk estimates and fractional Kelly-style scaling to keep worst-case losses bounded.

Protocol Implication

For comparability, we recommend reporting a **Position Sizing Checklist**: (1) **Risk Metric**: What is minimized? (Vol, CVaR, MaxDD), (2) **Constraints**: What are the boundaries? (Long-only, Leverage $<$ 1.5, Sector $<$ 20%),

and (3) **Scaling Logic**: How is conviction mapped to size? (Fractional Kelly, Volatility Targeting).

When LLMs contribute qualitative risk signals, they can be treated as features with strict timestamping and ablation tests (text signal on/off) to verify incremental value Kelly (1956); MacLean et al. (2011); Rockafellar and Uryasev (2000) [BG].

8.3. Rebalancing

Rebalancing adjusts portfolio weights to maintain target allocations as market prices move and new signals arrive. This operational decision trades the benefit of preserving the intended risk profile against the cost of frequent trading.

Across the mapped literature, agentic portfolio construction and allocation already appear in several applied settings. Agentic AI systems have been applied to dynamic crypto-asset allocation, where adaptability matters most under high volatility Castelli et al. (2025); as background examples, generative-AI pipelines construct sector portfolios from LLM selections combined with classical optimization Voronina et al. (2025) [BG], and Poly-Model theory coupled with iTransformer architectures targets hedge-fund portfolio construction Zhao et al. (2024) [BG]. On the transparency side, explainable DRL techniques aim to make financial policy decisions interpretable de la Rica Escudero et al. (2025) [BG]. Strategy-guided exploration mechanisms can broaden agent robustness for portfolio workflows, though they are best read here as a generic robustness idea rather than direct evidence for rebalancing Szot et al. (2026) [BG].

Representative Approaches. Canonical policies include threshold rebalancing (act only when weights drift beyond bands), time-based schedules, and cost-aware optimization that trades off tracking error vs. transaction/impact/tax costs.

Learning-based methods then refine how these policies are decided. Heuristic-guided reward learning has been proposed for portfolio optimization, showing how learned heuristics can improve rebalancing decisions Zhang et al. (2025), and multi-objective gradient descent enables simultaneous optimization of competing portfolio objectives during rebalancing Oliva et al. (2025).

Agentic workflows often add *event awareness* on top of these schedules (e.g., going risk-off before FOMC), which creates a second leakage surface. Comparisons are most credible when evaluations distinguish between **Calendar Events** (scheduled ahead of time, e.g., Earnings Calendar) and **Shock Events** (unexpected news): rebalancing driven by shocks should rely on real-time proxies (e.g., VIX spike, spread widening) observable at t, rather than future-annotated event labels Li et al. (2023); Unnikrishnan (2024). FinVision Fatemi and Hu (2024) [BG] is better read as a multimodal stock-prediction example than as direct evidence for this taxonomy.

News-driven rebalancing has emerged as a particularly active application: one line of work combines financial news analysis with reinforcement learning for portfolio management, enabling dynamic rebalancing in response to real-time sentiment shifts Unnikrishnan (2024). Liquidity-aware selection and rebalancing strategies Abensur (2024); Nguyen (2025) further incorporate machine learning-based liquidity classification to guide allocation and execution timing.

Technical Mechanisms. No-trade regions (bands) reduce churn by ignoring small drift until it matters for risk, while execution models (VWAP/TWAP/implementation shortfall (IS); Section 6.2) and explicit slippage assumptions determine whether a policy survives outside a backtest. For agentic trading, the key protocol caveat is separating *decision timing* (signal/optimizer) from *fill timing* (execution) to avoid overstating performance. Where tax-aware logic is included, the evaluation should match the account type and jurisdiction that the tax model assumes Constantinides (1986); Davis and Norman (1990); Almgren and Chriss (2001); Perold (1988) [BG].

Extension-window additions. Building on agentic crypto-asset allocation Castelli et al. (2025), additional studies include screening pipelines that feed portfolio construction Caner et al. (2026), LLM-driven small-capitalization trading with integrated allocation Kargarzadeh et al. (2026), recursively self-optimizing multi-agent portfolio systems Yang et al. (2026), and a market-regime council that performs dynamic credit assignment across specialized agents Pei et al. (2026). These systems make the portfolio layer an explicit, inspectable component, though their evaluation conventions (rebalancing frequency, constraints, cost models) remain heterogeneous.

Synthesis and Practical Considerations. Portfolio management translates alpha into diversified holdings under allocation, sizing, and rebalancing constraints. LLM-generated views can be paired with classical optimization to connect qualitative context with quantitative risk bounds; credible evaluation depends on reporting the asset universe, rebalancing schedule, and turnover costs. Section 9 examines controls across the trading lifecycle.

9. Risk Management Capability

The preceding sections examined alpha generation and portfolio management. This section addresses the question: "How do agents control financial risks?"

We treat risk management here as the set of constraints, monitors, and post-trade controls that shape allowable actions. The discussion stays tied to the evidence base because many risk mechanisms are specified at the architecture level but rarely isolated in empirical evaluation.

Systemic Role and Risk Governance Architecture. Risk management forms the essential protective foundation of an autonomous trading system, ensuring capital preservation, regulatory compliance, and resilience against market shocks. Because LLMs can exhibit hallucination, overconfidence, or vulnerability to adversarial input, real-world deployment requires independent, deterministically governed risk guardrails. Across the trading lifecycle, risk architecture is structured across three defensive stages:

Table 15: Portfolio management method comparison: traditional methods, candidate LLM enhancements, and key challenges at each decision level.

Level	Traditional Methods	LLM Enhancement	Key Challenges
Asset Allocation	Mean-variance optimization, Black-Litterman, Risk parity, Hierarchical clustering	Regime detection from narratives, View generation, Market sentiment analysis	Covariance estimation errors, Regime misclassification
Position Sizing	Kelly criterion, Risk parity equalization, Convex optimization (VaR/CVaR)	Qualitative risk assessment from news, Constraint handling with reasoning	Estimation error sensitivity, Over-aggressive sizing
Rebalancing	Threshold-based, Time-based, Cost-aware optimization	Strategic timing around events, Transaction cost estimation, Tax-aware harvesting	Transaction costs, Market impact, Tax optimization complexity

- **Pre-Trade Risk Controls** (Section 9.1): Hard gatekeeping filters verifying instrument whitelists, maximum order sizes, leverage ceilings, and liquidity availability prior to order submission.
- **In-Trade Monitoring and Dynamic Exposure** (Section 9.2): Real-time surveillance tracking aggregate portfolio value-at-risk (VaR), sector concentration limits, trailing stop-losses, and emergency circuit breakers.
- **Post-Trade Attribution and Compliance** (Section 9.3): Systematic trade reconciliation, slippage analysis, performance attribution, and regulatory audit logging.

From an engineering standpoint, the key challenge lies in establishing non-bypassable, deterministic overrides that can intercept errant LLM actions before market transmission without introducing unacceptable execution delays. These safety mechanisms apply across the pre-trade, in-trade, and post-trade stages. The stages are detailed in Section 9.1, Section 9.2, and Section 9.3; cross-stage guardrails are discussed in Section 9.4.

Classical Baseline: VaR, CVaR, and Drawdown Control

Traditional risk management quantifies tail risk through Value-at-Risk (VaR)—the quantile of the loss distribution—and Conditional VaR (CVaR), the expected loss beyond VaR. Drawdown control limits peak-to-trough declines via dynamic programming or convex optimization. These metrics provide computationally tractable risk bounds but assume stationary distributions and known correlation structures. LLM agents can instead produce natural-language explanations of risk exposures and generate scenarios; to reach an auditability standard comparable to classical frameworks, however, those explanations must be grounded in verifiable metrics and immutable logs Khatchadourian (2026) [BG].

Figure 11 illustrates the lifecycle overview.

Comparability note. Protocol-comparable risk claims typically report MR-3 (risk action semantics + priorities), MR-4 (execution/cost assumptions that risk controls depend on), and MR-6 (artifacts & logs); see Section 13.

9.1. Pre-trade Risk Control

Pre-trade risk control prevents excessive risk by checking orders against risk limits before they are submitted to the market. This preventive stage is the first line of defense against catastrophic losses.

Representative Approaches. Value-at-Risk (VaR) limits cap the maximum potential loss at a specified confidence level (e.g., 99% one-day VaR of $1M). Conditional VaR (CVaR) extends this by estimating the expected loss in the tail beyond the VaR threshold Rockafellar and Uryasev (2000) [BG]. Exposure constraints limit concentrations to single assets, sectors, or counterparties (e.g., no more than 10% of portfolio in any single stock), and compliance checks ensure that trades respect regulatory requirements and internal policies. On the agentic side, some prototypes use LLMs to translate human-readable policies into executable constraints Cosler et al. (2023) [BG], while others employ structured, model-based risk representations to estimate market risk transparently. This translation step, however, injects non-determinism into a layer that is expected to behave deterministically—a tension the protocol implication below addresses directly.

Protocol Implication

For protocol comparability, we recommend reporting: (1) **VaR Config** (Confidence Level, Holding Period, Window), (2) **Scenario Source** (Standard vs Custom), and (3) **LLM Verification**: policy-to-code translations are best validated against deterministic test sets rather than raw LLM chat outputs for compliance.

Technical Mechanisms. Portfolio risk models estimate the risk contribution of each potential trade, considering both the standalone risk of the position and its correlation with existing holdings. Let Σ denote the covariance matrix of asset returns with entries σ_{ij}, and let w denote portfolio weights. The portfolio variance is $V = w^\top \Sigma w$, and the *marginal contribution to variance* is $\frac{\partial V}{\partial w_i} = 2(\Sigma w)_i$. A commonly used *risk contribution* decomposition is $RC_i = w_i(\Sigma w)_i$, while the *marginal contribution to volatility* is $\frac{\partial \sigma_p}{\partial w_i} = (\Sigma w)_i/\sigma_p$ with $\sigma_p =$

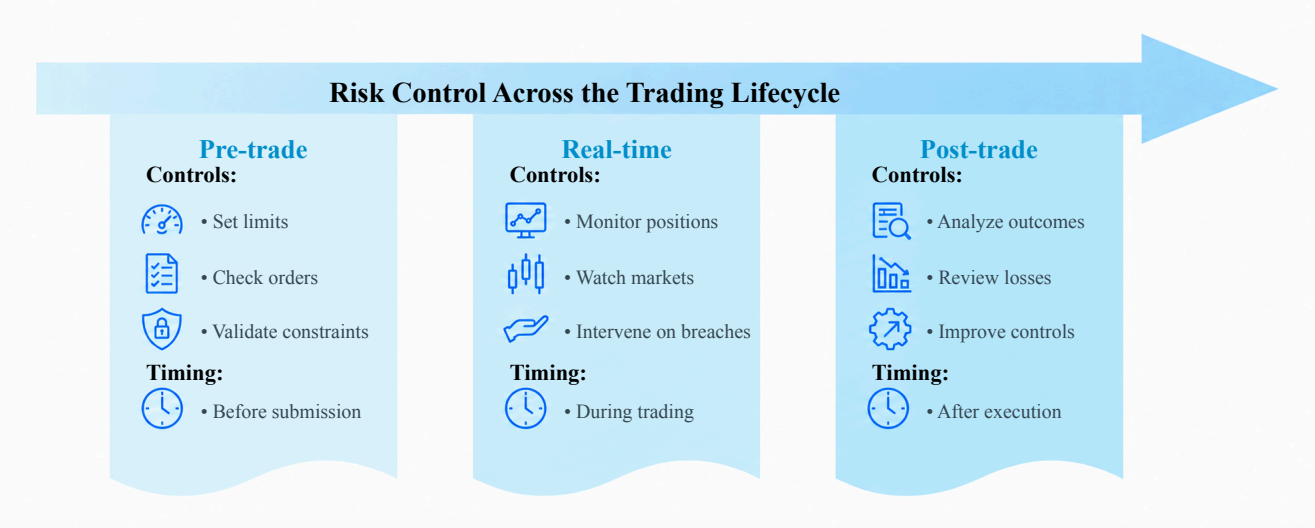

Figure 11: Risk control timeline across the trading lifecycle. Pre-trade risk control sets limits and checks before orders are submitted. Real-time risk control monitors positions and markets during trading, intervening when thresholds are breached. Post-trade risk analysis analyzes outcomes to improve future risk management.

$\sqrt{w^\top \Sigma w}$ Roncalli (2013) [BG]. Scenario analysis simulates portfolio performance under stress scenarios (market crash: -30% equities, interest rate spike: +200bps, volatility surge: $\sigma \times 3$) to identify hidden risks that might not manifest under normal conditions Basel Committee on Banking Supervision (2009). Rule engines encode regulatory and internal policies as executable rules (e.g., Reg T margin requirements, Reg NMS order protection rules); here LLMs can parse natural-language policies into such rules automatically, which is precisely where the non-determinism caveat above applies.

Adjacent work supports this layer at two different levels. Agent-based market models generate realistic transaction costs and slippage for liquidity risk assessment Vytelingum et al. (2025) [BG], while the STRIDE framework offers guidance on when a risk-sensitive application calls for agentic AI, an AI assistant, or a direct LLM call Asthana et al. (2025) [BG].

9.2. *Real-time Risk Control*

Real-time risk control monitors positions and market conditions during trading, intervening when risk thresholds are breached or anomalous conditions emerge. This reactive stage catches risks that pre-trade models miss due to changing market conditions or model error.

Representative Approaches. Position monitoring tracks portfolio metrics (greeks for options, duration for bonds, beta for equities) in real time, alerting when they exceed thresholds. Drawdown stops halt trading when portfolio losses reach specified levels (e.g., stop trading if down 5% from peak), and volatility scaling reduces position sizes as market volatility increases, maintaining constant risk exposure. Text streams can extend this monitoring with early warnings Tetlock (2007) [BG]; slower-moving macro uncertainty indices complement them Baker et al. (2016) [BG], though these indices are not real-time signals. Because real-time control is also a first-order design concern for cross-market robustness, the division of labor matters: LLMs are generally unsuitable for the sub-millisecond **Critical Path**, so a robust architecture separates a **Gating Layer** (deterministic algorithms such as circuit breakers for sub-millisecond stops) from an asynchronous **Explanation Layer** (LLMs for cause attribution). Evaluations should then disclose **Latency Budgets** and **False Positive Rates** for all triggers.

Technical Mechanisms. Stream processing systems update risk metrics from tick data, trades, and order-book changes. Anomaly detection can flag price spikes, volume surges, and order-book imbalance as stress signals Easley et al. (2012); Kirilenko et al. (2017) [BG]. Circuit breakers halt trading when drawdown or single-position loss limits are breached. Manual review before restart helps prevent stress-driven decisions Lo and Repin (2002); Kahneman and Tversky (1979) [BG].

Protocol Implication

We suggest reporting any dynamic exposure rule in display form rather than embedding it in prose. A simple example is

$$w_i(t) = \min\left(w_i^{\max}, \frac{\text{risk_budget}}{\text{risk_per_unit}_i}\right)$$

which caps concentration as prices and volatility change.

9.3. *Post-trade Risk Analysis*

Post-trade risk analysis learns from trading outcomes to improve future risk management. This retrospective stage identifies what risks were missed, how effective controls were, and where improvements are needed.

Representative Approaches. Performance attribution decomposes returns into component sources (alpha return, beta, timing,

stock selection), identifying which risks contributed to gains or losses; LLM-assisted attribution can narrate how market factors propagated through the portfolio structure. Stress testing simulates portfolio performance under historical crisis scenarios (2008 financial crisis and other major stress episodes) to assess resilience Basel Committee on Banking Supervision (2009). Regime analysis examines how portfolio performance varied across market regimes (bull, bear, high volatility), identifying conditions where risk management failed. LLM-enhanced systems can summarize these risk outcomes, but guarding against post-hoc rationalization requires every narrative claim to be grounded in immutable logs and replayable evidence Tan et al. (2025a); Khatchadourian (2026) [BG]. For the same reason, **Counterfactual Reasoning** (e.g., "what if" scenarios) is best labeled *Hypothetical Inference*—an assumption-dependent construction, not an empirical fact.

Technical Mechanisms. Causal analysis identifies the root causes of risk events, distinguishing between market risk (systemic factors affecting all assets), idiosyncratic risk (asset-specific events), and model error (incorrect assumptions or predictions). Techniques include Granger causality tests, Shapley value decomposition for attribution, and structural break detection to identify when risk models failed. Counterfactual reasoning estimates what would have happened under different risk management decisions (as an illustrative example, "if we had reduced position size by 50%, drawdown would have been 3% instead of 8%"), enabling learning from near-misses. Report generation produces human-readable risk summaries for compliance and management review, with LLMs translating quantitative metrics ("VaR exceeded by 20% on March 15th") into narrative explanations ("Risk was elevated due to concentrated tech exposure during the AI sector correction; position limits were triggered but too late to prevent significant losses"), making risk analysis interpretable for non-technical stakeholders Granger (1969); Shapley (1953); Bai and Perron (2003); Pearl (2009) [BG].

Table 16 summarizes risk management approaches across the three stages. Effective risk management depends on all three stages working in concert: pre-trade controls prevent excessive risk, real-time controls catch emerging risks, and post-trade analysis enables continuous improvement.

9.4. Safety and Constraint Mechanisms

Deployable trading agents require safety layers that sit outside the model's generative reasoning loop. These include pre-trade exposure checks, hard position limits, concentration caps, drawdown triggers, and kill-switch logic that can halt order flow when market conditions or model behavior leave the approved operating envelope. In the current literature, such mechanisms are often mentioned in passing but rarely specified at reproducible detail. We therefore treat safety architecture as part of the risk-management interface, not as an optional deployment afterthought.

Extension-window additions. The extension cohort begins to fill this gap: parameterized stop-loss/take-profit layers for autonomous strategies Li et al. (2026c) make exit-rule design an explicit, tunable risk interface, and dynamic objective selection under safeguard constraints with LLM oversight Sakurai et al. (2026) treats risk bounds as first-class control inputs. A complementary background analysis systematizes how trading agents fail under adversarial and regime-shifted conditions Wang and Saxena (2026) [BG]; it is not primary evidence for this gap.

Synthesis and Governance Considerations. Risk governance spans pre-trade limits, in-trade monitoring, and post-trade analysis, with deterministic safety mechanisms constraining actions across all three stages. FinRS Liu and Dang (2025b) incorporates risk sensitivity into the trading objective; dedicated limits, gating, and monitoring provide complementary safeguards at the execution and governance layers. Part III turns to adaptation and evolution, beginning with adaptive learning paradigms in Section 10.

10. Learning Paradigms

The preceding parts of this survey established the architectural foundations of agentic traders and the capabilities these architectures enable. This section addresses the question: "How do agents learn from market feedback?"

Learning enables agents to adapt their behavior based on experience and changing market conditions. This section covers learning mechanisms; governance of persistent state and update cycles is treated in Section 12.

Adaptive Mechanisms and Learning Spectrum. Financial markets are intrinsically non-stationary: statistical patterns decay, volatility regimes shift abruptly, and trading signals erode as market participants adapt. Consequently, static trading agents face rapid performance degradation unless equipped with mechanisms for continuous adaptation. Across the surveyed literature, agent learning spans three foundational paradigms along the spectrum of parameter persistence and adaptation speed:

- **In-Context Learning (ICL)** (Section 10.2): Adaptation through prompts that conditions future decisions on retrieved market exemplars and reflective feedback without modifying model weights.
- **Supervised Fine-Tuning (SFT)** (Section 10.3): Domain adaptation through weight updates that trains base LLMs on specialized financial instructions, historical trade rationales, and domain corpora.
- **Reinforcement Learning (RL)** (Section 10.4): Policy optimization that trains agents to maximize risk-adjusted sequential rewards (e.g., cumulative return, Sharpe ratio) within interactive trading environments.

The primary methodological trade-offs involve balancing sample efficiency and adaptation agility against catastrophic forgetting and distribution-shift vulnerability.

Table 16: Risk management method comparison: mechanisms and candidate LLM enhancements across the three lifecycle stages.

Stage	Mechanisms	LLM Enhancement
Pre-trade	VaR/CVaR limits; exposure constraints; compliance checks	Rule interpretation; policy translation; natural-language constraints
Real-time	Position monitoring; drawdown stops; volatility scaling	Anomaly detection; news monitoring; early-warning signals
Post-trade	Attribution; stress testing; regime analysis	Narrative explanation; report generation; learning insights

10.1. Reward and Feedback Design

Reward specification is a first-order design choice rather than an implementation detail. Trading environments impose delayed, noisy, and partially confounded feedback: realized P&L arrives after execution, inventory risk accumulates across steps, and reward proxies can be gamed through simulator artifacts. We therefore distinguish immediate execution feedback (fill quality, slippage, rejected orders) from delayed portfolio feedback (P&L, drawdown, turnover, and risk-adjusted return). Papers claiming adaptive improvement should report which of these signals are optimized, over what horizon, and with what safeguards against reward hacking.

Figure 12 provides a conceptual comparison of the three paradigms.

10.2. In-context Learning

In-context learning (ICL) enables agents to adapt from examples provided in the prompt without updating model parameters, following the GPT-3-style few-shot paradigm Brown et al. (2020) [BG]. This rapid adaptation mechanism is particularly valuable in dynamic markets where conditions change faster than models can be retrained.

Representative Approaches. FinAgent Zhang et al. (2024) uses ICL by retrieving relevant trading episodes from memory and inserting them as examples in the prompt, enabling the LLM to adapt its reasoning to similar situations without parameter updates.

Technical Mechanisms. ICL selects and orders examples for the prompt, using retrieval and context-management choices described in Section 4 Brown et al. (2020) [BG], Zhang et al. (2024). RAG also adds current financial documents from external stores. Because neither updates model parameters, any adaptation is temporary and bounded by the context window.

10.3. Supervised Fine-tuning

Supervised fine-tuning (SFT) updates model parameters on labeled demonstrations of desired behavior, teaching agents to imitate expert trading decisions. This approach transfers knowledge from expert traders or historical successful strategies into the agent.

Representative Approaches. The examples below distinguish finance-domain tuning methods from end-to-end trading systems. The extension cohort begins to move this boundary: post-training of local quantitative research agents Wang et al. (2026d) and cognitive fine-tuning for stable financial reasoning Pan and Liew (2026) bring weight updates closer to the trading loop, though fully end-to-end SFT evidence for trading agents remains thin. The FinGPT instruction-tuning benchmark Wang et al. (2023a) [BG] illustrates finance-domain instruction tuning on financial NLP tasks. Two supervised variants matter most for trading agents: *behavior cloning*, which maps market states to expert actions so that trader demonstrations become usable policy targets Ross et al. (2011) [BG]; and *multi-task adaptation*, which trains on sentiment, portfolio, and risk tasks jointly and produces representations that may transfer across finance subtasks Caruana (1997) [BG].

Technical Mechanisms. SFT relies on instruction and demonstration datasets with prompt-response pairs (market context → expert decision or analysis). Training minimizes a supervised loss (e.g., cross-entropy for discrete actions, regression for sizing/allocation), with standard regularization and careful held-out evaluation to check robustness to unseen regimes. Parameter-efficient methods (e.g., LoRA-style methods) reduce cost while preserving much of the base model; finance-oriented benchmarking work has explored these trade-offs Wang et al. (2025a). General parameter-efficient variants are background mechanisms Dettmers et al. (2023); Xu et al. (2024b) [BG]. SFT can produce persistent behavioral changes but is sensitive to label quality and distribution shift.

10.4. Reinforcement Learning

Reinforcement learning (RL) is used in trading agents when the objective is inherently sequential (portfolio rebalancing, execution, or iterative factor construction), the feedback is delayed, and the policy needs to trade off return, risk, and costs under constraints.

Representative Approaches. FinRL-DeepSeek Benhenda (2025) combines LLM-based priors with policy optimization, reporting risk-adjusted return improvements over baselines in simulated environments[1]. MacroHFT Zong et al. (2024) introduces memory-augmented context-aware RL for high-frequency trading, while EarnHFT Qin et al. (2024) explores hierarchical RL for HFT; both are conventional deep-RL trading systems rather than LLM agents and are retained as non-agentic baselines. Deep RL for alpha discovery Xu et al. (2024a) frames factor construction as a program-generation task, optimizing an

[1]Specific performance metrics vary by configuration (e.g., CPPO-DeepSharpe vs. CPPO-DeepVol) and market regime; readers should consult the original paper for detailed protocol assumptions, cost models, and train/test split discipline.

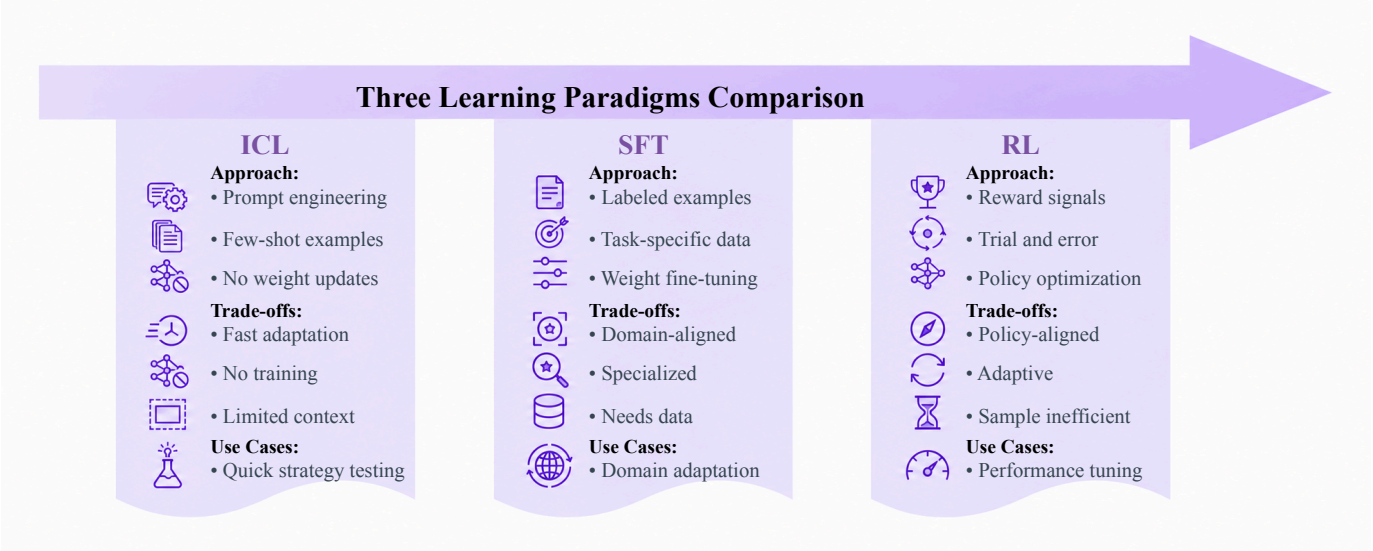

Figure 12: Conceptual comparison of three learning paradigms. The panels summarize what each loop updates, the main trade-offs emphasized in the design literature, and illustrative use cases.

evaluation metric over the search space while balancing alpha performance and diversity.

Adjacent work marks the boundary between learning rules and persistent cross-time change: modular memory architectures retain experience across tasks Dorovatas et al. (2026) [BG], explicit state dynamics improve temporal coherence Subaharan (2026) [BG], and code-based alpha evolution refines strategies from execution traces and natural-language feedback Liu et al. (2025) [BG]. The latter update-governance questions are taken up in Section 12.

What should be reported (and why it matters). Because RL results are highly sensitive to experimental setup, trading papers should explicitly specify (i) chronological data splits (train/validation/test or walk-forward) and how hyperparameters are chosen to avoid leakage; (ii) the cost model (commissions, spread/slippage, and market impact assumptions); and (iii) simulator fidelity (fill/latency assumptions, action/state/reward definitions, constraints, and whether the simulator matches the instrument/microstructure being claimed). Key risks include simulator overfitting and reward hacking: policies can exploit artifacts of the backtester/cost model and fail catastrophically out-of-sample, especially under non-stationarity and regime shifts.

Synthesis and Algorithmic Considerations. In-context learning conditions decisions on prompt examples and retrieved experience; supervised fine-tuning updates parameters from demonstrations; reinforcement learning optimizes sequential policies through reward feedback. FinAgent illustrates prompt-context adaptation Zhang et al. (2024), while FinRL-DeepSeek and Alpha^2 use RL Benhenda (2025); Xu et al. (2024a). These approaches differ in persistence and feedback, and their evaluation depends on example and label coverage, temporal splits, simulator fidelity, and reward design. Section 11 next examines collective coordination.

11. Multi-Agent Coordination

Note: Multi-agent coordination is a cross-cutting organizational dimension: role structure is architectural, while coordination dynamics can shape capabilities and adaptation. This section focuses on those interactions and appears in Part III because its emphasis is on emergent coordination.

The preceding sections examined the architecture and capabilities of agentic traders. This section addresses the question: "How do multiple agents collaborate in markets?"

Multi-agent systems enable specialization and decomposition, with different agents focusing on different aspects of the trading problem (analysis, execution, risk management). What remains unsettled is whether coordination itself delivers superior trading performance once computational budget, information access, and protocol design are held constant.

Coordination Patterns and Market Ecology. Financial decision-making in institutional environments is inherently collaborative, distributed across macroeconomic analysts, sector specialists, quantitative modelers, risk officers, and execution traders. Multi-agent trading frameworks mirror this division of labor by deploying specialized agents that interact, deliberate, and coordinate. In the literature, multi-agent organizations are structured along three principal paradigms:

- **Role-Based Collaboration** (Section 11.1): Flat or peer-to-peer agent topologies where specialized personas (e.g., fundamental bull/bear debaters, sentiment analysts) exchange structured messages to reach trading consensus.
- **Hierarchical Networks** (Section 11.2): Multi-tiered command structures where supervisory portfolio manager

Table 17: Conceptual design trade-offs of learning paradigms. These descriptions characterize adaptation mechanisms; training cost and data requirements depend on the task and implementation.

Paradigm	Adaptation mechanism	Data source	Typical strengths	Design considerations
ICL	Context updates	Prompts and examples	Immediate task conditioning	Context capacity; no persistent weight update
SFT	Supervised weight updates	Labeled demonstrations	Persistent domain specialization	Label coverage; retraining; overfitting
RL	Reward-driven policy updates	Interactive trajectories	Optimizing sequential decisions	Simulator fidelity; reward design; sample cost

agents aggregate tactical recommendations from subordinate asset specialists.

- **Synthetic Market Ecologies** (Section 11.3): Agent-based market simulations where heterogeneous populations of autonomous buyers, sellers, and liquidity providers interact, allowing the study of emergent price dynamics, market impact, and reflexivity.

The central research challenges involve attributing performance gains to specific communication topologies (via rigorous role ablations), managing communication latency overheads, and preventing groupthink or herd dynamics.

Comparability note. Protocol-comparable coordination claims should report MR-7 (roles/permissions, message protocol, consensus, shared state, and evaluation); see Section 13.

Figure 13 illustrates these stylized archetypes.

11.1. Role-based Collaboration

Role-based collaboration assigns specialized roles to different agents, each focusing on a specific aspect of the trading problem while coordinating to achieve collective goals. This division of labor enables deep specialization while maintaining coherent decision-making Wooldridge (2009) [BG].

Representative Approaches. We distinguish three evidence layers. *Primary trading systems* in the current section-level subset show role specialization most directly: TradingAgents Xiao et al. (2024) assigns analyst/trader/risk/portfolio roles; FINCON Yu et al. (2024) uses a manager–analyst architecture with Conceptual Verbal Reinforcement (CVRF) to update investment beliefs; TradingGroup Tian et al. (2025) combines role specialization with reflection; and the REIT pipeline Li (2026) makes an analysis–prediction–decision–execution workflow explicit. These papers make decomposition visible, yet complex role assignments can still obscure the source of gains. Supporting *attribution* therefore requires logging decision contributions (which agent proposed or vetoed an action) and running role/communication ablations that verify gains stem from coordination rather than from increased computational budget. General LLM multi-agent optimization work such as HiveMind supplies relevant background on contribution-guided credit assignment and online prompt refinement Xia et al. (2026) [BG].

Benchmarks and simulated arenas form a second layer: TraderBench, MarS, and related multi-agent market arenas Lopez-Lira (2025); Li et al. (2024a); Yuan et al. (2026a) [BG] stress specialization and interaction patterns under their respective simulation protocols. Trading-performance comparisons across these layers require matched execution semantics and computational budgets.

Background MAS/ABM literature forms a third layer, including social or behavioral market models Bohorquez and Cartlidge (2024); Wheeler and Varner (2023) [BG], RL-plus-ABM interaction studies Dicks et al. (2023); Oort et al. (2023) [BG], workspace delegation Nie et al. (2026) [BG], and reciprocity-oriented foundations Diau (2025); Gao et al. (2022) [BG]. These works inform mechanism design through models of delegation, reciprocity, and market interaction.

Communication Protocols. Role-based collaboration rests on communication protocols that coordinate specialized agents effectively. Broadcasting allows an agent to share information with all team members simultaneously, which suits market-wide alerts or portfolio status updates. Peer-to-peer communication enables direct exchange between specific agents, such as the risk manager warning the trader about excessive exposure. Shared memory provides a common knowledge base where agents can read and write information, enabling asynchronous coordination without requiring all agents to be available simultaneously. These protocols need to be designed to balance information richness against communication overhead, as excessive messaging can slow decision-making in fast-moving markets Wooldridge (2009) [BG].

Consensus Mechanisms. When specialized agents disagree, consensus mechanisms aggregate diverse opinions into coherent decisions. Voting treats all agents equally, with the majority or plurality determining the final action. Expert weights assign different influence to agents based on their historical accuracy or domain expertise, allowing the analyst to dominate stock selection while the risk manager influences position sizing. Debate-based consensus enables agents to present arguments and counterarguments, as illustrated by the bull–bear researchers in TradingAgents Xiao et al. (2024). FinCon instead refines investment beliefs across episodes through CVRF Yu et al. (2024). The choice of consensus mechanism involves trade-offs between speed (voting is fast) and quality (debate can

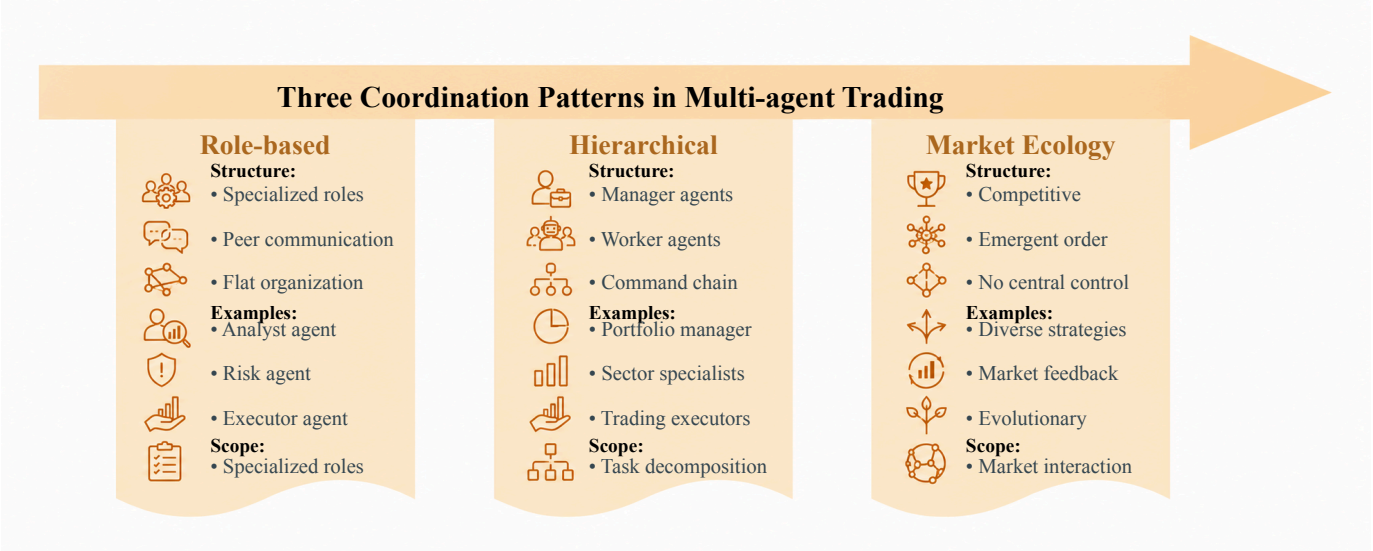

Figure 13: Three coordination patterns in multi-agent trading. Role-based collaboration assigns specialized roles (analyst, trader, risk manager) to different agents. Hierarchical organization layers agents from strategic (asset allocation) to tactical (position sizing) to execution (individual trades). Market ecology models agents interacting through markets with feedback and emergence. The structure/example/scope panels are stylized archetypes of these organizational designs.

uncover blind spots but takes longer). Dynamic mechanisms that adapt based on market conditions or agent confidence are particularly valuable in volatile markets Hu et al. (2024); Wooldridge (2009) [BG].

Task Allocation. Effective role-based collaboration requires task allocation that matches work to agent capabilities and current workload. Capability-based allocation assigns tasks based on specialized expertise, such as routing earnings analysis to the fundamental analyst while the technical analyst handles technical signals. Workload-based allocation considers current load, preventing bottlenecks when one agent is overwhelmed. Hybrid approaches combine both factors, ensuring that urgent but non-specialized tasks (such as emergency risk reduction) can be handled by any available agent. Task allocation may be centralized (a supervisor agent assigns work, as in TradingAgents) or decentralized (agents self-assign based on capability and availability). Dynamic reallocation enables the system to adapt when agents fail or become overloaded, improving robustness Wooldridge (2009) [BG].

Conflict Resolution. Disagreements between specialized agents are inevitable, particularly when agents have different perspectives (optimistic analyst vs pessimistic risk manager) or when their roles create tensions (trader seeking profit vs risk manager seeking safety). Negotiation allows agents to propose compromises, such as reducing position size to satisfy both profit targets and risk limits. Arbitration delegates dispute resolution to a higher authority (the supervisor in TradingAgents), ensuring that critical decisions are not deadlocked. Authority hierarchies establish which agent's view dominates in which domain (the risk manager has veto power over position sizing, while the analyst controls stock selection). Learning from past conflicts improves resolution over time, with agents developing heuristics for when to escalate disagreements and when to accept compromises Wooldridge (2009) [BG].

11.2. Hierarchical Organization

Hierarchical organization structures agents into layers, with higher-level agents making strategic decisions and lower-level agents executing tactical actions. This structure enables coherence across time scales, from long-term asset allocation to short-term execution.

Representative Approaches. FinMem Yu et al. (2023b) illustrates hierarchy within a single agent's memory rather than a multi-agent command structure. It employs a working memory for summarization, observation, and reflection, together with long-term memory divided into shallow, intermediate, and deep processing layers. These layers use different retention and decay schedules to support daily trading decisions; they are not separate working, episodic, and semantic stores. H-MEM Sun and Zeng (2025) extends this with semantic hierarchy, reporting efficiency improvements through hierarchical memory organization. In multi-agent settings, strategic agents set high-level targets ("reduce tech exposure by 20%") and tactical agents determine specific trades to achieve these targets. General-purpose frameworks provide complementary organizational examples: MetaGPT uses predefined roles and standard operating procedures for software engineering Hong et al. (2024) [BG], while DyLAN and HiveMind explore adaptive team selection or prompt refinement Liu et al. (2023); Xia et al. (2026) [BG]. These mechanisms inform the design of trading hierarchies without establishing trading performance themselves.

Benchmarks and Protocols. Coordination claims are difficult to compare without common evaluation protocols and cost and execution semantics. Recent work introduces live and

replay-based benchmarks for trading agents that stress sequential decision-making under real or event-driven market dynamics, including AI-Trader Fan et al. (2025) [BG], Agent Market Arena Qian et al. (2025) [BG], LiveTradeBench Yu et al. (2025) [BG], ContestTrade Zhao et al. (2025) [BG], and PredictionMarketBench Arora and Malpani (2026) [BG]. These benchmarks help separate architecture effects (coordination design, memory, workflow) from model-backbone effects, but remain sensitive to market selection, episode construction, and transaction cost assumptions.

Protocol Implication

For multi-agent systems, we recommend explicit reporting of: (i) *cross-play* evaluation (swap individual agents to test interoperability vs. team overfitting), (ii) explicit *communication constraints* (channel bandwidth and tool-call budgets), (iii) *crowding and impact* metrics (e.g., impact amplification under correlated policies), and (iv) *role ablations* to attribute gains to coordination rather than increased information access.

Goal Decomposition. Hierarchical organization requires breaking high-level objectives into actionable sub-goals that lower-level agents can execute. Top-down decomposition starts with strategic objectives ("target 15% annual return and 10% volatility") and recursively breaks them into smaller targets ("allocate 60% to equities, 30% to bonds, 10% to cash" then "select 50 technology stocks with equal weights"). Constraint-based decomposition helps ensure that sub-goals respect risk limits, liquidity constraints, and regulatory requirements. Dynamic decomposition adjusts targets based on market conditions, such as reducing return targets during bear markets or tightening risk limits when volatility spikes. The effectiveness of goal decomposition depends on how well strategic intent can be translated into tactical guidance without micromanaging lower-level agents; layers should specify clear I/O contracts (e.g., strategic targets vs. tactical orders) to prevent semantic drift Wooldridge (2009) [BG].

Feedback Mechanisms. Hierarchies rely on upward feedback to close the loop between strategy and execution. Bottom-up reporting enables tactical agents to inform strategic agents about outcomes, constraints, and opportunities. For example, execution agents report that liquidity is insufficient for the desired position size, prompting strategic agents to revise the allocation. Exception escalation triggers when lower-level agents encounter problems they cannot resolve, such as unexpected market events or violations of risk limits. Performance feedback provides information about which strategies are working, enabling strategic agents to reallocate resources to successful approaches. Effective feedback requires appropriate aggregation: raw data from execution must be summarized into metrics that inform strategic decisions without overwhelming strategic agents with details Wooldridge (2009) [BG].

Exception Handling. Lower-level agents inevitably encounter situations that exceed their authority or capability, requiring escalation to higher-level decision-makers. Predefined escalation triggers determine when to elevate issues, such as position limits exceeded, unusual market conditions, or conflicting signals. Time-sensitive exceptions (market crashes, circuit breakers) require immediate escalation with minimal delays. Resolution strategies range from fully automated (circuit breakers automatically reduce exposure) to human-in-the-loop (unusual patterns reported to human supervisors). Learning from exceptions improves future handling, with systems developing better triggers and resolution strategies over time. The design of exception handling involves trade-offs between false positives (escalating too often, creating noise) and false negatives (missing critical issues that require intervention) Wooldridge (2009) [BG].

Temporal Abstraction. Different layers of the hierarchy operate at different time scales, enabling separation between long-term planning and short-term reaction. Strategic agents operate at daily, weekly, or monthly frequencies, making decisions about asset allocation, factor tilts, and overall portfolio construction. Tactical agents operate at intraday or minute-level frequencies, managing individual positions, entry and exit timing, and execution quality. This separation prevents short-term noise from disrupting long-term strategy: a temporary price spike should not trigger a reallocation of the entire portfolio. However, temporal abstraction creates challenges for coordination: strategic agents need to set bounds (risk limits, position limits) that constrain tactical agents without being overly restrictive. Multi-scale learning enables agents to learn at different time horizons, with strategic agents learning from long-term patterns while tactical agents adapt to short-term dynamics Sutton et al. (1999) [BG].

11.3. Market Ecology

Market ecology models agents as participants in a market environment, interacting through trading and prices rather than direct communication. In the current literature, this pattern is best read as an exploratory setting for studying collective behavior, not as mature evidence that ecological coordination improves live trading outcomes.

Representative Approaches. Market simulators Hashimoto et al. (2025) populate markets with heterogeneous agents to generate synthetic market data and to stress test strategies under controlled conditions. Related work studies how agent-environment feedback can produce boom-bust dynamics and how repeated interaction may lead to undesirable emergent coordination, motivating regulatory and monitoring considerations. FinEvo Zou et al. (2026) proposes an ecological market-game framing to study the evolution of interacting financial strategies, emphasizing that competitive performance can depend on population composition and adaptive feedback rather than isolated backtests. Validating ecological models requires rigorous calibration against real market microstructure to avoid *simulator overfitting*, and protocols must further audit for *timestamp leakage* and *consensus smoothing* so that emergent coordination is not an artifact of look-ahead bias or noise suppression.

Pioneering work on generative agents Park et al. (2023) [BG] established the viability of LLM-based simulacra, and subsequent

financial-market simulations Hashimoto et al. (2025) use such agents to probe how collective behavior shapes market dynamics in controlled settings.

Market Mechanisms. Market ecology relies on established market structures to mediate agent interactions without requiring direct coordination. Continuous double auctions enable buyers and sellers to trade at any time, with prices emerging from the real-time balance of supply and demand. Call auctions aggregate orders over specific time intervals (open, close, volatility interruptions), determining prices through batch execution that reduces informational asymmetry. Dealer markets designate specific market makers who provide liquidity by continuously quoting bid and ask prices, with agents trading against these quotes rather than directly with each other. Each mechanism creates different incentives and dynamics: continuous double auctions favor fast reactions to information, call auctions reduce timing risk, and dealer markets provide guaranteed liquidity but at a cost (the bid-ask spread). The choice of market mechanism shapes the ecology by determining which strategies are viable and how information propagates through prices Kyle (1985); Glosten and Milgrom (1985) [BG].

Price Formation. In market ecology, no single agent sets prices; instead, prices emerge from the aggregate behavior of all agents. Order aggregation combines individual buy and sell orders into a collective order book, with the intersection of supply and demand determining the market-clearing price. Supply-demand balance shifts as agents react to news, portfolio changes, or risk management needs, causing prices to adjust. Informational efficiency measures how quickly prices incorporate new information, with efficient markets preventing agents from consistently profiting from public information. However, when agents are heterogeneous (different beliefs, time horizons, risk tolerances), prices may deviate from fundamental values for extended periods, creating opportunities for some agents at the expense of others. The dynamics of price formation depend on agent diversity: homogeneous agents tend to produce rapid convergence to consensus prices, while heterogeneous agents create more complex patterns with mispricings that persist until corrected by informed agents Kyle (1985); Glosten and Milgrom (1985) [BG].

Reflexivity and Feedback Loops. Market ecology creates reflexive relationships where agent actions affect prices, which in turn affect agent perceptions and actions. This two-way feedback can produce complex dynamics not present in systems with one-way causality. Positive feedback amplifies initial movements: price increases attract trend-following agents who buy, pushing prices higher and attracting more trend followers, potentially creating bubbles. Negative feedback stabilizes: price increases trigger profit-taking by mean-reversion agents who sell, pushing prices back toward fundamentals. The balance between positive and negative feedback determines market stability. Reflexivity becomes particularly powerful when agents learn and adapt: successful strategies attract imitators, changing market dynamics and potentially rendering the original strategy unprofitable. This co-evolution of agents and markets creates continuously changing patterns, making long-term prediction difficult even as short-term patterns may be exploitable Shiller (1981); Kirilenko et al. (2017) [BG].

Emergent Phenomena. The interaction of heterogeneous agents through market mechanisms can generate collective phenomena that are not apparent from studying individual agents in isolation. Bubbles may occur when positive feedback dominates, with rising prices attracting speculative buying disconnected from fundamentals. Crashes may follow when bubbles burst or when negative feedback triggers panic selling, with prices plummeting as agents rush to exit simultaneously. Flash crashes—extreme short-term collapses in which prices plummet and recover within minutes—are mechanistically plausible when high-frequency agents interact with market structure. Tacit collusion is another plausible risk when agents learn to coordinate behavior implicitly through repeated interaction without explicit communication, raising regulatory concerns about market manipulation. Because these phenomena depend on the collective behavior of many agents and are difficult to predict, agent-based simulation is more appropriately viewed as a hypothesis-generation tool for understanding market dynamics and stress cases Kirilenko et al. (2017); Shiller (1981) [BG].

Strategy Crowding and Alpha Decay. As LLM-based agents proliferate, similar training data and reasoning patterns may lead to *strategy homogenization*. When multiple agents:

- Use similar foundation models (GPT-4, Claude)
- Fine-tune on similar financial corpora
- Employ comparable CoT or RAG patterns

they may generate correlated trading signals, creating mechanistically plausible risks such as:

1. **Alpha decay**: Common signals are arbitraged away faster
2. **Self-reinforcing trends**: Similar entry points amplify momentum
3. **Correlated exits**: Simultaneous stop-loss triggering increases volatility
4. **Flash crash risk**: Rapid unwinding of crowded positions

The evidence base currently lacks studies of multi-agent market impact at scale—a critical gap that requires simulation-based research before deployment.

Regulatory Constraints. Regulators shape market ecology by constraining agent behavior to maintain market integrity and stability. Position limits prevent any single agent from dominating the market or taking excessive risks. Circuit breakers halt trading during extreme volatility, providing time for information to diffuse and potentially preventing panic selling. Market maker obligations require designated liquidity providers to quote prices continuously, ensuring that other agents can always trade. Reporting rules may oblige agents to disclose large positions or trades, reducing information asymmetry. Manipulation rules prohibit agents from engaging in behavior

that artificially affects prices (spoofing, layering, pump-and-dump schemes). The design of these regulations must balance competing goals: too little regulation enables manipulation and instability, while too much regulation reduces liquidity and efficiency. Agent-based simulation enables regulators to test proposed rules before implementation, identifying unintended consequences and optimizing regulatory parameters European Parliament and Council of the European Union (2014, 2016); Article 29 Working Party (2018).

TradingAgents is coded R2, whereas FinCon and TradingGroup are R0; all three report order-level execution semantics. The hierarchical investment pipeline reports target weights, and FinEvo uses market-game execution. These tags record reporting categories rather than equivalent execution realism.

Table 18 compares their communication, authority, and design considerations.

Latency Constraints on Coordination. The coordination mechanisms discussed above must operate within the latency budget established in Section 6. Coordination cost is dominated by communication, consensus formation, and decision aggregation, so voting is typically cheaper than debate while expert-weighted aggregation sits in between. In latency-sensitive regimes, systems may need to prefer the simplest mechanism that preserves acceptable decision quality. This latency-aware coordination design remains an open research direction for financial multi-agent systems.

The representative systems above illustrate several coordination mechanisms. Extension-window examples include selective-consensus trading (Li et al., 2026b), co-evolving MARL agents (Jia et al., 2026), live deliberative analysis (Letteri, 2026), and a belief-to-trade execution layer (Wang et al., 2026e). Alpha-mining pipelines are discussed in Section 7. Background market ecosystems examine self-organization (Hashimoto et al., 2026; Saxena et al., 2026), specialized asset markets (Casas and Montero, 2026; Passos et al., 2026), mechanism design (Sheng and Shen, 2026; Qian, 2026; Varghese et al., 2026), and behavioral dynamics (Ouyang and Sui, 2026; Park, 2026). Language-grounded testbeds (Shi et al., 2026) and prediction-market arenas (Barot and Borkhatariya, 2026) provide further settings for studying interaction. These background works inform coordination design and market behavior; the ledger separates them from evaluated, action-producing trading systems.

Synthesis and Coordination Trade-offs. Role-based collaboration, hierarchy, and market ecology differ in how agents communicate and where coordination occurs. Communication cost, consensus quality, and attribution shape these designs; role ablations and compute-matched single-agent baselines help distinguish coordination effects. The final section examines persistent adaptation and structural self-evolution in Section 12.

12. Adaptation and Self-Evolution

Adaptation is the umbrella term for updates to an agent's state or behavior. Here, self-evolution denotes agent-managed changes retained across decision episodes; structural or architectural self-evolution is the narrower case in which strategy code, update procedures, modules, interfaces, workflows, or system topology change. Memory consolidation changes what experience is retained, while meta-learning changes how learning is performed; either can support persistent adaptation without changing the architecture. Reflection within a single decision episode does not by itself qualify as self-evolution. Building on the stores and retrieval operations in Memory Architecture and the ICL, SFT, and RL update modes in Learning Paradigms, this section examines memory consolidation, meta-learning, and self-reflection in that order: "How are agent updates carried across decision episodes and evaluated?"The conceptual update loop is shown in Figure 15.

Note: Many mechanisms described in this section (e.g., memory consolidation, meta-learning, self-reflection) originate in general AI research; in trading, they remain experimental and should be interpreted through explicit *update governance* rather than as industry-standard practice. Studies marked [BG] are background literature without end-to-end trading evaluation. Adaptation/self-evolution claims are only comparable when the update loop is reproducible (MR-6; see Section 13): papers should report what can change (prompts, memory summaries, retrieval indices, weights), when updates trigger, and what historical window is used. They should preserve immutable logs, frozen execution and cost semantics, and a rollback path, and evaluate the post-update state on sealed data.

12.1. Memory Consolidation

Building on the stores and retrieval operations described in Section 4, memory consolidation changes what those stores retain over time: it compresses experience into reusable patterns while discarding details that no longer aid retrieval.

Representative Approaches. Within the section-level primary subset, memory-consolidation-like behavior appears in systems that retain episodic traces and derived summaries for later retrieval, including FinMem and related retrieval-centered trading agents Yu et al. (2023b); Zhang et al. (2024). Background work considers replay, summarization, abstraction, and forgetting as ways to select or compress those traces. Broader systems such as TradingGPT remain in the wider citation set and do not enter the section-level primary-study count Li et al. (2023).

Technical Mechanisms. Background methods score memories by recency, outcome, or retrieval frequency and use replay, compression, or abstraction to retain useful signals Lin (1992); Schaul et al. (2016); McClelland et al. (1995); Hinton and Salakhutdinov (2006) [BG]. When market conditions change, temporal or contextual rules can resolve conflicting memories McClelland et al. (1995); Kirkpatrick et al. (2017) [BG]. These operations may run periodically, respond to events such as regime shifts, or combine both schedules Kirkpatrick et al. (2017) [BG]; their value depends on whether the resulting state remains temporally valid and auditable.

Audit Trail Requirement: Summarization can hallucinate details not present in the original log or lose the link to the trade decision.

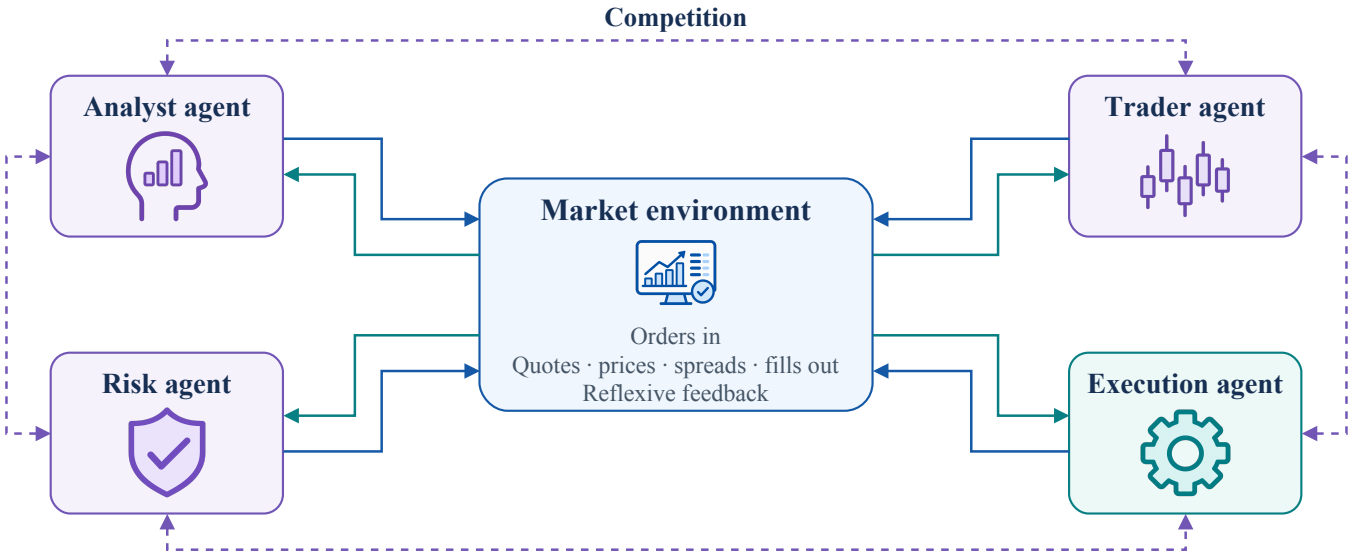

Figure 14: Market ecology interaction network. Agents interact both with one another and through the market environment, where prices, spreads, quotes, and fills feed back into subsequent behavior. The figure highlights reflexive feedback and competition channels that can support exploratory study of crowding, instability, and adaptation.

Table 18: Multi-agent coordination patterns and design considerations. **Tags**: R=Reproducibility level (R0–R3), E=Execution semantics reported. Tags apply to the named trading systems in each row (see Section 2 for full coding rules).

Pattern	Example	Comm.	Decision making	Coordination considerations	Tags
Role-based	TradingAgents Xiao et al. (2024), FINCON Yu et al. (2024), TradingGroup Tian et al. (2025)	Direct, rich communication	Debate; belief updates	Communication and decision aggregation	`R0-R2,E`
Hierarchical	Multi-agent investment pipeline Li (2026)	Vertical, layered feedback	Top-down commands, bottom-up reporting	Layered reporting and decision aggregation	`R0,E`
Market Ecology	Agent simulations; FinEvo Zou et al. (2026)	Indirect, via prices/trades	Decentralized, independent	Simulator calibration and market coupling	`R0,E`

To remain auditable, consolidated memories must retain an immutable pointer (hash or timestamp) to the raw transaction logs. Adaptation protocols should periodically validate that summaries do not contain factual errors or future information compared to the ground-truth logs.

12.2. Meta-Learning

Meta-learning enables agents to learn how to learn, improving the efficiency and effectiveness of the learning process itself. Finance-oriented prototype systems considered here primarily explore multi-agent evolutionary loops to generate and select strategies or factors under backtest feedback Yun et al. (2025); Luo et al. (2025); Zou et al. (2026). The extension window adds a distinct self-evolving cluster—experience-driven policy refinement Kim et al. (2026b), rubric-policy evolution Chen et al. (2026b), agent-to-agent alpha co-evolution Li et al. (2026e), meta-evolution of trading programs Sharma and Shroff (2026); Kvasiuk et al. (2026), and post-trained local research agents Wang et al. (2026d)—alongside regime-adaptive continual learning for portfolio management Pan et al. (2026). These results remain prototype-level and must be interpreted under each study's evaluation and cost/execution assumptions. Evolutionary simulation also provides an existence proof for trend-following behavior Chen (2026a), situating adaptation in a market-ecology context. Across these lines, the central trading challenge is to define tasks without temporal leakage and seal feedback against future outcomes.

Task Splitting Requirement. In trading, a "task" cannot simply be a random sample of assets or days. To prevent leakage, tasks must be split by time (e.g., 2010–2015 as source, 2016 as target)

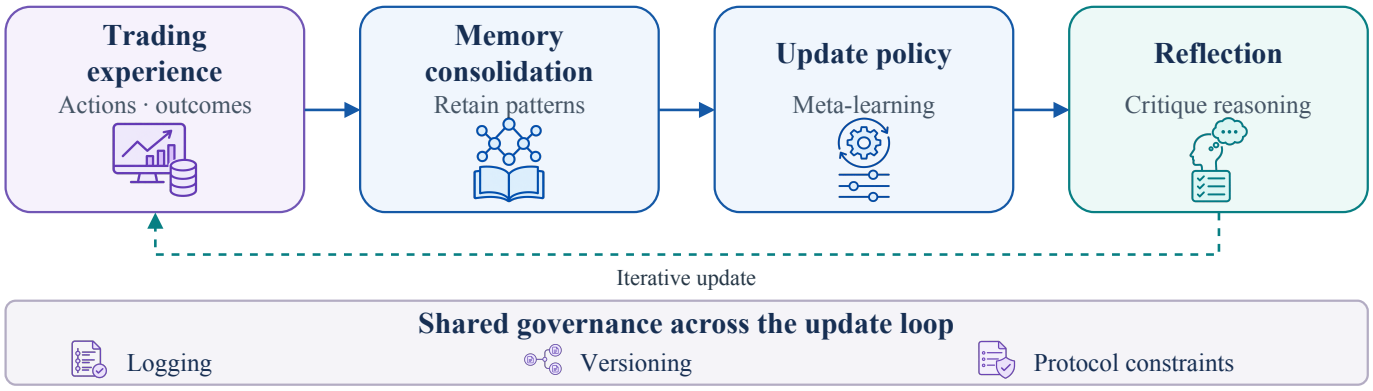

Figure 15: Conceptual governed update loop. Agents accumulate experience through trading, consolidate important patterns into memory, apply meta-learning-like update policies, and use reflection to critique reasoning under logging and rollback constraints.

or by distinct, non-overlapping asset classes. Random shuffling of episodes across time violates the causal structure of markets, creating false "adaptation" that is merely memorization of future regimes.

Representative Approaches. Background MAML learns an initialization that supports rapid adaptation, while learning-to-learn methods can update the learning-rate schedule, exploration strategy, or architecture Finn et al. (2017) [BG]. Finance-oriented prototypes apply iterative critique to factor or policy search: AlphaAgent regularizes factor generation, and TradingGroup combines reflection with data synthesis Tang et al. (2025); Tian et al. (2025). General multi-agent optimization provides a related refinement pattern, but remains background rather than finance-specific evidence Yuksel and Sawaf (2024) [BG].

Technical Mechanisms. Meta-learning evaluates adaptation across a task distribution, using support data for within-task updates and query data to measure post-update performance. MAML formalizes this as inner-loop adaptation and outer-loop optimization; few-shot and domain-generalization methods extend the same idea to low-data or unseen related tasks Finn et al. (2017); Vinyals et al. (2016); Snell et al. (2017); Muandet et al. (2013) [BG]. Memory or replay can support these updates, but concerns how learning uses stored experience rather than how that experience is represented Finn et al. (2017) [BG]. Transfer learning applies learned representations or behavioral strategies to new tasks, while domain adaptation fine-tunes them with target-specific data Pan and Yang (2010) [BG].

12.3. Self-Reflection

Self-reflection enables agents to critique their own reasoning and to catch mistakes before they propagate into actions, thereby improving decision quality.

Representative Approaches.

Reflexion Shinn et al. (2023) [BG] adds an explicit critique–revise loop: after proposing a decision, the agent diagnoses weaknesses in its own reasoning and regenerates an improved plan. In trading settings, such reflection can check for common failure modes (feature/label leakage, stale data, ignored costs, regime mismatch). Self-consistency and retrospective error analysis complement Reflexion by exposing unstable rationales and recurring mistake patterns. In a finance-specific setting, CryptoTrade uses a reflective mechanism to refine zero-shot cryptocurrency trading decisions by analyzing outcomes of prior trades Li et al. (2024c). Within the section-level primary subset, the main trading-facing anchor is TradingGroup, which combines role specialization with self-reflection and data synthesis Tian et al. (2025); other examples illustrate critique and refinement mechanisms.

Technical Mechanisms: Protocol Implication

Effective self-reflection is a *protocol*, not a free-form "thinking harder" step; we suggest five stages. (i) *Critique generation*: produce actionable checks tied to the trade proposal (data timestamps, assumptions, transaction costs, liquidity/slippage, and risk constraints). (ii) *Refinement*: run a small, bounded number of critique–revise iterations with early stopping; log the critique and the resulting change for auditability. (iii) *Consistency/uncertainty*: sample multiple reasoning paths; treat disagreement as a deferral/abstain signal to reduce overconfident errors, but validate thresholds empirically (consensus is not correctness). (iv) *Post-trade review*: categorize failures (prediction vs execution vs risk) and store compact lessons in memory keyed by instrument/regime so they can be retrieved at decision time. (v) *Pre-trade verification*: enforce hard constraints (risk limits, compliance rules, and operational feasibility) as deterministic checks.

Risk of Search Bias: Unbounded reflection acts as a search process. To prevent "testing into success" (retrying until a profitable trade appears by chance), protocols must enforce a **Reflection Budget** (max iterations) and refrain from using Out-of-Sample feedback to terminate the loop. Reflective traces are not a substitute for testing, monitoring, or out-of-sample evaluation Wang et al. (2022); Madaan et al. (2023); Geifman and El-Yaniv (2017) [BG].

Synthesis and Governance Dimensions. The subsections and Table 19 distinguish updates to retained experience, learning pro-

Table 19: Conceptual update mechanisms and uses of experience. All entries are background mechanisms (BG); the descriptions identify update operations and the role of retained experience.

Mechanism	Description	Key Mechanism	Update work	Experience use	Tags
Consolidation	Summarize, abstract, forget	Importance scoring, compression	Memory summarization, abstraction, and forgetting	Selects retained traces for later retrieval	BG
Meta-Learning	Learn to learn	MAML Finn et al. (2017), memory-based opt	Inner-loop adaptation and outer-loop optimization	Support/query data across tasks	BG
Self-Reflection	Critique and refine	Reflexion Shinn et al. (2023), self-consistency	Critique and revision of a proposed plan	Critique informs a regenerated plan	BG

cedures, and reasoning outputs; persistent changes to strategy code, update procedures, or system organization are structural self-evolution. Whether any update carries across later trading decisions is an empirical question under the governance conditions above. These distinctions frame the cross-cutting challenges and future directions in Section 13.

13. Technical Challenges, Methodological Guidelines, and Open Problems

The preceding sections have systematically synthesized the architectural foundations, functional capabilities, and adaptation pathways of agentic trading systems. This section examines the principal technical and practical bottlenecks that arise when deploying LLM agents in financial markets, consolidates methodological best-practice guidelines (MR-1–MR-7) to foster reliable empirical evaluation, and outlines key open research problems.

As observed across the empirical literature ($n = 80$), researchers adopt varying conventions regarding data splits, transaction frictions, asset universe definition, and execution timing. To promote cross-study comparability, mitigate data snooping, and facilitate real-world translation, we distill these recurring empirical considerations into a structured set of methodological recommendations:

Methodological Guidelines and Recommended Evaluation Practices (MR-1–MR-7)

To support rigorous benchmarking and reproducibility, future research on agentic trading is encouraged to document the following empirical dimensions:

- **MR-1 (Data & Universe)**: Specify asset universe construction criteria, survivorship bias treatment, and synchronized data availability timestamps across all perceived modalities.
- **MR-2 (Temporal Splitting)**: Enforce strict chronologically ordered train/validation/test partitions, walk-forward or rolling windows, and explicit embargo periods to prevent look-ahead bias; for iterative alpha search, maintain a frozen holdout test set.
- **MR-3 (Action Semantics)**: Explicitly define output format (discrete orders vs. continuous portfolio target weights), decision timestamps, order mapping rules, and portfolio constraints (leverage, gross/net exposure).

Methodological Guidelines (continued)

- **MR-4 (Execution & Costs)**: Explicitly state execution timing, fill price assumptions (e.g., next-open or volume-weighted prices), rebalancing intervals, and realistic friction models (brokerage fees, bid-ask spread, market impact).
- **MR-5 (Leakage Controls)**: Verify causal temporal integrity across feature normalization, text sentiment alignment, and target label construction.
- **MR-6 (Artifacts & Verification)**: Provide open-source code, reproducible configuration scripts, random seeds, and structured decision logs; for adaptive systems, record update frequencies and rollback criteria.
- **MR-7 (Multi-Agent Evaluation)**: In collaborative or competitive systems, document agent role topologies, communication protocols, and ablation experiments isolating the marginal contribution of inter-agent interactions.

Table 20 relates the guidelines to each study's task. For **end-to-end trading**, universe construction and cost models help interpret profitability; an **execution algorithm** receiving parent orders may take the universe as given. A standalone **component study**, such as sentiment analysis, needs time-consistent evaluation but may operate without a transaction cost model. The matrix distinguishes relevant evaluation dimensions from fields outside the study's scope.

13.1. Technical Challenges

The subsections below synthesize technical challenges, methodological guidance, and representative research directions.

13.1.1. Hallucination in Finance

LLM hallucination, which generates plausible but false information, poses *material* risk in finance because trading decisions may be executed, and losses realized, before the underlying claim can be verified. Only one primary study explicitly foregrounds this issue, so the discussion below combines that direct evidence with cross-section risk synthesis.

Propagation Mechanisms. In agentic workflows, hallucinated claims propagate through tool calls into downstream actions. A fabricated earnings report can trigger:

1. Erroneous position entry (immediate loss when corrected)
2. Stop-loss cascade (if other agents detect the "signal")

Table 20: Applicability of the MR-1–MR-7 evaluation guidelines by study type. **E**: Essential to interpreting the relevant result; **R**: Recommended for verification; **O**: Task-dependent; **–**: Generally outside the stated task. These designations summarize this survey's methodological recommendations.

MR Item	End-to-End *(Full Trading)*	Execution *(Ord/Wt Algo)*	Component *(Perception/Risk)*
MR-1 (Data & Universe)	**E**	O	**E**
MR-2 (Time Split & Leakage)	**E**	**E**	**E**
MR-3 (Action Semantics)	**E**	**E**	–
MR-4 (Exec & Costs)	**E**	**E**	–
MR-5 (Leakage Audit)	**E**	**R**	**R**
MR-6 (Artifacts)	**R**	**R**	**R**
MR-7 (Multi-Agent Eval)	O	–	–

Reading the matrix: Applicability follows the study's task and reported result. Action semantics and cost assumptions are essential to interpreting trading or execution outcomes, while a standalone perception study may be evaluated without generating orders. Universe selection can be exogenous to an execution algorithm. Artifacts and extraction checks support verification, and multi-agent evaluation applies when interaction is part of the design. The matrix guides interpretation; corpus-level reporting counts follow the separate coding rules in Section 2.

3. Confidence-based position scaling (larger losses due to false certainty)

Mitigation Limitations. Binding outputs to verifiable sources (FinMCP, HydraRAG) shifts rather than eliminates risk, because source selection itself can be erroneous. Cross-validation helps only where multiple independent sources exist, which is often not the case for time-sensitive trading signals; moreover, current fact-checking mechanisms operate at second-scale latency, which is incompatible with millisecond trading decisions. We therefore cite these systems as illustrative tooling, not as empirically validated mitigations.

Regulatory Implications. Firms deploying LLM agents may face heightened liability if hallucinated outputs cause client losses; documentation of source binding, validation checks, and human review may be needed to withstand regulatory scrutiny.

13.1.2. Look-ahead Bias

We treat look-ahead bias here as methodological guidance tied directly to MR-2 and MR-5. Recent work includes dedicated analyses of temporal leakage in LLM backtesting Zhang and Stadie (2026), look-ahead-freedom as a verifiable correctness property Fonseca (2026), deployment-evidence limits of reported alpha Ye et al. (2026c), and execution-assumption audits Yao et al. (2026), alongside leakage-masking benchmarks Zhu et al. (2026b); Qu and Chen (2026); Cai et al. (2026). Methodological work goes further by engineering the problem away: anonymization-first evaluation asks whether "blindfolded" LLMs can still trade when asset identities and dates are withheld Jeon and Lee (2026), directly operationalizing look-ahead-freedom as a test condition.

Look-ahead bias occurs when models inadvertently access future information during training or evaluation, producing unrealistic performance estimates that fail to generalize to real trading conditions. The risk is particularly insidious in time-series settings, where leakage can enter through improper cross-validation that shuffles temporal data, technical indicators computed with future information (e.g., centered moving averages), or textual sources that contain hindsight about market movements. Detection requires careful examination of model inputs and training procedures, including causal inference tests that check whether model performance degrades when strict temporal constraints are enforced: a model that appears strong under naive random splits can degrade sharply under proper time-consistent evaluation, revealing substantial look-ahead contamination López de Prado (2018). Prevention requires strict temporal splitting (train on past, test on future), walk-forward validation that simulates real-time deployment, and causal modeling so that information flows only from past to future. Without these controls, even the most sophisticated learning and adaptation procedures fail under evaluation leakage.

13.1.3. Latency vs Accuracy Tradeoff

Latency is a design constraint that links reasoning depth, model inference, communication, and execution timing. System 2 reasoning (reflective, multi-step) yields better decisions but is too slow for high-frequency trading, where decisions must be made in microseconds, while System 1 reasoning (reactive, fast) lacks the depth for complex decisions involving multi-hop reasoning or extensive analysis. In latency-sensitive regimes, the throughput advantage of faster decision policies can therefore outweigh modest accuracy gains from slower deliberation, especially when opportunities are fleeting and execution queues are competitive. Hybrid approaches address this tension with cascaded designs: fast System 1 handles routine decisions using Section 3 perceptual features and escalates difficult cases to slower System 2 reasoning that leverages Section 5 deliberative mechanisms. Model distillation offers a complementary route, compressing large reasoning models into smaller, faster models that retain much of their accuracy. Future work should develop tradeoff mechanisms that allocate computational budget dynamically according to decision importance, market volatility, and time pressure, potentially learned via reinforcement learning.

13.1.4. Explainability vs Performance

The explainability–performance tension is a cross-section governance concern. Representative approaches include debate-based explainable trading Huang et al. (2026b), post-hoc explanation

of learned portfolio policies de la Rica Escudero et al. (2025), and audits of asset-specific preferences embedded in financial LLMs Wu (2026). The discussion below accordingly provides background and author guidance linked to deployment governance.

More complex models (deep neural networks, ensemble methods, large language models) often achieve better performance but are less interpretable, creating a fundamental tension between explainability and predictive power. This tension has practical as well as academic weight: depending on jurisdiction and system role, regulatory and legal frameworks (e.g., the EU's MiFID II) may impose governance, record-keeping, and explainability expectations for algorithmic trading and decision support European Parliament and Council of the European Union (2014), and GDPR provisions with subsequent guidance can imply obligations to provide "meaningful information" about automated processing in certain contexts, though scope and applicability vary European Parliament and Council of the European Union (2016); Article 29 Working Party (2018). When an agent's alpha generation module (Section 7) triggers a large position, risk managers and regulators need to know why—whether the driver was a particular news event, a technical-indicator pattern, or a reasoning chain. Neuro-symbolic approaches pair neural components for pattern recognition with symbolic components for transparent reasoning, aiming to preserve both performance and interpretability. Attention visualization highlights which inputs most influenced the model (e.g., specific news articles or price movements), while counterfactual explanations specify what would have had to change to alter a decision (e.g., "the trade would not have occurred if the FOMC minutes had been less hawkish"). Regulatory and governance expectations around explainability are likely to motivate further advances in this area, particularly for the risk-management and compliance applications of Section 9.

Many academic systems demonstrate explanations over offline datasets or controlled prototypes; production deployments typically require additional engineering for auditability (decision logs, data lineage, and controlled model/prompt changes) to make explanations actionable for governance.

13.1.5. Computational Cost and Economic Feasibility

The economic viability of LLM-based trading agents depends on the relationship between inference costs and generated returns. A simple annualized estimate is

$$\text{Annual API Cost} = D \times N \times C_{req},$$

where D denotes trading days per year, N the number of decisions per day, and C_{req} the effective per-request cost implied by model pricing, prompt/completion token budgets, and any tool-specific charges. Because vendor pricing and model offerings change rapidly, empirical cost estimates should specify the model version, pricing snapshot date, token assumptions, tool usage, and the break-even gross P&L required for net profitability after inference and execution costs. Recent work directly examines whether agentic trading systems can cover their inference costs once those costs are charged against returns Duan et al. (2026).

13.1.6. Evaluation Protocols and Deployment Safeguards

This subsection translates the survey-wide minimum reporting framework (MR-1–MR-7) into concrete deployment safeguards, drawing on the recurring reporting gaps identified earlier. Evidence-backed guidelines and common anti-patterns for agentic trading research and deployment include:

- **Guard against temporal leakage**: use strictly time-ordered splits and walk-forward evaluation, and ensure feature engineering and scaling are fit only on past data Albelali and Ahmed (2025).
- **Avoid unrealistic cost models**: include fees, bid–ask spread, slippage, and market impact, and report sensitivity to cost assumptions Perold (1988); Olby et al. (2025).
- **Check universe construction**: document inclusion and exclusion rules and timestamp availability, and audit for survivorship/selection bias when building backtest datasets Brown et al. (1992).
- **Mitigate backtest overfitting**: separate model selection from final evaluation, use robust out-of-sample procedures, and be explicit about multiple-testing risk Arian et al. (2024).
- **Add deployment safeguards**: maintain immutable action logs that link each action to its inputs, evidence, and model versions, and use verifiable retrieval, citation checks, and cross-source consistency tests Khatchadourian (2026); Tan et al. (2025b).

The extension cohort supplies early empirical anchors for this guidance: backtest-to-live gaps measured directly in crypto markets Yu et al. (2026b), a live hybrid trading agent evaluated under a shared task protocol Rashid et al. (2026), prospective process-aware evaluation of adaptive agents Luo et al. (2026b), and a unified platform for tracking agentic trading performance over time Li et al. (2026a).

13.2. Human-in-the-Loop Oversight

Although this survey focuses on agentic systems, many realistic deployments retain human oversight at escalation points such as large notional changes, regime breaks, or tool failures. We therefore view human-in-the-loop design as a practical control boundary: stronger autonomy should imply stronger logging, clearer intervention rules, and explicit specification of when human approval is mandatory.

13.3. Negative Results and Failed Designs

The literature is likely affected by survivorship bias at the survey level as well as at the backtest level. Few papers report failed prompts, abandoned reward schemes, degraded live performance, or architectures that increased turnover without improving net returns. We therefore interpret the absence of negative results as a reporting limitation rather than as evidence that the design space is uniformly promising.

13.4. Future Directions

13.4.1. Neuro-Symbolic Integration

Current LLM-based agents rely almost entirely on neural computation, which is flexible but opaque and unreliable where exact calculation or logical guarantees are required. As discussed in Section 5, purely neural approaches struggle with tasks such as portfolio optimization under constraints or exact arithmetic. Neuro-symbolic integration addresses this by combining neural networks (for pattern recognition and learning from data) with symbolic reasoning (for logic, constraint satisfaction, and exact computation): hybrid architectures can use neural networks to extract structured information from text (prices, dates, entities) and symbolic reasoning to perform financial calculations, enforce budget constraints, and maintain logical consistency. Specific approaches include (1) **neural theorem provers** that use neural networks to guide proof search for financial constraints; (2) **differentiable logic** that integrates logical constraints into neural network loss functions; (3) **program synthesis**, in which LLM-generated code is formally verified; and (4) **neuro-symbolic ILP** (Inductive Logic Programming), which learns logical rules from neural representations. The research objective is to combine neural flexibility with symbolic precision while keeping the resulting systems auditable enough for financial use.

13.4.2. Longer-Term Directions: Coordination, Governance, and Infrastructure

Three longer-term directions extend these expert-system design considerations. First, larger multi-agent market simulations could serve as controlled "wind tunnels" for studying crowding, flash episodes, and systemic-risk propagation. The current evidence base is concentrated in small role-based teams, motivating evaluation of larger agent populations. Second, increasing agent autonomy motivates governance research on auditable records, intervention rules, and sandboxed evaluation that clarify who approved, monitored, or overrode each action. Third, low-latency infrastructure and model compression support deployment; evaluating their benefits requires end-to-end decision latency, execution timing, and cost/benefit analysis.

Table 21 connects the survey's design themes with representative studies and the evaluation dimensions needed to interpret them.

Across the empirical literature, rigorous evaluation methods—such as sealed out-of-sample testing, realistic market impact modeling, and latency-aware execution—are increasingly recognized as foundational pillars for advancing the field. Combining architectural innovation with disciplined empirical validation will be paramount as autonomous agent systems transition from experimental prototypes toward resilient, production-grade market deployment.

14. Conclusion

14.1. Synthesis of the Agentic Trading Landscape

The integration of Large Language Models into quantitative finance represents a fundamental paradigm shift from static, black-box forecasting models toward autonomous, cognitive trading agents. By unifying multimodal perception, structured memory retrieval, multi-step cognitive reasoning, and friction-aware order execution, agentic systems introduce an adaptive decision loop capable of operating in complex, non-stationary market environments.

This survey has provided a comprehensive, structured synthesis of this rapidly expanding domain. Through the **Architecture–Capability–Adaptation (A-C-A)** conceptual framework, we have systematically decoupled:

- **Architecture (Part I)**: The foundational cognitive and operational building blocks—perception across text, time-series, and charts; working memory, associative vector storage, and structured knowledge bases; reactive, reflective, and strategic reasoning paradigms; and order-routing action interfaces.
- **Capability (Part II)**: The primary quantitative functions realized by autonomous agents—symbolic and code-based alpha discovery, multi-asset portfolio optimization, and multi-tier pre-trade, in-trade, and post-trade risk governance.
- **Adaptation (Part III)**: Learning and cross-decision update mechanisms—in-context learning, persistent memory and reflection that can update decisions without changing the architecture, supervised fine-tuning, reinforcement learning, multi-agent coordination and adaptive interactions, and structural self-evolution.

Drawing on **185** systematically mapped studies from an initial multi-source ledger and an arXiv extension through 2026-09-30, our synthesis connects the core cohort of $n = 80$ closed-loop trading systems with background work on financial models, tools, and benchmarks.

14.2. Market Microstructure and Practical Implications

The proliferation of autonomous trading agents carries profound implications for both market dynamics and quantitative practice:

- **Market Dynamics and Ecology**: As autonomous agents interact at scale, their collective behaviors will increasingly shape market liquidity, price discovery, and volatility. Investigating emergent phenomena—such as strategic crowding, herding around shared LLM foundations, and algorithmic feedback loops—presents a fertile ground for synthetic market simulation.
- **Quantitative Engineering**: In practical asset management, agentic architectures provide a principled mechanism to bridge qualitative domain insights (earnings calls, regulatory filings) with rigorous quantitative optimizers, transforming quantitative workflow from manual factor engineering to autonomous research pipelines.

14.3. Empirical Evaluation and Methodological Practices

Empirical evaluation is an important dimension of the architectural landscape. Across the pooled 80-study empirical subset,

Table 21: Synthesis of recurring design themes and their evaluation contexts. Representative studies illustrate the mechanisms discussed in the technical chapters; the final column identifies dimensions for interpreting their implementations and reported results.

Design theme	Representative studies	Interpretive dimensions
Closed-loop pipelines and evaluation settings	FinAgent Zhang et al. (2024), TradingAgents Xiao et al. (2024), FinCon Yu et al. (2024), FinRL-DeepSeek Benhenda (2025), Agent Market Arena Qian et al. (2025), FinMem Yu et al. (2023b); evaluation examples include OpenPM Cai et al. (2026), CLQT Qu and Chen (2026), and LiveOption Luo et al. (2026a)	Connect observations, retained state, decisions, and market feedback; distinguish backtest, simulation, live, and benchmark settings.
Action and execution interfaces	FinAgent Zhang et al. (2024), TradingAgents Xiao et al. (2024), AI-Trader Fan et al. (2025), FinRL-DeepSeek Benhenda (2025)	Distinguish target weights from orders, and interpret outcomes using decision timing, fill rules, and cost assumptions.
Memory and adaptation mechanisms	FinAgent Zhang et al. (2024), FinMem Yu et al. (2023b), TradingGroup Tian et al. (2025), Alpha^2 Xu et al. (2024a)	Separate storage and retrieval from prompt, parameter, and strategy updates; identify which changes persist across decisions.
Multi-agent organization	TradingAgents Xiao et al. (2024), TradingGroup Tian et al. (2025), FinCon Yu et al. (2024), REIT multi-agent pipeline Li (2026)	Specify roles, communication, and decision authority; role ablations and matched resource budgets help assess coordination effects.
Evaluation practice and reproducibility	Primary-subset protocol tables in Section 2 and R0–R3 artifact descriptions	Interpret field coverage within the assembled cohorts, accounting for retrieval and coding differences; distinguish artifact availability from trading performance.

reported protocol-field coverage includes time-consistent split discipline (62.5%), explicit transaction-cost modeling (58.8%), universe survivorship controls (60.0%), and execution semantics (58.8%). The cohort breakdown in Table 4 gives the corresponding counts under the different retrieval and coding procedures. These pooled values summarize the assembled literature; they do not establish a temporal trend in reporting practices.

The methodological guidelines (MR-1–MR-7) organize practices that support interpretable comparisons and further validation. Walk-forward evaluation, realistic execution frictions, and transparent artifacts help assess generalization, while transfer to live markets also requires prospective evaluation under the relevant market and operational conditions.

14.4. Concluding Outlook

Looking forward, the frontier of agentic trading lies at the intersection of deep neural perception, symbolic reasoning, and institutional-grade governance. Promising avenues include architectures that combine neural hypothesis generation with symbolic reasoning and constraint satisfaction, large-scale agent-based market sandboxes for systemic risk testing, and verifiable execution contracts with full cryptographic auditability.

By providing a clear taxonomy, decomposing the end-to-end decision pipeline, and highlighting core methodological principles, this survey aims to serve as an enduring conceptual roadmap for researchers, quantitative practitioners, and system architects shaping the future of autonomous financial intelligence.

14.5. Supplementary Materials and Reproducible Resources

The survey's conceptual, empirical, and methodological contributions are tied to materials accompanying the manuscript. Parts I–III present the A-C-A synthesis, with illustrative system mappings in Appendix C; the empirical protocol summary is reported in Supplementary S2; and the methodological recommendations are stated as MR-1–MR-7 in Section 13. The supporting review materials are organized as follows:

- **Supplementary S1**: Search and screening records, including the current S0 v2 candidate registry with per-record decisions.
- **Supplementary S2**: The current n=80 protocol registry and coverage summary, alongside a preserved initial-window v1 evidence-mapping snapshot.
- **Supplementary S3**: Coding schema and extraction-audit materials, including a targeted 15-record sample from the legacy initial-window 41-record primary-role frame and a record of current amendments to historical codes.

Current per-study codes are recorded in S0 v2, with amendments documented separately from historical coding snapshots. These materials make the selection, protocol coding, and version changes traceable; repository availability is described in the Data availability statement.

CRediT authorship contribution statement

Yihan Xia: Methodology, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. Panpan You: Investigation, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. Taotao Wang: Conceptualization, Methodology, Supervision, Project administration, Resources, Validation, Writing – review & editing. Fang Liu: Conceptualization, Validation, Writing – review & editing. Han Qi: Conceptualization, Validation, Resources, Writing – review & editing. Xiaoxiao Wu: Conceptualization, Validation, Writing – review & editing.

Shengli Zhang: Conceptualization, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The registry, protocol tables, and coding materials accompany the manuscript as Supplementary Materials S1–S3. S1 contains the combined S0 v2 registry and initial-window search/screening records. S2 provides the current empirical protocol table and coverage summary alongside the historical mapping snapshot. S3 contains the coding schema, historical annotations, current amendments, and targeted extraction audit. An anonymized repository containing the same materials and the LaTeX source will be released upon acceptance, subject to journal policy and anonymization requirements.

References

Abbade, L.R., Costa, A.H.R., 2026. Realistic market impact modeling for reinforcement learning trading environments. arXiv:2603.29086.

Abensur, E., 2024. Machine learning for liquidity classification and its applications to portfolio selection. Brazilian Review of Finance / Revista Brasileira de Finanças 22, 1–14. URL: https://periodicos.fgv.br/rbfin/article/download/90713/85865/206675, doi:10.12660/rbfin.v22n2.2024.90713.

Albelali, S., Ahmed, M., 2025. Hidden leaks in time series forecasting: How data leakage affects LSTM evaluation across configurations and validation strategies. arXiv preprint arXiv:2512.06932 URL: https://arxiv.org/abs/2512.06932, doi:10.48550/arXiv.2512.06932.

Aldridge, I., An, J., Burke, R., Cao, M., 2026. Agentic artificial intelligence in finance: A comprehensive survey. arXiv:2604.21672.

Alexandrou, G., Pramatari, K., 2026. A multi-agent orchestration framework for venture capital due diligence. arXiv:2605.13110.

Alkhudaydi, M.H., 2025. An ode-based dynamic mean-variance portfolio optimization. Computational Economics URL: https://link.springer.com/article/10.1007/s10614-025-11152-3.

Almgren, R., Chriss, N., 2001. Optimal execution of portfolio transactions. The Journal of Risk 3, 5–39. URL: https://doi.org/10.21314/jor.2001.041, doi:10.21314/jor.2001.041.

Almon, B., Caliboso, M., Kim, A., Dutta, R., Raman, R., Srungarapu, M., Sharma, V., Zhu, K., 2025. RAPTOR: Reasoned agentic portfolio trading with orchestrated rebalancing, in: NeurIPS 2025 Workshop (NORA). URL: https://openreview.net/forum?id=ziuTkKhgT0.

Araci, D.T., 2019. Finbert: Financial sentiment analysis with pre-trained language models. arXiv preprint arXiv:1908.10063 URL: https://arxiv.org/abs/1908.10063, doi:10.48550/arXiv.1908.10063.

Arian, H., Norouzi Mobarekeh, D., Seco, L., 2024. Backtest overfitting in the machine learning era: A comparison of out-of-sample testing methods in a synthetic controlled environment. Knowledge-Based Systems 305, 112477. URL: https://www.sciencedirect.com/science/article/abs/pii/S0950705124011110, doi:10.1016/j.knosys.2024.112477.

Arora, A., Malpani, R., 2026. Predictionmarketbench: A swe-bench-style framework for backtesting trading agents on prediction markets. arXiv preprint arXiv:2602.00133 URL: https://arxiv.org/abs/2602.00133, doi:10.48550/arXiv.2602.00133.

Article 29 Working Party, 2018. Guidelines on Automated individual decision-making and Profiling for the purposes of Regulation 2016/679. Technical Report. European Data Protection Board (endorsed). URL: https://ec.europa.eu/newsroom/article29/items/612053. wP251 rev.01.

Asthana, S., Zhang, B., DeLuca, C., Mahindru, R., Patel, H., 2025. Stride: A systematic framework for selecting ai modalities – agentic ai, ai assistants, or llm calls. arXiv preprint arXiv:2512.02228 URL: https://arxiv.org/abs/2512.02228, doi:10.48550/arXiv.2512.02228.

Atkinson, R.C., Shiffrin, R.M., 1968. Human memory: A proposed system and its control processes, in: Spence, K.W., Spence, J.T. (Eds.), The Psychology of Learning and Motivation. Academic Press. volume 2, pp. 89–195. URL: https://psycnet.apa.org/record/1974-02719-003, doi:10.1016/S0079-7421(08)60422-3.

Baddeley, A.D., Hitch, G., 1974. Working memory. Psychology of Learning and Motivation 8, 47–89. doi:10.1016/S0079-7421(08)60452-1.

Bai, J., Perron, P., 2003. Computation and analysis of multiple structural change models. Journal of Applied Econometrics 18, 1–22. URL: https://doi.org/10.1002/jae.659, doi:10.1002/jae.659.

Bai, S., Kolter, J.Z., Koltun, V., 2018. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. arXiv preprint arXiv:1803.01271 URL: https://arxiv.org/abs/1803.01271.

Bailey, D.H., Borwein, J.M., Lopez de Prado, M., Zhu, Q.J., 2014. Pseudo-mathematics and financial charlatanism: The effects of backtest overfitting. SSRN Electronic Journal URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2326253, doi:10.2139/ssrn.2326253.

Baker, S.R., Bloom, N., Davis, S.J., 2016. Measuring economic policy uncertainty. The Quarterly Journal of Economics 131, 1593–1636. URL: https://doi.org/10.1093/qje/qjw024, doi:10.1093/qje/qjw024.

Barot, R.M., Borkhatariya, A.S., 2026. Polyswarm: A multi-agent large language model framework for prediction market trading and latency arbitrage. arXiv:2604.03888.

Basel Committee on Banking Supervision, 2009. Principles for Sound Stress Testing Practices and Supervision. Technical Report. Bank for International Settlements. URL: https://www.bis.org/publ/bcbs155.htm.

Benhenda, M., 2025. Finrl-deepseek: Llm-infused risk-sensitive reinforcement learning for trading agents. arXiv preprint arXiv:2502.07393 URL: https://arxiv.org/abs/2502.07393, doi:10.48550/arXiv.2502.07393.

Bertsimas, D., Lo, A.W., 1998. Optimal control of execution costs. Journal of Financial Markets 1, 1–50. URL: https://doi.org/10.1016/S1386-4181(97)00012-8, doi:10.1016/S1386-4181(97)00012-8.

Black, F., Litterman, R., 1992. Global portfolio optimization. Financial Analysts Journal 48, 28–43. URL: https://rpc.cfainstitute.org/en/research/financial-analysts-journal/1992/global-portfolio-optimization, doi:10.2469/faj.v48.n5.28.

Bohorquez, G., Cartlidge, J., 2024. Simulation of social media-driven bubble formation in financial markets using an agent-based model with hierarchical influence network. arXiv preprint arXiv:2409.00742v1 URL: https://arxiv.org/pdf/2409.00742v1.

Bongiorno, C., Manolakis, E., Mantegna, R.N., 2025. End-to-end large portfolio optimization for variance minimization with neural networks through covariance cleaning. URL: https://arxiv.org/abs/2507.01918, doi:10.48550/arXiv.2507.01918, arXiv:2507.01918.

Borjigin, A., Stadnyk, I., Bilski, B., Chikita, M., 2026. Absorbing complexity: An interaction-native knowledge harness for financial llm agents. arXiv:2606.01886.

Boyd, S., Johansson, R., Kahn, R.N., Schiele, P., Schmelzer, T., 2024. Markowitz portfolio construction at seventy. URL: https://arxiv.org/pdf/2401.05080, arXiv:2401.05080.

Briola, A., Bartolucci, S., Aste, T., 2024. Deep limit order book forecasting: A microstructural guide. arXiv preprint arXiv:2403.09267 URL: https://arxiv.org/abs/2403.09267, doi:10.48550/arXiv.2403.09267.

Brown, S.J., Goetzmann, W.N., Ibbotson, R.G., Ross, S.A., 1992. Survivorship bias in performance studies. The Review of Financial Studies 5, 553–580. doi:10.1093/rfs/5.4.553.

Brown, T.B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J.D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D.M., Wu, J., Winter, C., Hesse, C., Chen, M., Sigler, E., Litwin, M., Gray, S., Chess, B., Clark, J., Berner, C., McCandlish, S., Radford, A., Sutskever, I., Amodei, D., 2020. Language models are few-shot learners, in: Advances in Neural Information Processing Systems. URL: https://arxiv.org/abs/2005.14165.

Budish, E., Cramton, P., Shim, J., 2015. The high-frequency trading arms race: Frequent batch auctions as a market design response. The Quarterly Journal of Economics 130, 1547–1621. URL: https://doi.org/10.1093/qje/qjv027, doi:10.1093/qje/qjv027.

Byun, W., 2023. Practical application of deep reinforcement learning to optimal trade execution. FinTech 2, 414–429. URL: https://www.mdpi.com/2674-1032/2/3/23, doi:10.3390/fintech2030023.

Cai, X., Guo, M., Liu, J., Han, J., 2026. Openpm: Auditable point-in-time evaluation for llm portfolio-management agents. arXiv:2608.09988.

Caner, M., Capponi, A., Sun, N., Tan, J.Y., 2026. Designing agentic ai-based screening for portfolio investment. arXiv:2603.23300.

Caner, M., Fan, Q., 2024. Portfolio analysis in high dimensions with tracking error and weight constraints. URL: https://arxiv.org/abs/2402.17523, arXiv:2402.17523.

Cao, Y., Zhang, H., Keung, J.W., Chen, Y., 2026. Systradebench: An iterative build-test-patch benchmark for strategy-to-code trading systems with drift-aware diagnostics. arXiv:2604.04812.

Caruana, R., 1997. Multitask learning. Machine Learning 28, 41–75. doi:10.1023/A:1007379606734.

Casas, P.F.i., Montero, A.M., 2026. Conceptualization and experimentation of asset market with price manipulation. arXiv:2609.13304.

Castelli, A., Giudici, P., Piergallini, A., 2025. Building crypto portfolios with agentic ai. arXiv preprint arXiv:2507.20468v1 URL: https://arxiv.org/pdf/2507.20468v1.

Chai, J., Zheng, B., Liu, X., Lu, H., 2026. Agentic time machine as an infrastructure for future-event forecasting. arXiv:2606.21013.

Chang, S., Zhu, P., Chen, Y., 2026. Valueblindbench: Agreement-gated stress testing of llm-judged investment rationales before returns are observable. arXiv:2604.25224.

Chen, D., Zhu, Y., Liu, M., Li, J., 2022. Cost-efficient reinforcement learning for optimal trade execution on dynamic market environment, in: Proceedings of the ACM International Conference on AI in Finance (ICAIF). URL: https://dl.acm.org/doi/10.1145/3533271.3561761, doi:10.1145/3533271.3561761.

Chen, J., Cong, S.L., 2025. Agentguard: Repurposing agentic orchestrator for safety evaluation of tool orchestration. arXiv preprint arXiv:2502.09809v1 URL: https://arxiv.org/pdf/2502.09809v1.

Chen, J., Feng, A., Verma, H., Gu, S., 2026a. Reasoning through verifiable forecast actions: Consistency-grounded rl for financial llms. arXiv:2605.21975.

Chen, X., Zhu, W., Zheng, S., Rasul, K., 2026b. Sharp: A self-evolving human-auditable rubric policy for financial trading agents. arXiv:2605.06822.

Chen, Y., 2026a. Be water: An evolutionary proof for trend-following. arXiv:2603.29593.

Chen, Y., Yao, Z., Liu, Y., Xin, A., Ye, J., Yu, J., Hou, L., Li, J., 2025. Stockbench: Can LLM agents trade stocks profitably in real-world markets? arXiv preprint arXiv:2510.02209 URL: https://arxiv.org/abs/2510.02209, doi:10.48550/arXiv.2510.02209.

Chen, Z., 2026b. Decomposing financial market dynamics via mechanism analysis in an evolutionary multi-agent simulation. arXiv:2606.23158.

Cheng, L., Cheng, X., Liu, S., Oosterlee, C.W., 2026a. Ridge: An autonomous framework for validation and method discovery in llm-generated option pricing. arXiv:2607.25199.

Cheng, Z., Lai, L., Liu, Y., 2026b. Sustainable hybrid document-routed retrieval for financial rag: Resolving the robustness-precision trade-off. arXiv:2603.26815.

Cheridito, P., Weiss, M., 2025. Reinforcement learning for trade execution with market and limit orders. arXiv preprint arXiv:2507.06345 URL: https://arxiv.org/abs/2507.06345, doi:10.48550/arXiv.2507.06345.

Constantinides, G.M., 1986. Capital market equilibrium with transaction costs. Journal of Political Economy 94, 842–862. URL: https://doi.org/10.1086/261412, doi:10.1086/261412.

Cook, T., Osuagwu, R., Tsatiashvili, L., Vrynsia, V., Ghosal, K., Masoud, M., Mattivi, R., 2025. Retrieval augmented generation (rag) for fintech: Agentic design and evaluation. arXiv preprint arXiv:2510.25518 URL: https://arxiv.org/abs/2510.25518, doi:10.48550/arXiv.2510.25518.

Coriat, B., Benhamou, E., 2025. Harlf: Hierarchical reinforcement learning and lightweight llm-driven sentiment integration for financial portfolio optimization. URL: https://arxiv.org/abs/2507.18560, arXiv:2507.18560.

Cosler, M., Hahn, C., Mendoza, D., Schmitt, F., Trippel, C., 2023. nl2spec: Interactively translating unstructured natural language to temporal logics with large language models, in: Computer Aided Verification (CAV 2023), pp. 383–396. URL: https://arxiv.org/abs/2303.04864, doi:10.1007/978-3-031-37703-7_18.

Davis, M.H.A., Norman, A.R., 1990. Portfolio selection with transaction costs. Mathematics of Operations Research 15, 676–713. URL: https://doi.org/10.1287/moor.15.4.676, doi:10.1287/moor.15.4.676.

Deng, K., 2025. Autoquant: An auditable expert-system framework for execution-constrained auto-tuning in cryptocurrency perpetual futures. arXiv preprint arXiv:2512.22476 URL: https://arxiv.org/abs/2512.22476, doi:10.48550/arXiv.2512.22476.

Deng, K., Cai, H., Lu, J., Qian, C.Z., 2026. From search to research: Exploring search scaling in autonomous quantitative factor mining. arXiv:2609.35559.

Dettmers, T., Pagnoni, A., Holtzman, A., Zettlemoyer, L., 2023. Qlora: Efficient finetuning of quantized llms URL: https://arxiv.org/abs/2305.14314, arXiv:2305.14314.

Diau, E., 2025. Finance as extended biology: Reciprocity as the cognitive substrate of financial behavior. arXiv preprint arXiv:2506.00099v2 URL: https://arxiv.org/pdf/2506.00099v2.

Dicks, M., Paskaramoorthy, A., Gebbie, T., 2023. Many learning agents interacting with an agent-based market model. arXiv preprint arXiv:2303.07393v4 URL: https://arxiv.org/pdf/2303.07393v4.

Ding, H., Li, Y., Wang, J., Chen, H., 2024. Large language model agent in financial trading: A survey. arXiv preprint arXiv:2408.06361 URL: https://arxiv.org/abs/2408.06361, doi:10.48550/arXiv.2408.06361.

Dong, Y., Wu, F., Zhang, K., Dai, Y., Zhang, S., Ye, W., Chen, S., Cheng, Z.Q., 2025. Large language model agents in finance: A survey bridging research, practice, and real-world deployment, in: Findings of the Association for Computational Linguistics: EMNLP 2025, Association for Computational Linguistics, Suzhou, China. pp. 17889–17907. URL: https://aclanthology.org/2025.findings-emnlp.972/, doi:10.18653/v1/2025.findings-emnlp.972.

Dorovatas, V., Schwerin, M., Bagdanov, A.D., Caccia, L., Carta, A., Charlin, L., Hammer, B., Hayes, T.L., Hess, T., Kanan, C., Kudithipudi, D., Liu, X., Lomonaco, V., Mendez-Mendez, J., Patil, D., Prabhu, A., Ricci, E., Tuytelaars, T., van de Ven, G.M., Wang, L., van de Weijer, J., Choi, J., Mundt, M., Aljundi, R., 2026. Modular memory is the key to continual learning agents. arXiv preprint arXiv:2603.01761 URL: https://arxiv.org/abs/2603.01761, doi:10.48550/arXiv.2603.01761.

Du, K., Xing, F., Mao, R., Cambria, E., 2024. Financial sentiment analysis: Techniques and applications. ACM Computing Surveys 56, 1–42. doi:10.1145/3649451.

Duan, Q., Li, C., Wang, C., Zhang, F., 2026. Can agentic trading systems pay for their own intelligence? arXiv:2607.10286.

Duan, Y., Zhang, C., Li, J., 2025. Factormad: A multi-agent debate framework based on large language models for interpretable stock alpha factor mining, in: Proceedings of the 6th ACM International Conference on AI in Finance. URL: https://dl.acm.org/doi/10.1145/3768292.3770377.

Easley, D., López de Prado, M., O'Hara, M., 2012. Flow toxicity and liquidity in a high-frequency world. The Review of Financial Studies 25, 1457–1493. URL: https://doi.org/10.1093/rfs/hhs053, doi:10.1093/rfs/hhs053.

European Parliament, Council of the European Union, 2014. Directive 2014/65/EU of the European Parliament and of the Council of 15 May 2014 on markets in financial instruments and amending Directive 2002/92/EC and Directive 2011/61/EU (consolidated text). Technical Report. Official Journal of the European Union. URL: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A02014L0065-20240328. consolidated version available on EUR-Lex.

European Parliament, Council of the European Union, 2016. Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation). Technical Report. Official Journal of the European Union. URL: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32016R0679.

Fama, E.F., French, K.R., 1993. Common risk factors in the returns on stocks and bonds. Journal of Financial Economics 33, 3–56. URL: https://doi.org/10.1016/0304-405X(93)90023-5, doi:10.1016/0304-405X(93)90023-5.

Fan, T., Yang, Y., Jiang, Y., Zhang, Y., Chen, Y., Huang, C., 2025. Ai-trader: Benchmarking autonomous agents in real-time financial markets. arXiv preprint arXiv:2512.10971 URL: https://arxiv.org/abs/2512.10971, doi:10.48550/arXiv.2512.10971.

Fatemi, S., Hu, Y., 2024. Finvision: A multi-agent framework for stock market prediction, in: Proceedings of the 5th ACM International Conference on AI in Finance (ICAIF '24), pp. 582–590. URL: https://dl.acm.org/doi/10.1145/3677052.3698688, doi:10.1145/3677052.3698688.

Finn, C., Abbeel, P., Levine, S., 2017. Model-agnostic meta-learning for fast adaptation of deep networks, in: International Conference on Machine Learning (ICML), pp. 1126–1135. URL: https://arxiv.org/abs/1703.03400.

Fonseca, X., 2026. Look-ahead-freedom as temporal non-interference: A verifiable correctness property for backtesting and agentic trading pipelines. arXiv:2607.04958.

Foucault, T., 1999. Order flow composition and trading costs in a dynamic limit order market. Journal of Financial Markets 2, 99–134. URL: https://doi.org/10.1016/S1386-4181(98)00012-3, doi:10.1016/S1386-4181(98)00012-3.

Gao, K., Vytelingum, P., Weston, S., Luk, W., Guo, C., 2022. Understanding intra-day price formation process by agent-based financial market simulation: calibrating the extended chiarella model. arXiv preprint arXiv:2208.14207v1 URL: https://arxiv.org/pdf/2208.14207v1.

Geifman, Y., El-Yaniv, R., 2017. Selective classification for deep neural networks. URL: https://arxiv.org/abs/1705.08500, arXiv:1705.08500.

Genet, R., 2025. Deep learning for vwap execution in crypto markets: Beyond the volume curve. arXiv preprint arXiv:2502.13722 URL: https://arxiv.org/abs/2502.13722, doi:10.48550/arXiv.2502.13722.

Giglio, S., Kelly, B., Xiu, D., 2022. Factor models, machine learning, and asset pricing. Annual Review of Financial Economics 14, 337–368. URL: https://www.annualreviews.org/doi/full/10.1146/annurev-financial-101521-104735, doi:10.1146/annurev-financial-101521-104735.

Glosten, L.R., Milgrom, P.R., 1985. Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. Journal of Financial Economics 14, 71–100. URL: https://doi.org/10.1016/0304-405X(85)90044-3, doi:10.1016/0304-405X(85)90044-3.

Go, E., Deb, R., Banerjee, A., 2026. Plan before you trade: Inference-time optimization for rl trading agents. arXiv:2605.12653.

Gong, J., Jiang, M., Liu, Z., Qiao, Y., 2026. Shijianbench: From dialogue to decision for long-horizon evaluation of investment advisors. arXiv:2608.01204.

Gould, M.D., Porter, M.A., Williams, S., McDonald, M., Fenn, D.J., Howison, S.D., 2013. Limit order books. Quantitative Finance 13, 1709–1742. URL: https://doi.org/10.1080/14697688.2013.803148, doi:10.1080/14697688.2013.803148.

Granger, C.W.J., 1969. Investigating causal relations by econometric models and cross-spectral methods. Econometrica 37, 424–438. URL: https://www.jstor.org/stable/1912791, doi:10.2307/1912791.

Grinold, R.C., Kahn, R.N., 2000. Active Portfolio Management: A Quantitative Approach for Producing Superior Returns and Controlling Risk. McGraw-Hill. URL: https://www.mheducation.com/highered/product/active-portfolio-management-grinold-kahn/M9780070248820.html.

Hafsi, Y., Vittori, E., 2024. Optimal execution with reinforcement learning. arXiv preprint arXiv:2411.06389 URL: https://arxiv.org/pdf/2411.06389, doi:10.48550/arXiv.2411.06389.

Hamilton, J.D., 1989. A new approach to the economic analysis of nonstationary time series and the business cycle. Econometrica 57, 357–384. URL: https://www.jstor.org/stable/1912559, doi:10.2307/1912559.

Hansen, P.R., 2005. A test for superior predictive ability. Journal of Business & Economic Statistics 23, 365–380. doi:10.1198/073500105000000063.

Hashimoto, R., Takata, R., Suzuki, M., Tanaka, Y., 2026. Financial market as a self-organized ecosystem: Simulation via learning with heterogeneous preferences. arXiv:2604.23975.

Hashimoto, R., Takayanagi, T., Suzuki, M., Izumi, K., 2025. Agent-based simulation of a financial market with large language models. Proceedings of the International Conference on Principles and Practice of Multi-Agent Systems (PRIMA) URL: https://arxiv.org/abs/2510.12189, doi:10.48550/arXiv.2510.12189.

Hinton, G.E., Salakhutdinov, R.R., 2006. Reducing the dimensionality of data with neural networks. Science 313, 504–507. URL: https://doi.org/10.1126/science.1127647, doi:10.1126/science.1127647.

Hogan, B.R., Chen, X., Wilson, J.T., Rasul, K., 2026. Alphalab: Autonomous multi-agent research across optimization domains with frontier llms. arXiv:2604.08590.

Hong, S., Zhuge, M., Chen, J., Zheng, X., Cheng, Y., Zhang, C., Wang, J., Wang, Z., Yau, S.K.S., Lin, Z., Zhou, L., Ran, C., Xiao, L., Wu, C., Schmidhuber, J., 2024. Metagpt: Meta programming for a multi-agent collaborative framework, in: International Conference on Learning Representations (ICLR). URL: https://arxiv.org/abs/2308.00352, doi:10.48550/arXiv.2308.00352.

Hu, X., Xiong, Y., Zhang, Z., Li, C., 2024. Consensus of a new multi-agent system via multi-task, multi-control mechanism and multi-consensus strategy. Neurocomputing doi:10.1016/j.neucom.2024.127586.

Huang, C., Qiao, Y., Zhang, R., Ye, J., 2026a. Fintoolsyn: A forward synthesis framework for financial tool-use dialogue data with dynamic tool retrieval. arXiv:2603.24051.

Huang, J., Alrassan, A., Harischandra, D., Wu, T., 2026b. Multi-agent debate for explainable trading: Reasoning, consensus, and performance in simulated markets. arXiv:2609.29701.

Huang, W., Yu, Z., 2026. Strat-llm: Stratified strategy alignment for llm-based stock trading with real-time multi-source signals. arXiv:2605.06024.

Huang, Y., Fan, Z., Hu, K., Ye, Y., 2026c. From hypotheses to factors: Constrained llm agents in cryptocurrency markets. arXiv:2604.26747.

Hwang, Y., Kong, Y., Zohren, S., Lee, Y., 2025. Decision-informed neural networks with large language model integration for portfolio optimization. URL: https://arxiv.org/pdf/2502.00828, arXiv:2502.00828.

Iacovides, G., Konstantinidis, T., Xu, M., Mandic, D., 2024. Finllama: Llm-based financial sentiment analysis for algorithmic trading, in: Proceedings of the 5th ACM International Conference on AI in Finance, ACM. pp. 134–141. doi:10.1145/3677052.3698696.

Jagannathan, R., Ma, T., 2003. Risk reduction in large portfolios: Why imposing the wrong constraints helps. The Journal of Finance 58, 1651–1684. URL: https://doi.org/10.1111/1540-6261.00580, doi:10.1111/1540-6261.00580.

Jeon, J., Lee, H., 2026. Can blindfolded llms still trade? an anonymization-first framework for portfolio optimization. arXiv:2603.17692.

Jia, C., Luo, Y., Han, S., Li, P., 2026. Evonash-marl: A closed-loop multi-agent reinforcement learning framework for medium-horizon equity allocation. arXiv:2604.10911.

Jiang, J., Zhou, K., Zhao, W.X., Song, Y., Zhu, C., Zhu, H., Wen, J.R., 2024. Kg-agent: An efficient autonomous agent framework for complex reasoning over knowledge graph. arXiv preprint arXiv:2402.11163v1 URL: https://arxiv.org/pdf/2402.11163v1.

Johnson, J., Douze, M., Jégou, H., 2017. Billion-scale similarity search with GPUs. arXiv preprint arXiv:1702.08734 URL: https://arxiv.org/abs/1702.08734.

Kahneman, D., Tversky, A., 1979. Prospect theory: An analysis of decision under risk. Econometrica 47, 263–291. URL: https://www.jstor.org/stable/1914185, doi:10.2307/1914185.

Kang, Z., Gong, J., Hu, W., Yin, S., Jiang, K., Fang, Z., He, Y., Meng, C., Fu, R., Chen, D., Zheng, L., Jiang, E.H., Feng, Y., Leng, Y., Zhu, J., Chen, X., Yang, X., Xuan, R., 2026. Quanteval: A benchmark for financial quantitative tasks in large language models. arXiv preprint arXiv:2601.08689 URL: https://arxiv.org/abs/2601.08689, doi:10.48550/arXiv.2601.08689.

Kargarzadeh, A., Khaledian, N., Parvini, N., Khaledian, A., 2026. Large language model-driven small-capitalization trading: Integrating financial news sentiment, macroeconomic indicators, and technical signals. arXiv:2608.12283.

Kato, M., 2026. Ai economist agent: An agentic framework for evidence-based economic and financial analysis with rag, knowledge graphs, and large language models. arXiv:2606.20041.

Kelly, J.L., 1956. A new interpretation of information rate. Bell System Technical Journal 35, 917–926. URL: https://onlinelibrary.wiley.com/doi/10.1002/j.1538-7305.1956.tb03809.x, doi:10.1002/j.1538-7305.1956.tb03809.x.

Khatchadourian, R., 2026. Replayable financial agents: A determinism-faithfulness assurance harness for tool-using llm agents. arXiv preprint arXiv:2601.15322v2 URL: https://arxiv.org/pdf/2601.15322v2.

Khezresmaeilzadeh, T., Razmara, P., Azizi, S., Sadeghi, M.E., Potraghloo, E.B., 2025. Vista: Vision-language inference for training-free stock time-series analysis. arXiv preprint arXiv:2505.18570 doi:10.48550/arXiv.2505.18570.

Kim, H., Kim, M., Jung, S., Pyo, S., 2026a. Favor: Llm-based agentic framework for factor mining via empirical validation. arXiv:2608.30192.

Kim, S., Choi, Y., Kang, M., Hwang, S.J., 2026b. Evolvetrade: Experience-driven policy refinement for self-evolving llm trading agents. arXiv:2609.17632.

Kirilenko, A.A., Kyle, A.S., Samadi, M., Tuzun, T., 2017. The flash crash: High-frequency trading in an electronic market. The Journal of Finance 72, 967–998. URL: https://doi.org/10.1111/jofi.12498, doi:10.1111/jofi.12498.

Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Hadsell, R., 2017. Overcoming catastrophic forgetting using elastic weight consolidation. Proceedings of the National Academy of Sciences (PNAS) 114, 3521–3526. URL: https://www.pnas.org/doi/10.1073/pnas.1611835114, doi:10.1073/pnas.1611835114.

Kirtac, K., Germano, G., 2025. Leveraging llm-based sentiment analysis for portfolio optimization with proximal policy optimization, in: Proceedings of the 1st Workshop for Research on Agent Language Models (REALM 2025). URL: https://aclanthology.org/2025.realm-1.12/.

Kocsis, L., Szepesvári, C., 2006. Bandit based monte-carlo planning, in: Machine Learning: ECML 2006, pp. 282–293. URL: https://doi.org/10.1007/11871842_29, doi:10.1007/11871842_29.

Kolonin, A., Glushchenko, A., Bochkov, E., Saxena, A., 2026. Time series augmented generation for financial applications. arXiv:2604.19633.

Kou, Z., Yu, H., Luo, J., Peng, J., Li, X., Liu, C., Dai, J., Chen, L., Han, S., Guo, Y., 2025. Automate strategy finding with llm in quant investment, in: Findings of the Association for Computational Linguistics: EMNLP 2025, Suzhou, China. URL: https://aclanthology.org/2025.findings-emnlp.1005.pdf.

Kvasiuk, Y., Li, T., Colegrove, O., Münchmeyer, M., 2026. Madevolve: Evolutionary optimization of trading systems with large language models. arXiv:2605.23007.

Kyle, A.S., 1985. Continuous auctions and insider trading. Econometrica 53, 1315–1335. URL: https://www.jstor.org/stable/1913210, doi:10.2307/1913210.

Lau, E., Dücker, M., Chaudhary, R., Goh, H.W., 2026. Bankertoolbench: Evaluating ai agents in end-to-end investment banking workflows. arXiv:2604.11304.

Ledoit, O., Wolf, M., 2004. Honey, i shrunk the sample covariance matrix. The Journal of Portfolio Management 30, 110–119. URL: https://doi.org/10.3905/jpm.2004.110, doi:10.3905/jpm.2004.110.

Lee, H., Park, S., Lee, S., Seo, J., 2026. When summaries distort decisions: Information fidelity in llm-compressed financial analysis. arXiv:2606.29251.

Lee, Y., Kim, Y., Kim, J., Kim, S., Lee, Y., 2025. Llm-enhanced black-litterman portfolio optimization. URL: https://arxiv.org/abs/2504.14345, doi:10.48550/arXiv.2504.14345, arXiv:2504.14345.

Letteri, I., 2026. Agenticaita: A proof-of-concept about deliberative multi-agent reasoning for autonomous trading systems. arXiv:2605.12532.

Li, C., Ma, P., Duan, Q., Lin, Z., 2026a. Nextfund: A unified performance tracking platform for agentic portfolio management. arXiv:2607.11141.

Li, H., Cao, Y., Yu, Y., Javaji, S.R., Deng, Z., He, Y., Jiang, Y., Zhu, Z., Subbalakshmi, K.P., Xiong, G., Huang, J., Qian, L., Peng, X., Xie, Q., Suchow, J.W., 2025. Investorbench: A benchmark for financial decision-making tasks with llm-based agent, in: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 2509–2525. URL: https://aclanthology.org/2025.acl-long.126/, doi:10.18653/v1/2025.acl-long.126.

Li, J., Liu, Y., Liu, W., Fang, S., Wang, L., Xu, C., Bian, J., 2024a. Mars: a financial market simulation engine powered by generative foundation model. arXiv preprint arXiv:2409.07486 URL: https://arxiv.org/abs/2409.07486, doi:10.48550/arXiv.2409.07486.

Li, M., Gonsalves, R., Li, W., Yoon, S., 2026b. Trusttrade: Human-inspired selective consensus reduces decision uncertainty in llm trading agents. arXiv:2603.22567.

Li, N., Laryea, A., Ihlamur, Y., 2026c. Optimal stop-loss and take-profit parameterization for autonomous trading agent swarm. arXiv:2604.27150.

Li, S., Zhang, J., Yao, R., Qiu, W., 2026d. Evermine: Dissecting the self-evolution of research capabilities in long-horizon alpha research. arXiv:2609.33524.

Li, Y., Du, N., Song, X., Yang, X., Cui, T., Xue, N., Farjudian, A., Ren, J., Cheah, W.P., 2024b. Cardinality and bounding constrained portfolio optimization using safe reinforcement learning, in: Proceedings of the International Joint Conference on Neural Networks (IJCNN). URL: https://doi.org/10.1109/IJCNN60899.2024.10651491, doi:10.1109/IJCNN60899.2024.10651491.

Li, Y., Liu, S., Liu, B., 2026e. Ai trading's alpha singularity: Emergent market reasoning through agent-to-agent self-evolution. arXiv:2606.29194.

Li, Y., Liu, S., Liu, B., 2026f. Pandaai: A practical agent cq2 for neuro-symbolic data analysis and integrated decision-making in quantitative finance. arXiv:2606.06823.

Li, Y., Liu, S., Liu, B., 2026g. Vst: Verifiable structured transport for auditable agent-to-agent alpha discovery. arXiv:2609.07065.

Li, Y., Luo, B., Wang, Q., Chen, N., Liu, X., He, B., 2024c. CryptoTrade: A reflective LLM-based agent to guide zero-shot cryptocurrency trading, in: Al-Onaizan, Y., Bansal, M., Chen, Y.N. (Eds.), Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Miami, Florida, USA. pp. 1094–1106. URL: https://aclanthology.org/2024.emnlp-main.63/, doi:10.18653/v1/2024.emnlp-main.63.

Li, Y., Yu, Y., Li, H., Chen, Z., Khashanah, K., 2023. Tradinggpt: Multi-agent system with layered memory and distinct characters for enhanced financial trading performance. URL: https://arxiv.org/pdf/2309.03736, doi:10.48550/arXiv.2309.03736, arXiv:2309.03736.

Li, Z., 2026. Design and empirical study of a large language model-based multi-agent investment system for chinese public reits. arXiv preprint arXiv:2602.00082 URL: https://arxiv.org/abs/2602.00082, doi:10.48550/arXiv.2602.00082.

Li, Z., Li, Y., Guo, Y., Zhang, M., 2026h. Autoscientist-quant: Self-evolving coding agents for automatic research in quantitative investment. arXiv:2608.28632.

Lin, L.J., 1992. Self-improving reactive agents based on reinforcement learning, planning and teaching. Machine Learning 8, 293–321. URL: https://doi.org/10.1023/A:1022628806385, doi:10.1023/A:1022628806385.

Liu, B., Dang, R., 2025a. Finpos: A position-aware trading agent system for real financial markets. URL: https://arxiv.org/abs/2510.27251, arXiv:2510.27251.

Liu, B., Dang, R., 2025b. Finrs: A risk-sensitive trading framework for real financial markets. arXiv preprint arXiv:2511.12599v1 URL: https://arxiv.org/pdf/2511.12599v1.

Liu, F., Chen, L., Zheng, Y., Feng, Y., 2022a. A prediction method with data leakage suppression for time series. Electronics 11, 3701. URL: https://www.mdpi.com/2079-9292/11/22/3701, doi:10.3390/electronics11223701.

Liu, F., Fu, Y., Wang, Y., Liu, Q., 2026. Xalpha: A memory-driven ai quant researcher for hypothesis-to-code alpha discovery. arXiv:2607.08332.

Liu, F., Yi, H., Luo, S., Wang, Y., Yang, Y., Li, X., Hu, Z., Feng, J., Liu, Q., 2025. Cognitive alpha mining via llm-driven code-based evolution. arXiv preprint arXiv:2511.18850 URL: https://arxiv.org/html/2511.18850v1.

Liu, N.F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., Liang, P., 2024. Lost in the middle: How language models use long contexts. Transactions of the Association for Computational Linguistics URL: https://arxiv.org/abs/2307.03172, doi:10.1162/tacl_a_00638.

Liu, X.Y., Xia, Z., Rui, J., Gao, J., Yang, H., Zhu, M., Wang, C.D., Wang, Z., Guo, J., 2022b. Finrl-meta: Market environments and benchmarks for data-driven financial reinforcement learning. URL: https://arxiv.org/abs/2211.03107, arXiv:2211.03107.

Liu, X.Y., Yang, H., Chen, Q., Zhang, R., Yang, L., Xiao, B., Wang, C.D., 2020. Finrl: A deep reinforcement learning library for automated stock trading in quantitative finance. URL: https://arxiv.org/abs/2011.09607, arXiv:2011.09607.

Liu, Z., Zhang, Y., Li, P., Liu, Y., Yang, D., 2023. A dynamic llm-powered agent network for task-oriented agent collaboration. arXiv preprint arXiv:2310.02170 URL: https://arxiv.org/abs/2310.02170.

Lo, A.W., Repin, D.V., 2002. The psychophysiology of real-time financial risk processing. Journal of Cognitive Neuroscience 14, 323–339. URL: https://doi.org/10.1162/089892902317361921, doi:10.1162/089892902317361921.

Lopez-Lira, A., 2025. Can large language models trade? testing financial theories with llm agents in market simulations. arXiv preprint arXiv:2504.10789 URL: https://arxiv.org/abs/2504.10789, doi:10.48550/arXiv.2504.10789.

Lu, J., Wang, K., Wang, Y., Tang, Q., Zeng, H., Chen, X., Pi, J., Deng, S., Chen, L., Fu, Y., Yang, K., Sun, X., 2026. Fintoolbench: Evaluating llm agents for real-world financial tool use. arXiv preprint arXiv:2603.08262v1 URL: https://arxiv.org/pdf/2603.08262v1.

Luan, B., Sun, R., Wang, S., Gu, Y., 2026. Finresearchbench ii: A deep research benchmark with consensus-derived gold rubrics for distinguishing financial report quality. arXiv:2607.12252.

Lundberg, S.M., Lee, S.I., 2017. A unified approach to interpreting model predictions, in: Advances in Neural Information Processing Systems (NeurIPS). URL: https://arxiv.org/abs/1705.07874.

Luo, H., Li, Y., An, B.M., Luo, X., 2026a. Liveoption: Evaluating llm agents in structured option trading with nonlinear payoffs. arXiv:2609.33470.

Luo, H., Zhang, Y., Liu, C., 2025. Efs: Evolutionary factor searching for sparse portfolio optimization using large language models. arXiv preprint arXiv:2507.17211 URL: https://arxiv.org/abs/2507.17211, doi:10.48550/arXiv.2507.17211.

Luo, X., Hong, C., Li, H., Xie, Y., 2026b. Foresight-9: Prospective and process-aware evaluation of adaptive trading agents. arXiv:2608.29372.

MacLean, L.C., Thorp, E.O., Ziemba, W.T., 2011. The Kelly Capital Growth Investment Criterion: Theory and Practice. World Scientific. URL: https://doi.org/10.1142/7598, doi:10.1142/7598.

Madaan, A., Tandon, N., Gupta, P., Hallinan, S., Gao, L., Wiegreffe, S., Alon, U., Dziri, N., Prabhumoye, S., Yang, Y., 2023. Self-refine: Iterative refinement with self-feedback. URL: https://arxiv.org/abs/2303.17651, arXiv:2303.17651.

Malkov, Y.A., Yashunin, D.A., 2020. Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. IEEE Transactions on Pattern Analysis and Machine Intelligence 42, 824–836. URL: https://doi.org/10.1109/TPAMI.2018.2889473, doi:10.1109/TPAMI.2018.2889473.

Markowitz, H., 1952. Portfolio selection. Journal of Finance 7, 77–91. URL: https://www.jstor.org/stable/2975974, doi:10.2307/2975974.

McClelland, J.L., McNaughton, B.L., O'Reilly, R.C., 1995. Why there are complementary learning systems in the hippocampus and neocortex: Insights from the successes and failures of connectionist models of learning and memory. Psychological Review 102, 419–457. URL: https://doi.org/10.1037/0033-295X.102.3.419, doi:10.1037/0033-295X.102.3.419.

Muandet, K., Balduzzi, D., Schölkopf, B., 2013. Domain generalization via invariant feature representation, in: Proceedings of the 30th International Conference on Machine Learning (ICML), pp. 10–18. URL: https://proceedings.mlr.press/v28/muandet13.html.

Nechepurenko, M., Shuvalov, P., 2026. Foresight arena: An on-chain benchmark for evaluating ai forecasting agents. arXiv:2605.00420.

Ng, C.C., Lim, J.Y., Low, W.Z., 2025. Prism: Prompt-refined in-context system modelling for financial retrieval. arXiv preprint arXiv:2511.14130 URL: https://arxiv.org/abs/2511.14130, doi:10.48550/arXiv.2511.14130.

Nguyen, P., Pham, T., 2026. Toward reliable evaluation of llm-based financial multi-agent systems: Taxonomy, coordination primacy, and cost awareness. arXiv:2603.27539.

Nguyen, T., 2025. Sharpe-driven stock selection and liquidity-constrained portfolio optimization. URL: https://arxiv.org/pdf/2511.13251, arXiv:2511.13251.

Nie, X., Guo, Z., Chen, Y., Zhou, Y., Zhang, W., 2026. Awcp: A workspace delegation protocol for deep-engagement collaboration across remote agents. arXiv preprint arXiv:2602.20493v1 URL: https://arxiv.org/pdf/2602.20493v1.

Nitarach, N., Sirichotedumrong, W., Pitchayarthorn, P., Taveekitworachai, P., Manakul, P., Pipatanakul, K., 2025. Fincot: Grounding chain-of-thought in expert financial reasoning, in: Proceedings of the 10th Workshop on Financial Technology and Natural Language Processing (FinNLP) at EMNLP. URL: https://arxiv.org/abs/2506.16123, doi:10.48550/arXiv.2506.16123. accepted at FinNLP 2025 (EMNLP).

Ntale, G., 2026. Ai trading: Evaluating large language models for technical market analysis. arXiv:2607.15414.

Olby, O., Bacalum, A., Baggott, R., Stillman, N., 2025. Right place, right time: Market simulation-based rl for execution optimisation, in: Proceedings of ICAIF 2025. URL: https://arxiv.org/abs/2510.22206, doi:10.48550/arXiv.2510.22206.

Oliva, C., Ventura, P.R., Lago-Fernández, L.F., 2025. Multi-objective portfolio optimization via gradient descent. URL: https://arxiv.org/html/2507.16717v1, arXiv:2507.16717.

Oort, C.M.V., Ratliff-Crain, E., Tivnan, B.F., Wshah, S., 2023. Adaptive agents and data quality in agent-based financial markets. arXiv preprint arXiv:2311.15974v1 URL: https://arxiv.org/pdf/2311.15974v1.

Ouyang, S., Sui, P., 2026. Dissecting ai trading: Behavioral finance and market bubbles. arXiv:2604.18373.

Packer, C., Wooders, S., Lin, K., Fang, V., Patil, S.G., Stoica, I., Gonzalez, J.E., 2023. Memgpt: Towards llms as operating systems. arXiv preprint arXiv:2310.08560 URL: https://arxiv.org/abs/2310.08560, doi:10.48550/arXiv.2310.08560.

Pan, C., Ren, L., Xiong, L., Li, Y., 2026. Regime-adaptive continual learning for portfolio management. arXiv:2606.00143.

Pan, S.J., Yang, Q., 2010. A survey on transfer learning. IEEE Transactions on Knowledge and Data Engineering 22, 1345–1359. URL: https://ieeexplore.ieee.org/document/5288526, doi:10.1109/TKDE.2009.191.

Pan, Y., Liew, S.C., 2026. Learning to trade like an expert: Cognitive fine-tuning for stable financial reasoning in language models. arXiv:2604.16862.

Papadakis, C., Filandrianos, G., Dimitriou, A., Lymperaiou, M., Thomas, K., Stamou, G., 2025. Stocksim: A dual-mode order-level simulator for evaluating multi-agent llms in financial markets. URL: https://arxiv.org/abs/2507.09255, doi:10.48550/arXiv.2507.09255, arXiv:2507.09255.

Park, E., 2026. Reflexivity as prompt: Does awareness of self-reinforcing market dynamics improve llms as financial market forecasters? arXiv:2606.00061.

Park, J.S., O'Brien, J.C., Cai, C.J., Ringel Morris, M., Liang, P., Bernstein, M.S., 2023. Generative agents: Interactive simulacra of human behavior, in: Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems (CHI). URL: https://arxiv.org/abs/2304.03442, doi:10.1145/3544548.3581516.

Parlour, C.A., 1998. Price dynamics in limit order markets. Review of Financial Studies 11, 789–816. URL: https://doi.org/10.1093/rfs/11.4.789, doi:10.1093/rfs/11.4.789.

Passos, G.G.F.G., Schmitz, E.A., Ribeiro, S.A., 2026. An artificial market for brazilian real estate investment funds: An agent-based proposal. arXiv:2607.25098.

Pearl, J., 2009. Causality: Models, Reasoning, and Inference. 2 ed., Cambridge University Press. URL: https://www.cambridge.org/core/books/causality/E62B1C761BC88EF7A8FE13A25FDFBBCD.

Pei, Y., Ge, Z., Zheng, J., Cartlidge, J., 2026. Market regime council for dynamic credit assignment in multi-agent llm decision systems. arXiv:2605.24490.

Perold, A.F., 1988. The implementation shortfall: Paper versus reality. The Journal of Portfolio Management 14, 4–9. URL: https://doi.org/10.3905/jpm.1988.409150, doi:10.3905/jpm.1988.409150.

López de Prado, M., 2016. Building diversified portfolios that outperform out of sample. The Journal of Portfolio Management 42, 59–69. URL: https://doi.org/10.3905/jpm.2016.42.4.059, doi:10.3905/jpm.2016.42.4.059.

López de Prado, M., 2018. The 10 reasons most machine learning funds fail. The Journal of Portfolio Management 44, 120–133. URL: https://doi.org/10.3905/jpm.2018.44.6.120, doi:10.3905/jpm.2018.44.6.120. also circulated as an SSRN working paper.

Qian, C., 2026. Information limits and attractor dynamics in economies of frontier llm agents: A pre-registered test. arXiv:2607.06001.

Qian, L., Peng, X., Wang, Y., Zhang, V.J., He, H., Smith, H., Han, Y., He, Y., Li, H., Cao, Y., Yu, Y., Lopez-Lira, A., Lu, P., Nie, J.Y., Xiong, G., Huang, J., Ananiadou, S., 2025. When agents trade: Live multi-market trading benchmark for llm agents. arXiv preprint arXiv:2510.11695 URL: https://arxiv.org/abs/2510.11695, doi:10.48550/arXiv.2510.11695.

Qin, M., Sun, S., Zhang, W., Xia, H., Wang, X., An, B., 2024. Earnhft: Efficient hierarchical reinforcement learning for high frequency trading, in: Proceedings of the AAAI Conference on Artificial Intelligence. URL: https://arxiv.org/abs/2309.12891, doi:10.1609/aaai.v38i13.29384.

Qu, B., Chen, M., 2026. Clqt: A closed-loop, cost-aware, strategy-consistent benchmark for diagnostic evaluation of llm portfolio-management agents. arXiv:2606.29771.

Qu, B., Chen, M., Wang, L., 2026. Propose, don't judge: An anytime-valid referee for llm agents that mine investment factors. arXiv:2609.27051.

Rashid, M., Hong, L., Ding, J., Hossain, K.S.M.T., 2026. Fin-analyst at finmmeval 2026 task 3: A live hybrid trading agent with llm specialists and rule-based signals. arXiv:2607.12233.

Reis, P., Serra, A.P., Gama, J., 2025. A deep learning framework for medium-term covariance forecasting. arXiv preprint arXiv:2503.01581 URL: https://arxiv.org/abs/2503.01581, doi:10.48550/arXiv.2503.01581.

de la Rica Escudero, A., Garrido-Merchán, E.C., Coronado-Vaca, M., 2025. Explainable post hoc portfolio management financial policy of a deep reinforcement learning agent. PLOS ONE 20, e0315528. URL: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0315528, doi:10.1371/journal.pone.0315528.

Rockafellar, R.T., Uryasev, S., 2000. Optimization of conditional value-at-risk. Journal of Risk 2, 21–41. doi:10.21314/JOR.2000.037.

Rodriguez Inserte, P., Nakhl'e, M., Qader, R., Caillaut, G., Liu, J., 2024. Large language model adaptation for financial sentiment analysis. arXiv preprint arXiv:2401.14777 doi:10.48550/arXiv.2401.14777.

Roncalli, T., 2013. Introduction to Risk Parity and Budgeting. CRC Press. URL: https://www.thierry-roncalli.com/RiskParityBook.html.

Ross, S., Gordon, G.J., Bagnell, J.A., 2011. A reduction of imitation learning and structured prediction to no-regret online learning, in: Proceedings of the International Conference on Artificial Intelligence and Statistics. URL: http://proceedings.mlr.press/v15/ross11a.html.

Sakurai, K., Ogawa, T., Haseyama, M., Anan, A., 2026. Dynamic objective selection with safeguards and llm oversight for financial decision-making. arXiv:2606.03704.

Salo, A., Doumpos, M., Liesiö, J., Zopounidis, C., 2024. Fifty years of portfolio optimization: A survey and outlook. European Journal of Operational Research 318, 1–18. URL: https://www.sciencedirect.com/science/article/pii/S0377221723009827, doi:10.1016/j.ejor.2023.12.031.

Saxena, M., Pangallo, M., Hommes, C., Caccioli, F., 2026. Machine spirits: Speculation and adaptation of llm agents in asset markets. arXiv:2604.18602.

Schaul, T., Quan, J., Antonoglou, I., Silver, D., 2016. Prioritized experience replay, in: International Conference on Learning Representations (ICLR). URL: https://arxiv.org/abs/1511.05952.

Shapley, L.S., 1953. A value for n-person games. Contributions to the Theory of Games 2, 307–317. URL: https://www.rand.org/pubs/papers/P295.html.

Sharma, D., Shroff, G., 2026. Algoevolve: Llm-driven meta-evolution of algorithmic trading programs. arXiv:2606.26173.

Sheng, E.N.Y., Shen, B., 2026. Formal mechanisms for market stability in self-interested agent societies: A marketplace simulation study. arXiv:2607.08652.

Shi, Y., Duan, Y., Li, J., 2025. Navigating the alpha jungle: An llm-powered mcts framework for formulaic factor mining. arXiv preprint arXiv:2505.11122 URL: https://arxiv.org/abs/2505.11122, doi:10.48550/arXiv.2505.11122.

Shi, Y., Luo, K., Tang, N., Luo, Y., 2026. Cstrader: A testbed for language-grounded trading in a community-driven virtual asset market. arXiv:2606.31461.

Shiller, R.J., 1981. Do stock prices move too much to be justified by subsequent changes in dividends? American Economic Review 71, 421–436. URL: https://www.aeaweb.org/aer/top20/71.3.421-436.pdf.

Shinn, N., Cassano, F., Berman, E., Gopinath, A., Narasimhan, K., Yao, S., 2023. Reflexion: Language agents with verbal reinforcement learning, in: Advances in Neural Information Processing Systems (NeurIPS). URL: https://papers.nips.cc/paper_files/paper/2023/hash/1b44b878bb782e6954cd888628510e90-Abstract-Conference.html.

Snell, J., Swersky, K., Zemel, R., 2017. Prototypical networks for few-shot learning, in: Advances in Neural Information Processing Systems (NeurIPS). URL: https://arxiv.org/abs/1703.05175.

Subaharan, S., 2026. Controlling long-horizon behavior in language model agents with explicit state dynamics. arXiv preprint arXiv:2601.16087v1 URL: https://arxiv.org/pdf/2601.16087v1.

Sun, H., Zeng, S., 2025. H-mem: Hierarchical memory for high-efficiency long-term reasoning in llm agents. arXiv preprint arXiv:2507.22925 URL: https://arxiv.org/abs/2507.22925, doi:10.48550/arXiv.2507.22925.

Sutton, R.S., Barto, A.G., 2018. Reinforcement Learning: An Introduction. 2 ed., MIT Press. URL: http://incompleteideas.net/book/the-book-2nd.html.

Sutton, R.S., Precup, D., Singh, S., 1999. Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning. Artificial Intelligence 112, 181–211. URL: https://doi.org/10.1016/S0004-3702(99)00052-1, doi:10.1016/S0004-3702(99)00052-1.

Szot, A., Kirchhof, M., Attia, O., Toshev, A., 2026. Expanding llm agent boundaries with strategy-guided exploration. arXiv preprint arXiv:2603.02045 URL: https://arxiv.org/abs/2603.02045, doi:10.48550/arXiv.2603.02045.

Tan, L., Huang, K.W., Wu, K., 2025a. Fred: Financial retrieval-enhanced detection and editing of hallucinations in language models. arXiv preprint arXiv:2507.20930 URL: https://arxiv.org/abs/2507.20930. financial retrieval-enhanced hallucination detection and editing.

Tan, X., Wang, X., Liu, Q., Xu, X., Yuan, X., Zhu, L., Zhang, W., 2025b. Hydrarag: Structured cross-source enhanced large language model reasoning, in: Conference on Empirical Methods in Natural Language Processing (EMNLP). URL: https://arxiv.org/abs/2505.17464, doi:10.48550/arXiv.2505.17464. accepted at EMNLP 2025 (Main Conference).

Tang, Z., Chen, Z., Yang, J., Mai, J., Zheng, Y., Wang, K., Chen, J., Lin, L., 2025. Alphaagent: Llm-driven alpha mining with regularized exploration to counteract alpha decay, in: Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '25). URL: https://dl.acm.org/doi/abs/10.1145/3711896.3736838.

Tetlock, P.C., 2007. Giving content to investor sentiment: The role of media in the stock market. The Journal of Finance 62, 1139–1168. URL: https://doi.org/10.1111/j.1540-6261.2007.01232.x, doi:10.1111/j.1540-6261.2007.01232.x.

Tian, F., Salim, F.D., Xue, H., 2025. Tradinggroup: A multi-agent trading system with self-reflection and data-synthesis, in: Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25). URL: https://arxiv.org/abs/2508.17565.

Tsai, Y.H.H., Bai, S., Yamada, M., Morency, L.P., Salakhutdinov, R., 2019. Multimodal Transformer for unaligned multimodal language sequences, in: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL), pp. 6558–6569. URL: https://aclanthology.org/P19-1656/, doi:10.18653/v1/P19-1656.

Tulving, E., 1972. Episodic and semantic memory, in: Tulving, E., Donaldson, W. (Eds.), Organization of Memory. Academic Press. URL: https://psycnet.apa.org/record/1973-08477-007.

Unnikrishnan, A., 2024. Financial news-driven llm reinforcement learning for portfolio management. URL: https://arxiv.org/pdf/2411.11059, arXiv:2411.11059.

Uysal, A.S., Li, X., Mulvey, J.M., 2024. End-to-end risk budgeting portfolio optimization with neural networks. Annals of Operations Research 339, 397–426. URL: https://doi.org/10.1007/s10479-023-05539-4, doi:10.1007/s10479-023-05539-4. originally circulated as arXiv:2107.04636.

Varghese, J., Bickmann, L., Sandmann, S., 2026. Information specialization and constrained synthesis in multi-agent llm forecasting: A prospective live-study of the 2026 fifa world cup. arXiv:2609.12495.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I., 2017. Attention is all you need, in: Advances in Neural Information Processing Systems (NeurIPS). URL: https://arxiv.org/abs/1706.03762.

Vinyals, O., Blundell, C., Lillicrap, T., Kavukcuoglu, K., Wierstra, D., 2016. Matching networks for one shot learning, in: Advances in Neural Information Processing Systems (NeurIPS). URL: https://arxiv.org/abs/1606.04080.

Voronina, A., Romanko, O., Cao, R., Kwon, R.H., Mendoza-Arriaga, R., 2025. Generative ai-enhanced sector-based investment portfolio construction. arXiv preprint arXiv:2512.24526v1 URL: https://arxiv.org/pdf/2512.24526v1.

Vytelingum, P., Baggott, R., Stillman, N., Zhang, J., Zhu, D., Chen, T., Lyon, J., 2025. Agent-based liquidity risk modelling for financial markets. arXiv preprint arXiv:2505.15296 URL: https://arxiv.org/abs/2505.15296, doi:10.48550/arXiv.2505.15296.

Wang, B., Zhang, X., 2026. Deep reinforcement learning for equity trading: Benchmarking actor-critic methods with forward retraining. arXiv:2609.31870.

Wang, B., Zhou, K., Guo, L., Chen, F., 2026a. Finperma: A theory-informed, event-grounded personalized-memory benchmark for llm agents. arXiv:2608.04095.

Wang, D., Patel, J., Zha, D., Yang, S.Y., Liu, X.Y., 2025a. Finlora: Benchmarking lora methods for fine-tuning llms on financial datasets. arXiv preprint arXiv:2505.19819 URL: https://arxiv.org/pdf/2505.19819, doi:10.48550/arXiv.2505.19819.

Wang, D., Xu, H., Xian, J., 2026b. Polygnosis 2.0: Enhancing llm reasoning via agentic harness engineering for polymarket and osint insight extraction. arXiv:2605.25958.

Wang, L., Ma, C., Feng, X., Zhang, Z., Yang, H., Zhang, J., Chen, Z., Tang, J., Chen, X., Lin, Y., Zhao, W.X., Wei, Z., Ji, H., 2024. A survey on large language model based autonomous agents. Frontiers of Computer Science 18, 186345. URL: https://arxiv.org/abs/2308.11432.

Wang, M., Saxena, N., 2026. Sok: Trading agents or market crashers? dissecting robustness and security failures in academic financial llm trading schemes. arXiv:2609.19705.

Wang, N., Yang, H., Wang, C.D., 2023a. Fingpt: Instruction tuning benchmark for open-source large language models in financial datasets, in: NeurIPS 2023 Workshop on Instruction Tuning and Instruction Following. URL: https://arxiv.org/abs/2310.04793, doi:10.48550/arXiv.2310.04793.

Wang, Q., Tong, Z., Chen, N., Wu, Z., 2026c. Crossalpha: An annual-report benchmark for cross-market factor research (with llm agents). arXiv:2605.29286.

Wang, Q., Wei, Z., Wei, Z., Shen, W., 2026d. Alphadiverse: Post-training local quantitative research agents for diverse exploration in alpha factor mining. arXiv:2609.29014.

Wang, X., Chi, J., Tai, Z., Kwok, T.S.T., Li, M., Li, Z., He, H., Hua, Y., Lu, P., Wang, S., Wu, Y., Huang, J., Tian, J., Mo, F., Cui, Y., Zhou, L., 2025b. Finsage: A multi-aspect rag system for financial filings question answering. arXiv preprint arXiv:2504.14493 URL: https://arxiv.org/abs/2504.14493, doi:10.48550/arXiv.2504.14493. accepted at CIKM 2025.

Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., 2022. Self-consistency improves chain of thought reasoning in language models. URL: https://arxiv.org/abs/2203.11171, arXiv:2203.11171.

Wang, Y., Wang, Y., Deng, J., Tang, H., 2026e. Beyond forecasting: The belief-to-trade layer in prediction-market agents. arXiv:2607.03015.

Wang, Z., Li, Y., Wu, J., Soon, J., Zhang, X., 2023b. Finvis-gpt: A multimodal large language model for financial chart analysis. arXiv preprint arXiv:2308.01430 doi:10.48550/arXiv.2308.01430.

Wang, Z., Yang, W., Wu, Q., Ma, L., 2026f. Backtestbench: Benchmarking large language models for automated quantitative strategy backtesting. arXiv:2605.17937.

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q., Zhou, D., 2022. Chain-of-thought prompting elicits reasoning in large language models, in: Advances in Neural Information Processing Systems (NeurIPS), pp. 24824–24837. URL: https://arxiv.org/abs/2201.11903, doi:10.48550/arXiv.2201.11903.

Wen, Y., Cai, Z., Mirzaei, I., Cai, X., 2026. Evage: Autonomous mev generation and adaptation via multi-agent harness. arXiv:2609.27424.

Weng, Z., Zhang, S., Wang, T., Xia, Y., 2026. Alphalogics: A market logic-driven multi-agent system for scalable and interpretable alpha factor generation. arXiv:2603.20247.

Wheeler, A., Varner, J.D., 2023. Scalable agent-based modeling for complex financial market simulations. arXiv preprint arXiv:2312.14903v2 URL: https://arxiv.org/pdf/2312.14903v2.

White, H., 2000. A reality check for data snooping. Econometrica 68, 1097–1126. URL: https://doi.org/10.1111/1468-0262.00152, doi:10.1111/1468-0262.00152.

Wooldridge, M., 2009. An Introduction to MultiAgent Systems. 2 ed., John Wiley & Sons. URL: https://www.wiley.com/en-us/An+Introduction+to+MultiAgent+Systems%2C+2nd+Edition-p-9780470519462.

Wu, S., Irsoy, O., Lu, S., Dabravolski, V., Dredze, M., Gehrmann, S., Kambadur, P., Rosenberg, D., Mann, G., 2023. Bloomberggpt: A large language model for finance. arXiv preprint arXiv:2303.17564 URL: https://arxiv.org/abs/2303.17564, doi:10.48550/arXiv.2303.17564.

Wu, W., 2026. Auditing asset-specific preferences in financial large language models: Evidence from bitcoin representations and portfolio allocation. arXiv:2606.02528.

Xia, Y., Wang, T., Liew, S.C., Zhang, S., 2025. A hierarchical deep learning framework for pair trading with attention and graph networks. Expert Systems with Applications 291, 128496. URL: https://www.sciencedirect.com/science/article/pii/S0957417425011183, doi:10.1016/j.eswa.2025.128496.

Xia, Y., Wang, T., Zhang, S., Weng, Z., Cao, B., Liew, S.C., 2026. HiveMind: Contribution-guided online prompt optimization of LLM multi-agent systems, in: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 29767–29774. URL: https://ojs.aaai.org/index.php/AAAI/article/view/40222, doi:10.1609/AAAI.V40I35.40222, arXiv:2512.06432.

Xiao, M., Jiang, Z., Qian, L., Chen, Z., He, Y., Xu, Y., Jiang, Y., Li, D., Weng, R.L., Peng, M., Huang, J., Ananiadou, S., Xie, Q., 2025a. Retrieval-augmented large language models for financial time-series forecasting. arXiv preprint arXiv:2502.05878 URL: https://arxiv.org/abs/2502.05878, doi:10.48550/arXiv.2502.05878.

Xiao, Y., Sun, E., Chen, T., Wu, F., Luo, D., Wang, W., 2025b. Trading-r1: Financial trading with llm reasoning via reinforcement learning. arXiv preprint arXiv:2509.11420 URL: https://arxiv.org/abs/2509.11420, doi:10.48550/arXiv.2509.11420.

Xiao, Y., Sun, E., Luo, D., Wang, W., 2024. Tradingagents: Multi-agents llm financial trading framework. arXiv preprint arXiv:2412.20138 URL: https://arxiv.org/abs/2412.20138, doi:10.48550/arXiv.2412.20138.

Xie, Q., Han, W., Chen, Z., Xiang, R., Zhang, X., He, Y., Xiao, M., Li, D., Dai, Y., Feng, D., Xu, Y., Kang, H., Kuang, Z., Yuan, C., Yang, K., Luo, Z., Zhang, T., Liu, Z., Xiong, G., Deng, Z., Jiang, Y., Yao, Z., Li, H., Yu, Y., Hu, G., Huang, J., Liu, X.Y., Lopez-Lira, A., Wang, B., Lai, Y., Wang, H., Peng, M., Ananiadou, S., Huang, J., 2024. Finben: A holistic financial benchmark for large language models, in: Advances in Neural Information Processing Systems (NeurIPS), pp. 95716–95743. URL: https://arxiv.org/abs/2402.12659.

Xie, Q., Han, W., Zhang, X., Lai, Y., Peng, M., Lopez-Lira, A., Huang, J., 2023. Pixiu: A large language model, instruction data and evaluation benchmark for finance. arXiv preprint arXiv:2306.05443 URL: https://arxiv.org/abs/2306.05443, doi:10.48550/arXiv.2306.05443.

Xing, F., 2025. Designing heterogeneous llm agents for financial sentiment analysis. ACM Transactions on Management Information Systems 16, 1–24. doi:10.1145/3688399.

Xiong, G., Deng, Z., Wang, K., Cao, Y., Li, H., Yu, Y., Peng, X., Lin, M., Smith, K.E., Liu, X.Y., Huang, J., Ananiadou, S., Xie, Q., 2025. FLAG-TRADER: Fusion LLM-agent with gradient-based reinforcement learning for financial trading, in: Che, W., Nabende, J., Shutova, E., Pilehvar, M.T. (Eds.), Findings of the Association for Computational Linguistics: ACL 2025, Association for Computational Linguistics, Vienna, Austria. pp. 13921–13934. URL: https://aclanthology.org/2025.findings-acl.716/, doi:10.18653/v1/2025.findings-acl.716.

Xu, F., Yin, Y., Zhang, X., Liu, T., Jiang, S., Zhang, Z., 2024a. Alpha^2: Discovering logical formulaic alphas using deep reinforcement learning. arXiv preprint arXiv:2406.16505 URL: https://arxiv.org/abs/2406.16505.

Xu, Y., Xie, L., Gu, X., Chen, X., Chang, H., Zhang, H., Chen, Z., Zhang, X., Tian, Q., 2024b. Qa-lora: Quantization-aware low-rank adaptation of large language models, in: Proceedings of the International Conference on Learning Representations (ICLR). URL: https://openreview.net/forum?id=WvFoJccpo8, doi:10.48550/arXiv.2309.14717.

Xue, W., 2026. Representation signatures and risk-feedback alignment in llm trading agents. arXiv:2605.28850.

Yang, H., Liu, X.Y., Zhong, S., Walid, A., 2025. Deep reinforcement learning for automated stock trading: An ensemble strategy. arXiv preprint arXiv:2511.12120 URL: https://arxiv.org/abs/2511.12120, doi:10.48550/arXiv.2511.12120.

Yang, J., Wu, Y., Liu, J., Liang, P., 2026. Recursive multi-agent trading system: Iterative optimized portfolio strategy under geopolitical uncertainty. arXiv:2605.25311.

Yang, Z., 2026. When valid signals fail: Regime boundaries between llm features and rl trading policies. arXiv:2604.10996.

Yao, J., Zheng, Z., Li, B., 2026. Beyond agent architecture: Execution assumptions and reproducibility in llm-based trading systems. arXiv:2606.08285.

Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T.L., Cao, Y., Narasimhan, K., 2023a. Tree of thoughts: Deliberate problem solving with large language models, in: Advances in Neural Information Processing Systems (NeurIPS). URL: https://arxiv.org/abs/2305.10601, doi:10.48550/arXiv.2305.10601.

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., Cao, Y., 2023b. React: Synergizing reasoning and acting in language models, in: International Conference on Learning Representations (ICLR). URL: https://arxiv.org/abs/2210.03629, doi:10.48550/arXiv.2210.03629.

Ye, W., Sun, Y., Ren, X., Yu, S., 2026a. Agonalpha: Autonomous alpha discovery via prompt economy and scalable agentic search. arXiv:2608.11250.

Ye, W., Xu, J., Wu, Y., 2026b. When ai agents meet mev: Cross-chain arbitrage in the agentic economy. arXiv:2609.17897.

Ye, Y., Han, J., Hu, A., Bu, J., 2026c. The alpha illusion: Reported alpha from llm trading agents should not be treated as deployment evidence. arXiv:2605.16895.

Yi, J., Yang, J., Jin, Y., Li, Y., 2026. Alphaschema: Exploring the space of trading semantics for llm-based alpha mining. arXiv:2607.26642.

Yu, H., Li, F., You, J., 2025. Livetradebench: Seeking real-world alpha with large language models. arXiv preprint arXiv:2511.03628 URL: https://arxiv.org/abs/2511.03628, doi:10.48550/arXiv.2511.03628.

Yu, H., Zheng, Z., Pan, J.Z., Liu, T., 2026a. Alphamemo: Structured search-process memory for self-evolving alpha mining agents. arXiv:2606.20625.

Yu, S., Xue, H., Ao, X., Pan, F., He, J., Tu, D., He, Q., 2023a. Generating synergistic formulaic alpha collections via reinforcement learning, in: Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. URL: https://arxiv.org/abs/2306.12964, doi:10.1145/3580305.3599831.

Yu, X., Zhou, J., Ding, G., Wei, W., 2026b. Can ai make money in crypto? measuring the gap from backtests to real markets. arXiv:2609.34510.

Yu, Y., Li, H., Chen, Z., Jiang, Y., Li, Y., Zhang, D., Liu, R., Suchow, J.W., Khashanah, K., 2023b. Finmem: A performance-enhanced llm trading agent with layered memory and character design. arXiv preprint arXiv:2311.13743 URL: https://arxiv.org/abs/2311.13743, doi:10.48550/arXiv.2311.13743.

Yu, Y., Yao, Z., Li, H., Deng, Z., Cao, Y., Chen, Z., Suchow, J.W., Liu, R., Cui, Z., Xu, Z., Zhang, D., Subbalakshmi, K., Xiong, G., He, Y., Huang, J., Li, D., Xie, Q., 2024. Fincon: A synthesized llm multi-agent system with conceptual verbal reinforcement for enhanced financial decision making, in: Advances in Neural Information Processing Systems (NeurIPS). URL: https://arxiv.org/abs/2407.06567, doi:10.48550/arXiv.2407.06567.

Yuan, X., Xu, H., Xu, S., Zou, C., Xiong, J., 2026a. Traderbench: How robust are ai agents in adversarial capital markets? arXiv preprint arXiv:2603.00285v1 URL: https://arxiv.org/abs/2603.00285.

Yuan, Y., Sheng, J., Zeng, S., Wang, J., 2026b. Alphacrafter: Harnessing multi-agent workflows for cross-sectional quantitative trading. arXiv:2605.05580.

Yuksel, K.A., Sawaf, H., 2024. A multi-ai agent system for autonomous optimization of agentic ai solutions via iterative refinement and llm-driven feedback loops. arXiv preprint arXiv:2412.17149v1 URL: https://arxiv.org/pdf/2412.17149v1.

Yun, J., Lee, H.J., Jeon, I., 2025. Quantevolve: Automating quantitative strategy discovery through multi-agent evolutionary framework. arXiv preprint arXiv:2510.18569 URL: https://arxiv.org/abs/2510.18569, doi:10.48550/arXiv.2510.18569.

Zhang, B., 2026a. Graph attention-based heterogeneous multi-agent deep reinforcement learning for adaptive portfolio optimization. Scientific Reports 16, 2674. URL: https://www.nature.com/articles/s41598-025-32408-w, doi:10.1038/s41598-025-32408-w.

Zhang, B., Yang, H., Liu, X.Y., 2023a. Instruct-fingpt: Financial sentiment analysis by instruction tuning of general-purpose large language models. arXiv preprint arXiv:2306.12659 URL: https://arxiv.org/abs/2306.12659, doi:10.48550/arXiv.2306.12659.

Zhang, B., Yang, H., Zhou, T., Babar, M.A., Liu, X.Y., 2023b. Enhancing financial sentiment analysis via retrieval augmented large language models, in: Proceedings of the 4th ACM International Conference on AI in Finance, pp. 349–356. doi:10.1145/3604237.3626866.

Zhang, K., Ge, W., Jiang, L., Yang, W., 2026a. Openfingym: A verifiable multi-task gym environment for evaluating quant agents. arXiv:2606.26350.

Zhang, L., 2026b. Llm-enhanced dynamic financial knowledge graphs for cross-entity signal propagation and alpha discovery. arXiv:2607.10932.

Zhang, O., Zhang, Z., 2026. A review of large language models for stock price forecasting from a hedge-fund perspective. arXiv:2605.05211.

Zhang, W., Jia, R., Wang, Y., Cheng, D., Zhao, M., Chen, C., 2025. Enhancing portfolio optimization via heuristic-guided inverse reinforcement learning with multi-objective reward and graph-based policy learning, in: Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI). URL: https://www.ijcai.org/proceedings/2025/1054.pdf.

Zhang, W., Zhao, L., Xia, H., Sun, S., Sun, J., Qin, M., Li, X., Zhao, Y., Zhao, Y., Cai, X., Zheng, L., Wang, X., An, B., 2024. A multimodal foundation agent for financial trading: Tool-augmented, diversified, and generalist, in: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24). URL: https://arxiv.org/abs/2402.18485, doi:10.1145/3637528.3671801. aCM KDD 2024. Full paper PDF available at the NTU author page.

Zhang, Y., Koyluoglu, O.O., Venkatesh, T., Martinez, R.D., 2026b. Frontierfinance: A challenging benchmark for measuring frontier intelligence of finance agents. arXiv:2608.11683.

Zhang, Z., Stadie, B.C., 2026. Temporal leakage in llm backtesting: Measurement, validation, and adjusted scores. arXiv:2608.02985.

Zhao, L., Sun, R., Jiang, Z., Yang, B., Bai, Y., Chen, M., Wang, X., Li, J., Bai, Z., 2025. Contesttrade: A multi-agent trading system based on internal contest mechanism. arXiv preprint arXiv:2508.00554 URL: https://arxiv.org/abs/2508.00554, doi:10.48550/arXiv.2508.00554.

Zhao, R., Raissi, M., 2026. Backtrader-bench: Benchmarking llm agents on algorithmic trading with self-generated mcqs. arXiv:2608.11232.

Zhao, S., Dong, Z., Cao, Z., Douady, R., 2024. Hedge fund portfolio construction using polymodel theory and itransformer. arXiv preprint arXiv:2408.03320 URL: https://arxiv.org/abs/2408.03320, doi:10.48550/arXiv.2408.03320.

Zhao, S., Sha, T., 2026. Goant: Quality-diversity multi-agent search for alpha factor discovery in market microstructure data. arXiv:2609.08719.

Zhao, Y., Yang, Z., Zhu, Z., 2026. Alphapareto: Formulaic alpha discovery with llm-guided multi-objective reinforcement learning. arXiv:2609.34188.

Zhu, L., Cai, M., 2026. Artificial intelligence in equity and crypto markets: Progress, profitability evidence, and the limits of automated investing. arXiv:2609.04917.

Zhu, Q., Lu, Z., Zhu, J., Chen, N., 2026a. Alphaseek: Trajectory-level self-iterative factor mining framework for multi-source financial data. arXiv:2608.13913.

Zhu, T., Zhao, W., Sun, R., Luan, B., 2026b. From knowing to doing: A memory-controlled benchmark for llm trading agents on stock markets. arXiv:2605.28359.

Zong, C., Wang, C., Qin, M., Feng, L., Wang, X., An, B., 2024. Macrohft: Memory augmented context-aware reinforcement learning on high frequency trading. arXiv preprint arXiv:2406.14537 URL: https://arxiv.org/abs/2406.14537, doi:10.48550/arXiv.2406.14537. accepted to KDD 2024.

Zou, M., Chen, J., Luo, A., Dai, J., Zhang, C., Sun, D., Xu, Z., 2026. Finevo: From isolated backtests to ecological market games for multi-agent financial strategy evolution. arXiv preprint arXiv:2602.00948 URL: https://arxiv.org/abs/2602.00948, doi:10.48550/arXiv.2602.00948.

Zouaoui, H., Naas, M.N., 2025. Portfolio optimization based on mpt-lstm neural networks: A case study of cryptocurrency markets. Finance, Accounting and Business Analysis (FABA) 7, 82–98. URL: https://faba.bg/index.php/faba/article/view/265/137, doi:10.37075/FABA.2025.1.07.

Appendix A. Systematic Search Summary

This appendix documents the information sources and query patterns that define the survey's search boundary and support reproduction of the search.

Table A.22: Information sources and search summary (submitted as Supplementary Material S1).

Source	Notes
ACM Digital Library	Keyword search over agentic trading, LLM agents, and financial decision-making.
IEEE Xplore	Keyword search focused on trading systems, execution, and risk control.
arXiv	Preprints in cs.AI/cs.LG/q-fin with agent/trading keywords.
SSRN	Working papers in quantitative finance and computational finance.
Google Scholar	Broad supplementary source for cross-venue discovery and citation chaining.

Table A.23: Example query patterns used in the search.

Target	Query pattern (example)
Agentic trading	(`agent` OR `agentic` OR `LLM agent`) AND (`trading` OR `portfolio` OR `execution` OR `risk`)
LLM finance agents	(`large language model` OR LLM) AND (`finance` OR `financial`) AND (`agent` OR `tool` OR `multi-agent`)
Evaluation / validity	(`trading` OR `backtest`) AND (`look-ahead` OR `leakage` OR `transaction cost` OR `slippage` OR `survivorship`)

Coverage note. This survey discusses literature through the formal registry cutoff of **2026-09-30**. The initial ledger was screened through **2026-03-09**; at revision, we queried arXiv in an extension window covering **2026-03-10 to 2026-09-30** with the same query families in Table A.23 supplemented by trading-agent, multi-agent-market, and alpha-mining phrase patterns, retrieving 1,539 unique records. Of these, 154 records entered eligibility assessment under the same inclusion boundary; the combined v2 registry records the per-record decisions. Supplementary S1 retains the initial-window registry, retrieval counts, and screening logs. Extension query families are described here and in Section 2; the extension retrieval dumps are separate working materials. Table A.23 lists representative example patterns; the full six query families are enumerated in Sec. 2. Evidence annotations, screening counts, and protocol-coded denominators all derive from the combined registry snapshot.

Appendix B. Corpus Snapshot of the Evidence Ledger

This appendix consolidates corpus-level descriptive findings that follow from the review protocol but are not themselves part of the protocol definition. Unless stated otherwise, denominator-based summaries here use the primary empirical subset (n = 80), while the full included ledger contains 185 studies partitioned into 80 primary empirical studies and 105 background studies.

The primary subset is concentrated in very recent work: within the current ledger, **79/80** primary studies were published between 2024 and 2026. As coded by venue type in the registry, **19/80** primary studies appear in peer-reviewed venues (journal or conference), **61/80** are preprints or otherwise venue-unrecorded, and **0/80** remain uncoded. The registry records the first public version year; a later journal version may therefore have a different publication year. These counts describe the publication landscape of the assembled corpus.

The primary subset reports different levels of protocol detail. In the pooled 80-study subset, **50/80** studies report an extractable split protocol, **47/80** specify a transaction-cost model, **48/80** document universe/survivorship handling, **47/80** report execution timing or semantics, and **35/80** meet the R1–R3 artifact-availability proxy, derived from the reproducibility code rather than a separate direct artifact-extraction field. Table 4 separates these pooled rates by cohort and records the retrieval and coding differences needed for interpretation. Evaluation types are distributed across backtest, simulation, live, and benchmark evaluations; narrative-only and unclear types are outside this empirical denominator.

Reading note. In the figures below, `Unknown` denotes information that is not reported clearly enough in the source for reliable extraction under our conservative coding policy; the panels are corpus-level reporting distributions within the current ledger.

Appendix B.1. Extension-Cohort Landscape

Table B.25 groups the extension-window background/context studies by the role they serve in the map. Market-level multi-agent simulation ecosystems, perception/retrieval infrastructure, and additional validity-oriented studies are discussed in the body text (Sections 3, 4, 11 and 13); the table collects representative studies from the remaining background cohort so that the extension window's coverage can be inspected in one place.

Appendix C. Representative A-C-A Reclassification Notes

This appendix provides reclassification notes for 19 representative works to illustrate how the Architecture-Capability-Adaptation (A-C-A) lens distinguishes system design choices. For each work, we document: (i) classification under prior frameworks (where available), (ii) A-C-A reclassification with coding rationale, and (iii) the specific analytical insight surfaced by the lens.

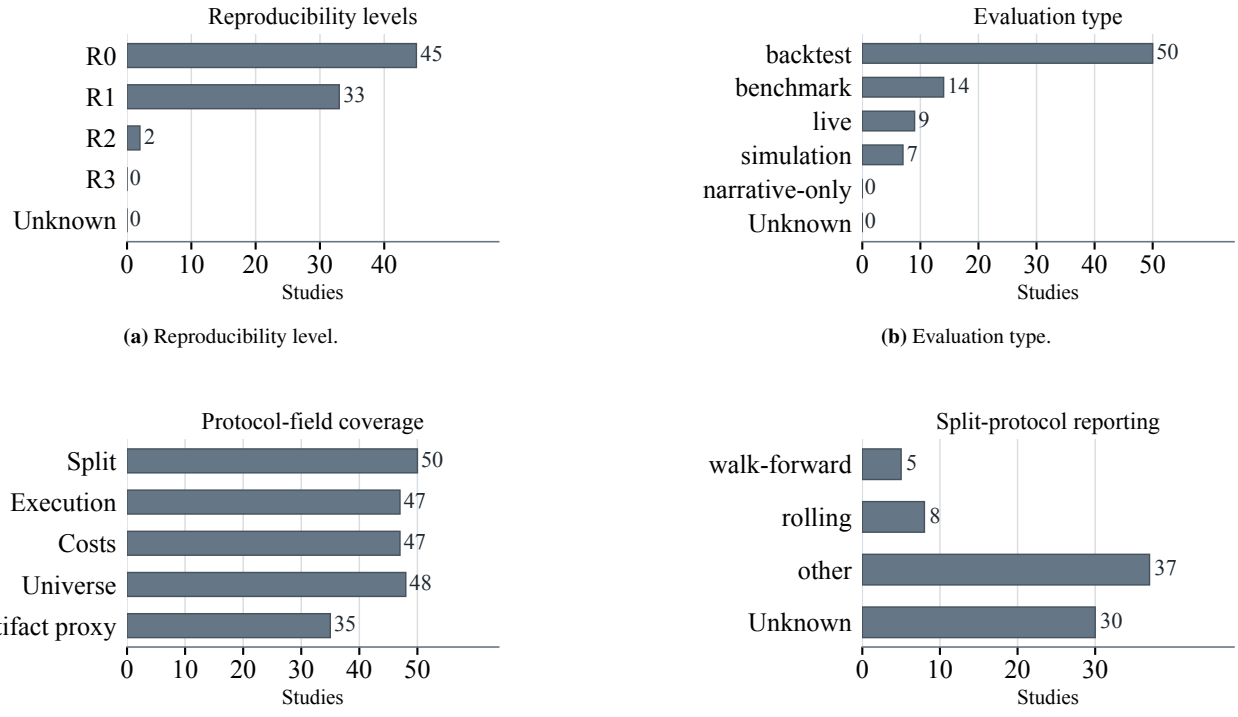
Appendix C.1. Reclassification Procedure

Paper Selection. We selected 19 works representing diverse architectural approaches, capabilities, and adaptation mechanisms. Selection criteria: (a) coverage of all three A-C-A dimensions, (b) citation count or recency (indicating influence), (c) availability of technical details sufficient for classification.

Coding Procedure. For each paper, one author prepared an initial A-C-A classification and coding rationale, and the authors then reconciled edge cases through discussion. The resulting notes document how the lens is applied to the selected works. Inter-rater agreement and external validation of these A-C-A classifications were not assessed.

Table B.24: Reproducibility levels within the primary empirical subset (n = 80).

Level	Definition	Criteria	Count	%
R0	No runnable artifacts	No publicly accessible code repository, or repository is broken/404, or it lacks essential files and cannot run	45	56.2
R1	Code available (not runnable)	Public repository exists and is accessible, but key components are missing (dependencies, data access, eval scripts)	33	41.2
R2	Runnable with gaps	Complete code is available and data is accessible or clearly documented, and evaluation scripts are provided, but pinned environment, complete documentation, or an end-to-end pipeline may still be missing	2	2.5
R3	Full reproduction package	Complete code with pinned environment, data snapshots or clear access instructions, complete documentation, and an end-to-end evaluation pipeline	0	0.0
R2 or above (runnable with gaps or better)			**2**	**2.5**



(a) Reproducibility level.

(b) Evaluation type.

(c) Protocol-field coverage.

(d) Split-protocol reporting.

Figure B.16: Corpus-level reporting distributions within the primary empirical subset (n = 80).

Prior Framework Identification. For each work, we identified its classification in existing surveys where available: Wang et al. 2024 (LLM-Agent Survey), Ding et al. 2024 (LLM-Finance Survey), or task-centric benchmarks (FinBen, InvestorBench). Where a work was not explicitly labeled in one of those sources, the shorthand in the Prior column should be read as our closest reconstruction of how that work would map to the earlier scheme.

Appendix C.2. Reclassification Dataset

Memory descriptors in the table retain terms used in individual systems. In the survey's operational taxonomy, experience retrieval is discussed under associative vector storage when vector indexing is specified, while explicit financial knowledge stores

Table B.25: Extension-cohort landscape: background/context studies included in the ledger (2026-03-10 to 2026-09-30), grouped by the role they play for readers. These studies inform design and evaluation practice but are not part of the primary empirical subset (n = 80).

Reader-facing role	Representative studies
Benchmarks & evaluation harnesses	BacktestBench Wang et al. (2026f), Backtrader-Bench Zhao and Raissi (2026), FinResearchBench II Luan et al. (2026), BankerToolBench Lau et al. (2026), Foresight Arena Nechepurenko and Shuvalov (2026), ValueBlindBench Chang et al. (2026), ShiJianBench Gong et al. (2026), FrontierFinance Zhang et al. (2026b), QuantEval Kang et al. (2026), reliable multi-agent evaluation Nguyen and Pham (2026)
Research tooling & agent infrastructure	AlphaLab Hogan et al. (2026), PolyGnosis 2.0 Wang et al. (2026b), interaction-native knowledge harnesses Borjigin et al. (2026), RIDGE Cheng et al. (2026a), PandaAI Li et al. (2026f), agentic forecasting infrastructure Chai et al. (2026)
Surveys & cross-cutting analyses	agentic AI in finance Aldridge et al. (2026), LLM stock forecasting review Zhang and Zhang (2026), AI in equity/crypto markets Zhu and Cai (2026)
Adjacent financial-agent applications	AI-economist frameworks Kato (2026), multi-agent venture-capital due diligence Alexandrou and Pramatari (2026), verifiable forecast actions Chen et al. (2026a)

are discussed under structured knowledge bases; these descriptors do not by themselves establish a particular implementation.

Table C.26 presents the complete reclassification notes, organized by architectural family.

Appendix C.3. Illustrative Patterns from Reclassification

Pattern 1: Architecture-Capability Non-Alignment. The same Reactive Architecture (FinAgent, AgentGuard) supports different capabilities (α vs R) depending on perceptual input and memory integration. This cross-cutting pattern is obscured by capability-centric categorizations.

Pattern 2: Adaptation Level Distinction. The adaptation dimension distinguishes prompt- and memory-mediated updates (FinAgent, RAG-Fintech, FinRS, and FinCon) from weight-based RL (Alpha^2, FinRL-DeepSeek) and structural search (AlphaAgent, FinEvo). MetaGPT provides a background example of SOP-guided prompting, rather than financial SFT. Each update mechanism has distinct evaluation requirements (Sec. 13).

Pattern 3: Multi-Agent Architectural Hierarchy. Prior frameworks often use broad "multi-agent" labels. The A-C-A lens distinguishes role-based, hierarchical, and market-ecology coordination patterns, making the coordination mechanism visible rather than treating agent count as the primary distinction.

Appendix C.4. Coding Status

The coding notes document qualitative author reconciliation and the rationale for boundary cases, providing a traceable account of applying the A-C-A lens to the selected works.

Appendix C.5. Limitations of the Reclassification Notes

1. **Selection Bias:** 19 works selected from a broader review corpus spanning included studies, background references, and a small number of boundary cases used to stress-test the A-C-A lens; they were not randomly sampled.
2. **Temporal Coverage:** 15 of these 19 illustrative works first appeared in 2024–2026 (79%), using the first public version year rather than a later publication year. This denominator is separate from the 80-study empirical subset and reflects field recency. Earlier works may reveal different architectural patterns.
3. **Prior Framework Availability:** Not all works classified in existing surveys. Where unavailable, we inferred likely categorization based on paper content and survey scope.

Across these examples, the A-C-A lens distinguishes design choices and surfaces architectural patterns that may be obscured by broader system- or task-level labels.

Appendix D. Excluded Studies and Rationale

To keep the evidence-mapping process auditable, this appendix distinguishes two decisions that are easily conflated. First, *final exclusions* are candidate-registry records outside the finance, trading, portfolio, or risk-management scope of this survey. Second, *background-tier boundary decisions* are finance-relevant records that remain in the included evidence ledger but do not enter the primary empirical subset because they fail at least one of three criteria: Action Output, Closed-Loop Evaluation, or an eligible empirical evaluation type. We document both decision types—using PRISMA-ScR as a reporting aid for scoping-style transparency rather than as a claim of a full systematic-review workflow—to facilitate audit and assessment of potential selection bias.

Appendix D.1. Exclusion Categories and Boundary Decisions

Candidate records were counted as final exclusions only when they fell outside the domain boundary of this survey. Finance-relevant records that were prediction models, benchmarks, tools, or conceptual background papers were retained as background or context where useful, but were excluded from the primary empirical denominator. The operational criteria are:

Table C.26: Representative A-C-A reclassification notes for 19 works, documenting architectural components, financial capabilities, adaptation mechanisms, and the rationale for each mapping.

Work	Prior	A-C-A Reclassification	P	M	R	Adapt	Key Insight from A-C-A
FinBERT Araci (2019)	W: "NLP Model", D: "Sentiment"	Arch: Text Perception only; Cap: BG (no Action); Adapt: BG	✓	–	–	BG	Excluded as BG: demonstrates Action Output criterion excludes pure prediction
FinGPT Wang et al. (2023a)	W: "LLM Tool", D: "Generation"	Arch: Text Perception + Static Memory; Cap: BG (no closed-loop); Adapt: BG	✓	✓	–	BG	Excluded as BG: Generation without execution integration
FinLlama Iacovides et al. (2024)	D: "Sentiment"	Arch: Text Perception; Cap: BG; Adapt: BG	✓	–	–	BG	Same architecture as FinBERT but instruction-tuned; capability depends on integration, not just model
AlphaAgent Tang et al. (2025)	D: "Multi-agent"	Arch: Reflective/Strategic hybrid + Episodic Memory; Cap: α; Adapt: Self-reflection	✓	✓	✓	SE	Multi-agent debate = validation mechanism for Alpha, not coordination capability
FinAgent Zhang et al. (2024)	W: "Multi-modal", D: "Agent"	Arch: Multi-modal P + Episodic M + Reflective R + Microstructure A; Cap: α (cross-modal); Adapt: ICL	✓	✓	✓	ICL	Core: cross-modal perception-to-action pipeline, not "multi-modality"
Alpha^2 Xu et al. (2024a)	B: "Alpha Mining"	Arch: Working Memory + Strategic Reasoning (MCTS); Cap: α (evolutionary); Adapt: RL	–	✓	✓	RL	MCTS is Strategic Reasoning; evolutionary = search-based adaptation
RAG-Fintech Cook et al. (2025)	D: "Retrieval"	Arch: Text P + Semantic M (KB) + Reflective R; Cap: α (retrieval-based); Adapt: ICL	✓	✓	✓	ICL	Retrieval = Memory architecture, not just augmentation technique
RAPTOR Almon et al. (2025)	D: "Portfolio"	Arch: Working M + Strategic R + Execution A; Cap: P (hierarchical); Adapt: ICL	–	✓	✓	ICL	Portfolio = Strategic + Execution integration, not separate capability
AgentGuard Chen and Cong (2025)	W: "Safety"	Arch: Reactive R + Action A with gating; Cap: R (real-time); Adapt: Rule-based	–	–	✓	–	Risk as gating mechanism in Action architecture, not post-hoc
FinRS Liu and Dang (2025b)	D: "Risk"	Arch: Text P + Episodic M + Reflective R; Cap: R (risk-sensitive); Adapt: Reward reflection	✓	✓	✓	ICL	Risk capability derives from Perception-Memory-Reasoning chain
TradingAgents Xiao et al. (2024)	W: "Multi-agent"	Arch: Hierarchical role-based Coordination; Cap: α+P+R distributed; Adapt: Hierarchical feedback	✓	✓	✓	ICL + SE	Analyst, research, trader, risk, and manager teams form a decision hierarchy
TradingGroup Tian et al. (2025)	D: "Multi-agent"	Arch: Role-based + Shared Memory; Cap: α; Adapt: Self-reflection	✓	✓	✓	SE	Self-reflection coordinates roles, not external mechanism
FINCON Yu et al. (2024)	D: "Consensus"	Arch: Role-based + Communication Protocol; Cap: P; Adapt: CVRF belief updates	✓	–	✓	ICL	CVRF updates investment beliefs across episodes
FinMem Yu et al. (2023b)	W: "Memory"	Arch: Working M + Layered long-term M + Strategic R; Cap: P; Adapt: Memory consolidation	–	✓	✓	SE	Memory hierarchy enables temporal abstraction for Portfolio
MetaGPT Hong et al. (2024)	W: "Multi-agent"	Arch: Hierarchical (Manager + Engineers); Cap: BG (software engineering); Adapt: SOP-guided prompting	✓	✓	✓	–	General workflow design; no evaluated trading capability
FinRL-DeepSeek Benhenda (2025)	B: "RL Trading"	Arch: Working M + Reactive/Strategic hybrid R + Execution A; Cap: α+P; Adapt: RL	–	✓	✓	RL	RL spans multiple capabilities with shared architecture
Reflexion Shinn et al. (2023)	W: "Reflection"	Arch: Working + Episodic M + Reflective R; Cap: BG (general); Adapt: Self-reflection	–	✓	✓	SE	Self-reflection as adaptation mechanism applicable to trading
Hashimoto Hashimoto et al. (2025)	B: "Simulation"	Arch: Market Ecology (no central coordination); Cap: α+P+R emergent; Adapt: Evolutionary	✓	✓	✓	RL + SE	Ecology = emergent capabilities from interaction, not design
FinEvo Zou et al. (2026)	D: "Evolution"	Arch: Market Ecology + Hierarchical; Cap: α; Adapt: Evolutionary + SE	✓	✓	✓	SE	Combines ecology (interaction) with hierarchy (structure)

Prior-column shorthand: W = Wang et al. 2024 (LLM-Agent Survey); D = Ding et al. 2024 (LLM-Finance Survey); B = task-centric benchmark (FinBen, InvestorBench). Where a work was not explicitly labeled in those sources, the shorthand is our closest reconstruction of how it would map to the earlier scheme. P/M/R mark the Perception–Memory–Reasoning architectural components.

E1 No Action Output: Pure prediction or analysis systems that do not generate tradable actions. Finance-relevant examples remain included as background.

E2 No Eligible Empirical Evaluation: Finance-relevant studies without a qualifying backtest, simulation, live-trading, or benchmark evaluation. Narrative-only or unclear evaluation reports remain background/context even when action output and a closed-loop claim are coded present.

E3 Non-Financial Domain: Systems applied to non-financial markets or generic agent settings without financial trading, portfolio, execution, or risk-management instantiation.

Table D.27: Representative final exclusions and background boundary cases.

Study (Representative)	Category	Decision rationale
Final excluded records in the current canonical registry		
Socio-Agentic LLMs for Socio-Technical Systems	E3	Generic socio-technical agent paper outside finance/trading/portfolio/risk management.
Learning Game-Playing Agents with Generative Code Optimization	E3	General game-playing agent study without financial-market instantiation.
NeSyCoCo	E3	Neuro-symbolic compositional generalization study outside the trading domain.
Agentic AI for Commercial Insurance Underwriting	E3	Insurance underwriting rather than trading, portfolio construction, execution, or market-risk management.
Included background boundary cases		
FinBERT (Araci 2019)	E1/BG	Finance-relevant sentiment model retained as background; no tradable action loop, so it is excluded from the primary empirical subset.
FinGPT (Wang 2023a)	E1/BG	Finance language-model/tooling work retained as background; no explicit closed-loop trading execution integration.
InvestorBench	E1/BG	Finance benchmark retained as background; the registry codes no tradable action output, so it does not enter the primary empirical subset.

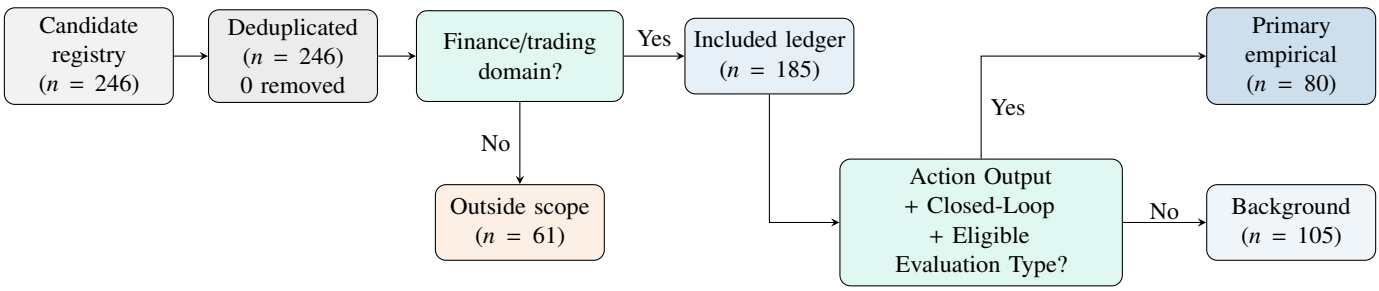


Figure D.17: Inclusion, exclusion, and evidence-tier decision flow. Final exclusions are outside the survey domain; finance-relevant records that fail Action Output, Closed-Loop Evaluation, or the eligible empirical evaluation-type rule remain in the included background tier.

These form the final excluded set in the current canonical registry.

- **E4 No LLM Component**: Traditional quantitative trading systems without a language-model component. These may be cited only as background context when needed for finance methodology.
- **E5 Duplicate**: Multiple versions of the same work; the most complete version is retained.
- **E6 Inaccessible**: Full text unavailable after retrieval attempts.

Appendix D.2. Examples of Final Exclusions and Background Cases

Table D.27 separates final exclusions from background boundary cases. The aggregate exclusion count in the current canonical registry is 61, all of which are outside the finance/trading scope; the background cases remain in the 185-record included ledger but do not enter the 80-record primary empirical subset. Under the current boundary, the 102 included records tagged `primary` in S0 v2 include 22 narrative-only records; these are assigned to the background denominator, yielding 80 empirical primary and 105 background records.

Appendix D.3. Inclusion/Exclusion Decision Flow

Figure D.17 illustrates the screening logic encoded in the canonical ledger and summarized in the main text, making the inclusion/exclusion workflow traceable.

Appendix D.4. Impact of Exclusion Criteria on Evidence Set

After deduplication, all 61 final exclusions fall outside the survey domain. Finance-relevant records failing the three-part empirical criterion remain background/context. Reassigning 22 `narrative-only` studies among 102 `primary`-tagged records yields 80 empirical primary and 105 background studies. Table 3 reports these macro-backed categories, rather than earlier screening-log totals.